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The Ocean's role in driving Antarctic Sea Ice Predictability

by

Stephy Ann Libera (M.Sc., B.Sc.)



UNIVERSITY of
TASMANIA

Submitted in fulfilment of the requirements for the Degree of Doctor of Philosophy

Institute for Marine and Antarctic Studies, University of Tasmania,

Hobart, Australia

December 2023

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Stratification dominates accurate simulation of sea ice predictability

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Table of Contents

List of figures.....	xiii
Abstract.....	xx
 Chapter 1: Introduction	1
1.1 Antarctic sea ice and its critical role in the climate system	2
1.2 Insights from Antarctic sea ice predictability studies	8
1.3 Thesis aim and objectives	14
 Chapter 2: Ocean-sea ice processes and their role on multi-month predictability of Antarctic sea ice.....	16
Chapter overview	16
Abstract	18
2.1 Introduction	18
2.2 Methods.....	20
2.3. Results	23
2.4 Discussions.....	28
2.5 Conclusions	34
 Chapter 3: Ocean-sea ice processes driving the variability in regional Antarctic sea ice predictability	36
3.1 Introduction	37
3.2 Methods.....	39
3.3 Results	40
3.4 Discussion	54
3.5 Conclusion.....	57

Chapter 4: Stratification dominates accurate simulation of sea ice predictability	59
4.1 Introduction.	59
4.2 Methods.....	61
4.3 Results	64
4.4 Discussion	72
4.5 Conclusion.....	74
Chapter 5: Discussion and Conclusions.....	77
Appendices.....	83
References:.....	100

List of figures

- 1.1 Summer (March) and Winter (September) Antarctic sea ice concentration, with the 1981-2010 average marked using the pink sold line. (Credit: National Snow and Ice Data Centre; Source: <https://nsidc.org/learn/parts-cryosphere/sea-ice>). 02

- 1.2 (a) Composite differences in the 500 hPa geopotential height field (units: geopotential meters) based on the phase of Zonal Wave 3. Positive minus negative phase. Arrows indicate the direction of anomalous airflow associated with circulation pattern. (b) Correlation of the Zonal Wave 3 index on sea ice concentration for April, May, and June. Extracted from (Raphael, 2007). 04

- 1.3 Schematic depicting the water masses and their transformation in the Southern Ocean. Upwelled Circumpolar Deep Water (CDW) modified to form Sub Antarctic Mode Water (SAMW) and Antarctic Intermediate Water (AAIW) forming the upper cell and lower [Antarctic Bottom Water (AABW)] cells of the meridional overturning circulation in the Southern Ocean; extracted from Abernathey et al. (2016). Various bulk contributions to the freshwater exchange at the ocean surface (positive downward) south of 50° S are given in units of freshwater sverdrups: 1 fwSv = 106 m³ freshwater s⁻¹. 06

- 1.4 Schematic representing the timescales of weather prediction, seasonal to interannual predictions and climate projections; extracted from Kirtman et al. (2013). 9

- 1.5 (a) Prognostic predictability of ice-edge locations as a function of longitude and lead time. (b) Correlation between sea surface temperature in September and lagged upper ocean temperature. (c) prognostic predictability of the upper ocean heat content (0-100 m) for the same model used for (a). (d) diagnostic sea ice predictability of observed sea ice in the Weddell Sea region. (a-c) extracted from Marchi et al. (2019) and (d) from Morioka et al. (2021). 13

- 2.1 (a) Map defining the Weddell sector in Antarctica along with the model climatological winter maximum (September) of sea ice extent (in shading) and 22

summer minimum (February, white contour). (b) Sea ice predictability: autocorrelation of sea ice area from observation, and (c) Sea ice predictability from model output. In (b) and (c), SIA from initial months (or lead) along the y-axis are correlated against the SIA in the future months (or lags) along the x-axis and statistically significant values (>95%) are hatched. Summer persistence (*) and spring reemergence (**) patterns are marked in (c).

2.2 Comparison of summer persistence (plots on the left) and spring reemergence 24

patterns (plots on the right) evident from sea ice (top row) and upper ocean predictability (0-200 m) (middle and bottom row). Sea ice predictability: correlation of SIA from January (a) and October (b) initial months with future months; Correlation between SIA in (c) January and (d) October with ocean temperature in future months; Correlation of ocean temperature in (e) January and (f) October with future months. Statistically significant values (>95%) are hatched in all panels. In the oceanic predictability results (c-f), the thick black line is the vertical temperature gradient (dT/dz) maximum, dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification.

2.3 Annual evolution of climatological (a) Sea ice area (red line) and 32

thermodynamic freeze and melt (blue line), (b) Temperature, (c) Salinity, (d) vertical temperature gradient (dT/dz), and (e) vertical density gradient (dp/dz), spatially averaged over the sea ice-covered ocean zone of the Weddell sector in the upper 200 m. The vertical temperature gradient maximum (thick black line), vertical temperature gradient (white lines), and vertical density gradient maximum (dashed line) are the same as used in Figure 2, and mixed layer depth (red line) is marked on all panels.

3.1 (a) Map of the six regions around Antarctica (a), also showing the 44

climatological winter sea ice extent maximum (in September; shading) and minimum (in February; white contour). (b) Graphical representation of the main seasonal sea ice predictability regimes and associated features; (1) Summer

persistence, (2) Winter persistence, (3) Spring re-emergence, (4) Predictability suppression, and (5) Predictability barrier. (c)-(h) Sea ice predictability results for the six regions, where the total sea ice areas from the initial months (along y-axis) are correlated against the total sea ice areas in the future months (along x-axis) and statistically significant values (>95 %) are hatched.

- 3.2** Ice-ocean correlations in the upper ocean (0-200 m) for the six sectors showing summer persistence from January initial month. Each of the panels also includes the sea ice predictability from January initial month for the respective sectors for comparison. The thick black line and the dashed black line over the ice-ocean correlations are the climatological vertical temperature gradient (dT/dz) maximum and vertical density gradient (dp/dz) maximum. The vertical temperature gradients are marked using white contours, white dashed contours bound the vertical temperature gradient values that are negative due to summer stratification. Statistically significant values (>95%) are hatched on all panels. **45**
- 3.3** Stratification strength at the permanent pycnocline (PP) and seasonal pycnocline (SP) for all six sectors. Annual evolution of the vertical temperature gradient at (a) the PP and (b) at the seasonal pycnocline. And annual evolution of vertical density gradient at (c) the PP and (d) the SP. (e) Map of climatological Surface Heat Flux (SHF) in January, showing the six sectors used in this study and highlighting SHF values equalling 60 Wm^{-2} (black line) and climatological sea ice extent (white line). **48**
- 3.4** Ice-ocean correlations in the upper ocean (0-200 m) for the six sectors showing the spring re-emergence pattern occurring in the six sectors from various initial months. Each of the panels includes corresponding sea ice predictability for respective sectors and initial months. Hatching and all contours used in the panels are described in Figure 2. **49**
- 4.1** Figure 4.1: Map of Antarctica showing the six regions used in the study King Haakon Sea (KH), East Antarctica (EA), Ross Sea (RS), Amundsen Sea (AS), Bellingshausen Sea (BS) and Weddell Sea (WS). (a) the number of available **63**

observed ocean profiles (for the period of study 1985-2015) in summer (January-March) and (b) in winter (July-September). The selected regions used for calculating upper ocean climatologies are marked using red polygons. (c) Total number of profiles per month for the three selected regions (for 1985-2015). The months where the climatological ocean observations have less than 25 profiles in (c) are hatched to indicate less confidence in Figures 4.3, 4.4, and 4.5.

- 4.2** Regional Sea ice predictability (autocorrelation of sea ice area) for the six sectors around Antarctica. Model sea ice predictability (LHS) and observed sea ice predictability (RHS) are presented side-by-side. In each panel, the sea ice area from the initial month (along the y-axis) is correlated against the future months or lags (along the x-axis) and statistically significant values (>95%) are hatched. The model sea ice predictability for the six sectors (a-1, b-1, c-1, d-1, e-1, and f-1) are the same as shown in Figure 3.1. **66**
- 4.3** Ocean climatology of the upper ocean in a region within the East Antarctic sector for (a) temperature from the model and (b) observation. (c) Salinity from the model and (d) observation, (e) the difference in vertical temperature gradient between the model and observation and (f) the difference in vertical density gradient between the model and observation. In panels (a-e), the solid black line is the climatological vertical temperature gradient maximum and the dashed black line is the climatological vertical density gradient maximum. In panels (e) solid and dashed black lines are the climatological vertical temperature gradient maximum in the model and observation respectively. In panel (f) solid and dashed black lines are climatological vertical density gradient maximum in the model and observation respectively. **68**
- 4.4** Same as Figure 4.3 but for the region in the Bellingshausen Sea sector. **69**
- 4.5** Same as Figure 4.3 but for the region in the Weddell Sea sector. **70**

Supplementary Figures:

- S1** Correlation between sea ice area (SIA) and ocean temperature in future months (ice-ocean correlations) for all 12 initial months in the Weddell Sea sector. The black line is the vertical temperature gradient (dT/dz) maximum, the dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification. Statistically significant values (>95%) are hatched in all panels. **85**
- S2** Hydrological profiles of spatially averaged monthly climatological (a) temperature, (b) salinity, (c) vertical temperature gradient, (d) vertical salinity gradient with markers of vertical temperature gradient maximum (black circles) and vertical density gradient maximum (red circles) in the Weddell Sea sector. **86**
- S3** Correlation between the total sea ice area and sea ice concentration in the Weddell Sea region. Longitude along the x-axis and latitude along the y-axis. **86**
- S4** Longitude-depth section plot of Weddell Sea ocean temperatures in January (representing summer) and October (representing winter). **87**
- S5** Latitude-depth section plot of Weddell Sea ocean temperatures in January (representing summer) and October (representing winter). **87**
- S6** Regional plot of Weddell Sea mixed layer depth distribution in January (representing summer) and October (representing winter). The winter mixed layer depths show the model producing considerably deeper mixed layers near the continental shelf edge (compared to the observed depth ranges, not shown). **88**
- S7** (a) Sea ice volume tendencies in the Weddell Sea sector, showing the contribution from various freeze and melt components; (b) the net freeze by net melt ratio for the Weddell Sea sector. **89**

Sx1	Correlation between initial sea ice area and ocean temperature in future months (ice-ocean correlations) for all 12 initial months for the King Haakon VII sea region. The black line is the vertical temperature gradient (dT/dz) maximum, the dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification. Statistically significant values (>95%) are hatched in all panels. The initial months for the sea ice area are marked as the title on each subplot).	90
Sx2	Same as Sx1, but for the East Antarctic sector.	91
Sx3	Same as Sx1, but for the Ross Sea sector.	92
Sx4	Same as Sx1, but for the Amundsen Sea sector.	93
Sx5	Same as Sx1, but for the Bellingshausen Sea sector.	94
Sx6	Same as Sx1, but for the Weddell Sea sector.	95
Sx7	Evaluation of Arctic and Antarctic sea ice simulated in the ACCESS-OM2 model against the observed sea ice record. (a–b) Sea ice extent, (c–d) volume, and (e–f) 1993–2017 mean annual cycle of the extent in the (a, c, e) Arctic and (b, d, f) Antarctic in the three configurations. Panels (a)–(d) show running 12-month minima, means, and maxima. Observational estimates are based on passive microwave retrievals (NSIDC Sea Ice Index version 3, Fetterer et al., 2017). Note that the three models and observations use different land masks, so the time series would not be expected to agree perfectly in magnitude. Figure extracted from (Kiss et al., 2020) with permission from the author.	96
Sx8	The 1993–2017 mean Antarctic sea ice thickness (ice volume per unit area; a, d) and concentration (ice area per unit area; b, e) in ACCESS-OM2-01 and concentration estimates from NOAA G02202 V3 passive microwave Goddard merged monthly data (Peng et al., 2013; Meier et al., 2017, c, f) for (a)–(c)	97

September and (d)–(f) February. The concentration scale is nonlinear to highlight differences at high concentrations. Contour lines show the sea ice extent (defined as the 15 % concentration contour) in all three model configurations and observations. Figure extracted from (Kiss et al., 2020) with permission from the author.

- Sz1** The spatial distribution of ocean profiles; (a) instrumented seal data, (b) CTD data, (c) drifting Buoy data, (d) glider data, (e) ocean station data, (f) Argo float data, (g) moored data and (h) Undulating Oceanographic Recorder (Towed CTD) data. **98**
- Sz2** Decomposition of the number of observed ocean profiles in the (a) Weddell Sea, (b) East Antarctic and (c) Bellingshausen Sea sectors for the three decades of the study period. **99**

List of tables

- 3.1** Table 3.1: Dynamic contribution of sea ice growth in the six sectors showing the net dynamic sea ice growth term integrated over a year and divided with the integrated annual sea ice volume. **53**

Abstract

Antarctic sea ice is a critical component of the climate system and acts as a vital habitat for Southern Ocean ecosystems. Understanding the drivers and physical processes influencing Antarctic sea ice predictability is of broad interest. This thesis presents a comprehensive investigation of the physical processes underlying Antarctic sea ice predictability.

This thesis investigated the predictability of sea ice and the upper ocean in the Weddell Sea region using outputs from a high-resolution global coupled sea ice ocean model; the Australian Community Climate and Earth System Simulator (ACCESS-OM2). Predictability in the ice and the ocean are consistent with each other and follow similar evolution. Oceanic predictability or ‘oceanic memory’ is the major source of memory underlying sea ice predictability. Oceanic memory in our analysis (i.e., the correlation between sea ice area and lagged upper ocean temperature), can be interpreted as residual thermal anomalies from sea ice processes lingering in the ice-ocean system at inter-seasonal timescales. This implies that sea ice predictability is a signature of local ice-ocean interaction. Oceanic memory of sea ice anomalies is largely confined to the Winter Water layer (WW) bounded by the permanent pycnocline (PP), which prevents deep water from mixing into the WW and helps sustain the ocean memory. Once the mixed layer deepens in winter to reach the PP, further deepening induces entrainment of warm Circumpolar Deep Water which has no sea ice-related memory into the WW and causes the dissipation of predictability (termed as the predictability barrier). Our analysis shows that sea ice and upper ocean predictability is closely tied to the vertical ocean structure and its seasonal evolution.

Further, the regional variability of sea ice and ocean predictability in six sectors around Antarctica is assessed using model data. Individual sectors produced unique sea ice and ocean predictability patterns and showed strong regional variability. Summer persistence and spring re-emergence are the two main predictability regimes found among the sectors. A major regional contrast in sea ice predictability is the lack of summer persistence in the eastern sectors (King Haakon Sea, East Antarctica, and Ross Sea). This contrast is also seen in the ocean predictability, where the western sectors produced high correlations extending deep (~200 m)

into the ocean and persisting longer, while in the eastern sectors, correlations form only in a shallow layer (~50 m) and dissipate earlier. It is inferred that the lack of summer persistence in the eastern sectors is due to the rapid dissipation of oceanic memory to the atmosphere due to weak stratification of the seasonal pycnocline and lack of summer sea ice cover in the eastern sectors.

Finally, the model sea ice predictability is compared with observed sea ice predictability. Sea ice predictability in the model (ACCESS-OM2) is found to be more persistent than observed. On evaluating the climatological ocean conditions in the model and observations, the model biases in sea ice predictability are traced back to systematic biases in the representation of the upper ocean vertical structure. The model has an overly stratified upper ocean, which preserves upper ocean memory over longer timescales and produces longer predictability skill. This over-stratified upper ocean is a consequence of the model ocean being too fresh above the permanent pycnocline, and too warm below. Our study shows that a better representation of the upper ocean in the sea ice zone of the Southern Ocean can produce sea ice predictability of better agreement with observation. Since ocean vertical structure, especially changes in stratification can have a strong influence on predictability, the disparity between observed and model predictability presents a priority area for future model improvements and developments.

This thesis builds on existing research on Antarctic sea ice predictability and presents further evidence on the dominant role of the ocean in retaining ice-ocean memory and driving sea ice predictability. It presents evidence on how the memory in the ocean follows the seasonal evolution of the upper ocean and sheds light on how variability in the upper ocean structure leads to regional differences in sea ice predictability. The findings in this thesis not only elucidate the physical processes in the upper ocean underlying Antarctic sea ice predictability but also present the nature of these processes in the form of an ‘ice-ocean interaction’. The investigation in this thesis is one of the first to provide a detailed assessment of the ocean’s role in modulating seasonal predictability. Our results also imply that changes to upper ocean conditions in a warming climate might alter sea ice predictability patterns in the future.

CHAPTER 1

Chapter 1: Introduction

Antarctic sea ice plays an important role in the global climate through its influence on local and global oceanic and atmospheric circulations, planetary radiative balance, and the crucial support it provides for the Southern Ocean ecosystem. Understanding the physical processes influencing Antarctic sea ice, the drivers of its change, and being able to predict Antarctic sea ice are therefore of broad interest.

Observations of Antarctic sea ice concentrations reveal high regional and seasonal variability (Zwally, 1983; Holland, 2014), and large interannual variability in total annual mean sea ice extent (Parkinson, 2019). Since sea ice forms from ocean-atmosphere exchanges of heat, freshwater, and momentum, and integrates to all major components of the Antarctic climate system, our understanding of observed sea ice changes also reflects our level of understanding of the Antarctic climate system as a whole (Hobbs et al., 2016). Due to a number of complex interactions and processes that influence Antarctic sea ice involving the atmosphere, ocean, and adjacent Antarctic ice shelves, presently there is low confidence in the projections of Antarctic sea ice (Meredith et al., 2019). In return, uncertainties in sea ice projections reduce the confidence of various components interacting with sea ice, such as confidence in projections of Antarctic ice-sheet mass balance (Meredith et al., 2019), which is of relevance to global sea level projections. Understanding the complex interaction between Antarctic sea ice and all of the components of the climate system, and how they relate to broader issues of climate change is an important scientific challenge.

This thesis contributes to the overall goal of advancing our knowledge of the complex interactions occurring in the Antarctic climate system. Using sea ice predictability as a tool, this thesis investigates Antarctic sea ice-ocean interaction.

1.1 Antarctic sea ice and its critical role in the climate system

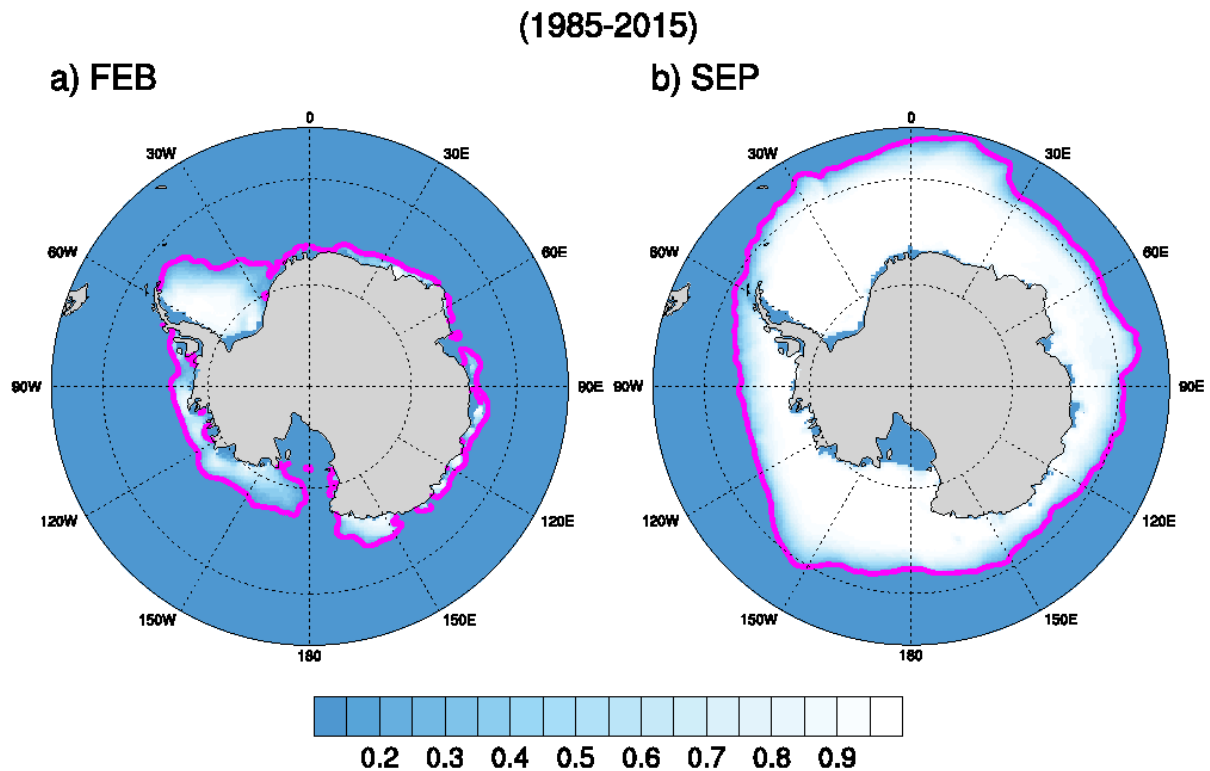


Figure 1.1: Summer (February) and Winter (September) Antarctic sea ice concentration ($\times 10^{-1}$ %), with the 1985-2015 average sea ice extent marked using the pink solid line.

The annual expansion and contraction of Antarctic sea ice produces arguably the largest seasonal cycle on the planet (Handcock and Raphael, 2020). The annual sea ice cover ranges from ~ 3 million km^2 in summer to ~ 19 million km^2 in winter (summer and winter sea ice cover from 2020 are shown in Figure 1.1), covering about 4% of the Earth's surface (Comiso and Nishio, 2008; Massom and Stammerjohn, 2010; Parkinson, 2019). The snow-covered sea ice reflects nearly all (~ 80 - 90%) incoming sunlight and shields the much darker ocean surface from radiative warming (Brandt et al., 2005; Zhou et al., 2019). Sea ice cover also modulates the exchange of heat, gases, and momentum between the ocean and the atmosphere, making sea ice a key agent in a spectrum of feedbacks in the polar regions (Hall, 2004; Goosse et al., 2018).

Stammerjohn et al. (2012) investigated ice-ocean feedbacks driving the changes in the timing of the advance and retreat of Antarctic sea ice for the 1980-2011 period. They found opposing

regional trends: sea ice retreated 1 month earlier and advanced 2 months later in the Antarctic Peninsula and Bellingshausen Sea region (3 months longer ice-free season), while in the western Ross Sea, sea ice retreated ~1 month later and advanced ~1 month earlier (>2 months longer sea ice season). Stammerjohn et al. (2012), then assessed the potential feedbacks leading to the rapid changes in sea ice and showed that there was a strong correspondence between the timing of sea ice retreat and subsequent advance. They linked this correspondence to ice-albedo and ocean heat feedbacks, where earlier sea ice retreat exposes the ocean to radiative warming for a longer period, increasing the solar ocean warming and resulting in later sea ice advance.

Ice-albedo studies have largely focused on changes happening in the Arctic, where sustained losses in sea ice (post-2000) contributed to increasing the strength of ice-albedo feedback (i.e., more solar warming due to more open water) (Curry et al., 1995). Until 2015, the expansion of Antarctic sea ice generated negative ice-albedo feedback in the Southern hemisphere, however, the precipitous loss of Antarctic sea ice since 2016 completely reversed the sign and impacted the planetary albedo (Riihelä et al., 2021). The mean snow and ice albedo feedback (SIAF) between the 1992-2018 period is equivalent to approximately 10% of anthropogenic carbon dioxide emissions over the same period. The contribution will rise considerably if the record-low Antarctic sea ice conditions become common. The recent changes in Antarctic sea ice cover and the reversal of global SIAF highlight the importance of Antarctic sea ice cover to global radiative feedback (Riihelä et al., 2021).

The role of the atmosphere in driving Antarctic sea ice variability is long established (Gordon and Taylor, 1975). Strong interactions occur between Antarctic sea ice and the atmosphere from synoptic to planetary scales (Lemke et al., 1980; Cavalieri and Parkinson, 1981; Kwok and Comiso, 2002; Raphael, 2007; Raphael et al., 2011; Holland et al., 2018). Antarctic sea ice cover impacts the atmosphere through its effect on the heat flux, surface temperature, surface winds and cyclone frequency near the Antarctic continent (Mitchell and Senior, 1989; Simmonds and Wu, 1993; Simmonds et al., 2003). Atmospheric processes influence Antarctic sea ice variability mainly through changes in the wind (Holland and Kwok, 2012; Raphael and Hobbs, 2014; Hobbs et al., 2016; Raphael et al., 2019). Warm and moist winds from the mid-latitudes move sea ice poleward, resulting in higher surface temperatures and reduced sea ice extent. While the cold dry winds from the Antarctic continent contribute to the equatorward drift and increased sea ice extent (Holland and Kwok, 2012; Hobbs et al., 2016).

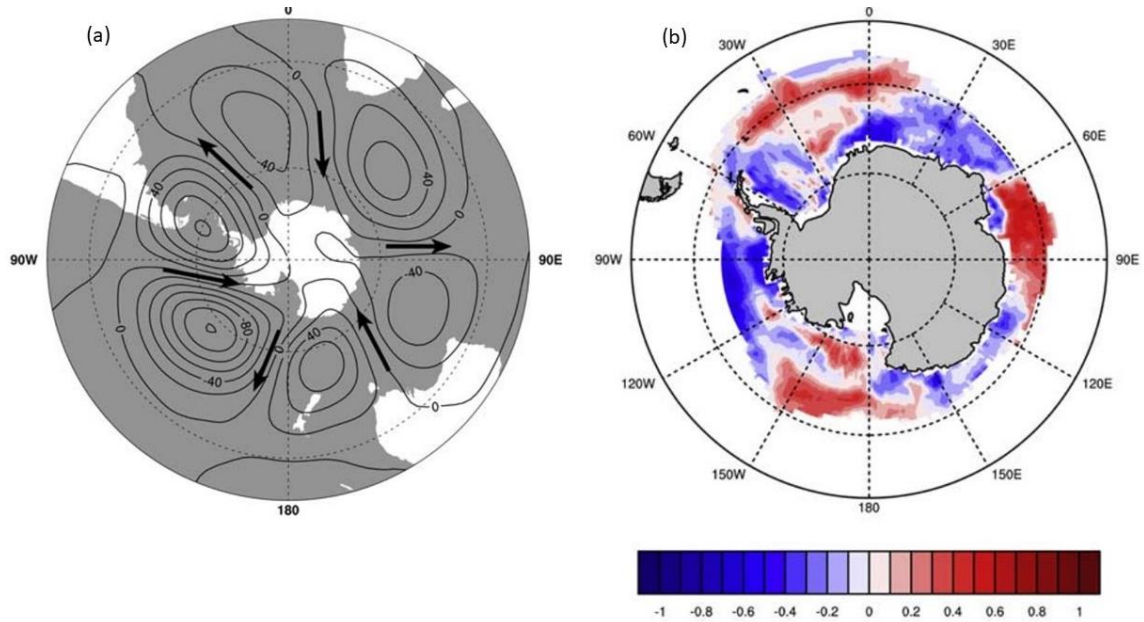


Figure 1.2: (a) Composite differences in the 500 hPa geopotential height field (units: geopotential meters) based on the phase of Zonal Wave 3. Positive minus negative phase. Arrows indicate the direction of anomalous airflow associated with circulation pattern. (b) Correlation of the Zonal Wave 3 index on sea ice concentration for April, May, and June. Extracted from (Raphael, 2007)

Large-scale atmospheric circulation systems and patterns acting at daily to interannual timescales such as the Southern Annular Mode (Hall and Visbeck, 2002; Lefebvre, 2004; Sen Gupta and England, 2006; Stammerjohn et al., 2008), Zonal Wave 3 (ZW3) (Raphael, 2004; Raphael, 2007; Raphael and Hobbs, 2014), and Amundsen Sea Low (Raphael et al., 2016; Raphael et al., 2019) largely modulate the high latitude wind system and thereby impact the variability of Antarctic sea ice. The ZW3 is a quasi-stationary atmospheric wave, it is different from the Antarctic Circumpolar Wave (ACW), which is a propagating feature coupled between the ocean and atmosphere and is non-stationary (circling the Southern Ocean in approximately eight years). The meridional airflow associated with the ZW3 pattern (Figure 1.2 a) is presented as an example. Beyond the large-scale atmospheric patterns in the southern high latitudes, tropical, and extra-tropical patterns are found to interact with Antarctic sea ice through atmospheric teleconnections (Carleton, 2003; Lachlan-Cope and Connolley, 2006; Stammerjohn et al., 2008; Matear et al., 2015; Chung et al., 2022).

Complex interactions occur between Antarctic sea ice and the ocean below: Sea ice is frozen seawater, although it is defined as a component of the cryosphere, it is inevitably a part of the ocean mixed layer and therefore regional variability in sea ice characteristics reflects the

variability of the mixed layer and part of the upper ocean that impact sea ice processes. In most of the Southern Ocean (offshore) sea ice forms and interacts with the winter water layer (Park et al., 1998; Dong et al., 2008) (WW: i.e., cold water formed during sea ice production and its summer remnant) in the upper ocean (~0-200 m) at near freezing temperature. In winter, the mixed layer and winter water layer are essentially the same (extending from the surface to the permanent pycnocline). During summer this layer is capped by the freshwater from sea ice melt and is sandwiched between the seasonal pycnocline above and the permanent pycnocline below. Below the WW, separated by a weak (permanent) pycnocline is the Circumpolar Deep Water (deep ocean layer, which is considerably warmer than WW and deeper) (Gordon and Huber, 1984). When sea ice forms, the brine released densifies the upper ocean and drives vertical mixing (Martinson and Iannuzzi, 1998). Once this mixing initiates the entrainment of CDW into the WW, the heat entering the WW exerts a thermal limit for sea ice growth, producing a negative ice-ocean feedback (Martinson, 1990). Increased sea ice production can occur from positive ice-ocean feedback in regions where stratification inhibits the entrainment of heat from CDW. The positive ice-ocean feedback is also found to cause warming of subsurface ocean from the increased heat storage in the CDW, i.e., the heat that would otherwise melt sea ice from below and limit sea ice growth (Goosse and Zunz, 2014a; Lecomte et al., 2017).

Sea ice is found to have a dominant role in the water mass transformation in the Southern Ocean and meridional overturning circulation (Abernathey et al., 2016; Evans et al., 2018; Pellichero et al., 2018). In some regions close to the Antarctic continent, the water column is destratified (i.e., the ocean properties are vertically homogenous throughout the water column). In these regions, the negative ice-ocean feedback does not occur (as there is no heat stored at a deeper layer). When unhindered sea ice growth occurs in these regions, the excessive brine rejection makes the water column very dense and forms Dense Shelf Water. The Dense Shelf Water cascades down the continental shelf, mixes with CDW and transforms to form the Antarctic Bottom Water (AABW; Figure 1.3) which is the densest global-scale water mass (Orsi et al., 1999), forming the lower branch of the meridional overturning circulation. Sea ice advection and melt also play a role in the water mass transformation occurring in the Southern Ocean. Sea ice which largely forms near the Antarctic continent, drifts and melts largely near the sea ice edge at a lower latitude, effectively acting as a pump removing freshwater from high latitudes and transporting it to lower latitudes (Pellichero et al., 2018). The meridional transport of freshwater (from the annual sea ice cycle) plays an important role in driving a large-scale circulation that upwells CDW to the surface (complemented by wind-driven forcings), which

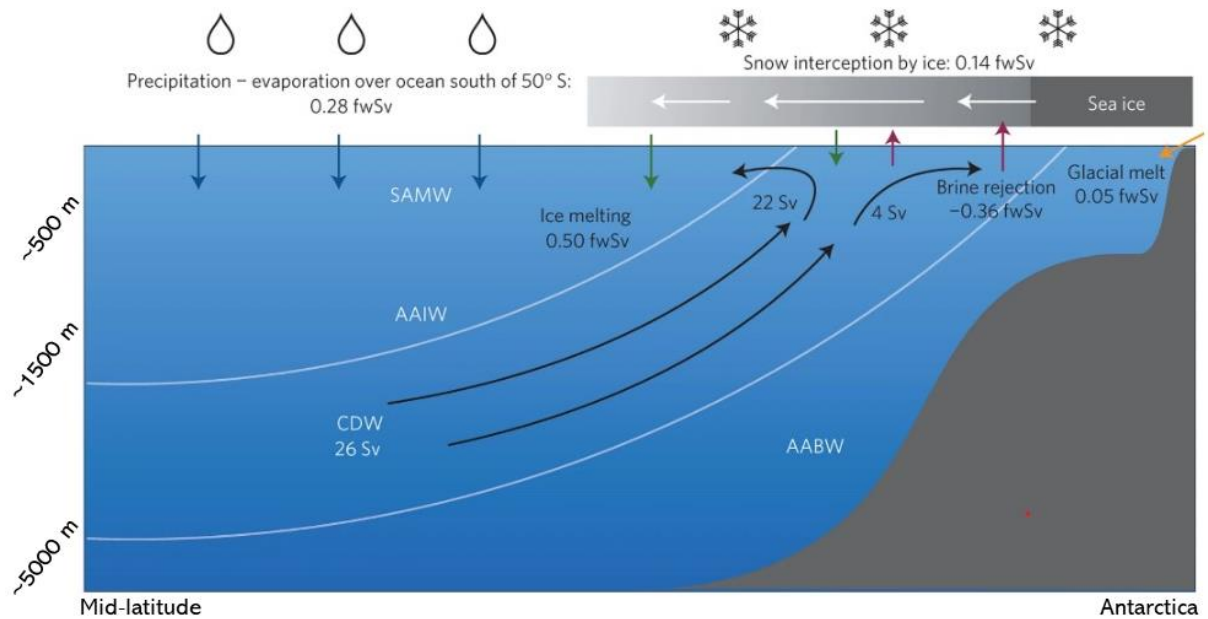


Figure 1.3: Schematic depicting the water masses and their transformation in the Southern Ocean. Upwelled Circumpolar Deep Water (CDW) modified to form Sub-Antarctic Mode Water (SAMW) and Antarctic Intermediate Water (AAIW) forming the upper cell and lower [Antarctic Bottom Water (AABW)] cells of the meridional overturning circulation in the Southern Ocean; extracted from Abernathy et al. (2016) and edited to show approximate depth and spatial orientations. Various bulk contributions to the freshwater exchange at the ocean surface (positive downward) south of 50° S are given in units of freshwater sverdrups: 1 fwSv = 106 m³ freshwater s⁻¹.

is then transformed into lighter and denser water that then becomes part of the upper and lower branches of the meridional overturning circulation (Figure 1.3). The freshwater exported by sea ice to the open ocean contributes to the production of fresh Antarctic Intermediate Water (AAIW), and the upper branch of the meridional overturning circulation (Saenko et al., 2002).

The meridional overturning circulation connecting the surface ocean to the interior ocean has profound impact on the global climate, largely due to its role in modulating the global heat and carbon budget (Rintoul, 2018). The ocean south of 40°S dominates the global ocean uptake of anthropogenic heat and carbon dioxide, accounting for ~70% of the anthropogenic ocean heat uptake and ~40% of anthropogenic ocean carbon uptake (Sabine et al., 2004; Mikaloff Fletcher et al., 2007; Watson et al., 2014; Frölicher et al., 2015; Gruber et al., 2019). In the Southern Ocean, the combination of cold surface waters and upwelled CDW from preindustrial times with low carbon concentrations allow high rates of absorption of atmospheric heat and anthropogenic carbon. The deep water (CDW) ventilation absorbs atmospheric heat and

anthropogenic carbon; the surface water transforms and sinks to the ocean interior (as AAIW and AABW) while new deep waters are drawn to the surface that continues to absorb atmospheric heat and anthropogenic carbon. (Sabine et al., 2004; Frölicher et al., 2015). Hence, the growth and melt of sea ice in the Southern Ocean contribute to modulating global anthropogenic heat and carbon budgets and subsequently impact global climate.

Sea ice-ocean interactions close to the Antarctic continent is found to impact the basal melt rates of Antarctic ice shelves that subsequently impact the stability of Antarctica's massive glacial ice sheets (Khazendar et al., 2013; Gwyther et al., 2014; Miles et al., 2016; Kusahara, 2021). Miles et al. (2016) associated the reduction in sea ice production to ice shelf melt from entrainment of warm CDW in the Wilkes Land Glacier in East Antarctica. They proposed that a reduction in sea ice growth and associated decrease in brine rejection reduces the formation of (cold) dense shelf waters in the vicinity of the ice shelf, changing the composition of water masses (or changing upper ocean stratification) in these regions. These changes in the ocean lead to increased intrusion of warm CDW towards the glacier and accelerate the rate of the glacier retreat. Freshwater input from glacial retreat and ice sheet loss may affect sea ice as it modifies the upper ocean conditions, and increased stratification can lead to higher sea ice production (Merino et al., 2018; Rye et al., 2020). Sea ice area biases in the model are found to have a large impact on Antarctic temperature and precipitation, which are important for ice-sheet mass balance (Agosta et al., 2015; Bracegirdle et al., 2015). Hence, although sea ice does not directly contribute to global sea level rise (as it is frozen sea water, which is already part of the global sea level), it contributes to global sea level changes through its interaction with the atmosphere that impacts adjacent ice sheets.

Antarctic sea ice plays a fundamental role in the sustenance of marine ecosystems in the Southern Hemisphere. Antarctic sea ice is one of the most extensive, seasonally variable, and productive habitats on Earth (Arrigo and Thomas, 2004) and a fundamental driver of Southern Ocean primary production (Lizotte, 2001). Sea ice provides a vital habitat, breeding platform, and source of food to many species in the high latitude Southern Ocean ranging from microbes (algae, bacteria, etc), herbivores (e.g., krill), fish, and seabirds (e.g., penguins), to mammals (seals and whales) (Brierley and Watkins, 2000; Nicol, 2006; Nicol et al., 2008; Thomas, 2017; Veytia et al., 2021). High concentrations of algae are found in the sea ice compared to the underlying water column, due to the combination of availability of sufficient key nutrients and high light availability (Meiners et al., 2012; Arrigo, 2017). Antarctic sea ice melt in austral spring to summer fuels the formation of intense phytoplankton blooms as it releases ice-algae,

freshwater, and nutrients specifically iron, an important limiting micronutrient in the Southern Ocean into the upper water column (Smith and Nelson, 1986). This abundance of food attracts and supports many of the species that are reliant on sea ice, including species that migrate (such as humpback whales) and the species that are reliant year-round (such as Adelie penguins). Changes in sea ice conditions are resulting in observable responses in the distribution and behaviour of several sea ice-dependant species with cascading effects across the Antarctic marine ecosystem (Massom and Stammerjohn, 2010). The changes in sea ice and the consequent changes in Southern Ocean ecosystem structure and dynamics are also found to have impacts on the biogeochemical cycling (Ducklow et al., 2007) and ocean acidification (Doney et al., 2009), that have cascading effects and implications that extend beyond the Antarctic/Southern Ocean (Massom and Stammerjohn, 2010).

1.2 Insights from Antarctic sea ice predictability studies

In the fields of weather and climate, the terms ‘prediction’, and ‘projection’ are sometimes used interchangeably, however, these terms have important distinctions. A prediction is a probabilistic statement that something will happen in the future, based on what is known today. (Kirtman et al., 2013). It assumes that the future is significantly influenced by the ‘initial conditions’, i.e., the situation from which the prediction is made, even though boundary/dynamic conditions (e.g., atmospheric events, for the specific case of sea ice prediction) may be a more direct driver of the future state. Projections, on the other hand, allow for significant changes to influence the prediction from the boundary conditions that might affect the prediction into the future (e.g., a projection on how global surface air temperatures will change from a doubling of atmospheric carbon dioxide). Both predictions and projections are used for process understanding, i.e., how the initial condition of x affects y in the future (prediction) and how changes in x in the future affect y in the future (projection). Predictions and projections also vary in the timescales considered (Figure 1.4). Predictions can range from a few months or years, and projections look at changes that are several decades or centuries ahead. ‘Weather forecasts’ refer to the “best” future prediction based on current conditions (at daily to monthly timescales). It is generally used in daily to weekly weather forecasts and not as a method for process understanding. Hence predictions (not weather forecasts) considered in this thesis describe conditions at timescales longer than a forecast (i.e., seasonal-to-

interannual timescales). They are less precise than a forecast, look at timescales shorter than projections, and have smaller uncertainties (or spread) than projections.

Predictions of Antarctic sea ice conditions ranging from a few weeks to a few years in advance are of interest to many stakeholders. As they are essential for the management of logistics shipping, fisheries management, conservation activities, tourism, and scientific research. Compared to the Arctic, research efforts on Antarctic sea ice prediction and predictability have been considerably lower (Holland et al., 2013; Ordoñez et al., 2018; Marchi et al., 2019; Zampieri et al., 2019; Marchi et al., 2020). The heightened focus on the Arctic sea ice is attributed largely to the proximity to highly populated areas in the northern hemisphere, leading to a large economic interest. Arctic sea ice predictability studies are also motivated by the opportunity of an open summer Arctic basin for transit shipping through the Arctic Ocean (Melia et al., 2017). Recent changes in Antarctic sea ice, which shifted from consecutive years of record highest ice cover to consecutive record lows have focused attention on Antarctic sea ice (Eayrs et al., 2021; Raphael and Handcock, 2022). The international scientific community has recognized the need to advance the field of sea ice prediction at both poles simultaneously (Jung et al., 2016). This has led to coordinated sea ice prediction research programs such as Sea Ice Outlook (SIO, focusing on the Arctic, since 2008) which is now within the Sea Ice Prediction Network (SIPN) for both poles (Massonnet et al., 2023).

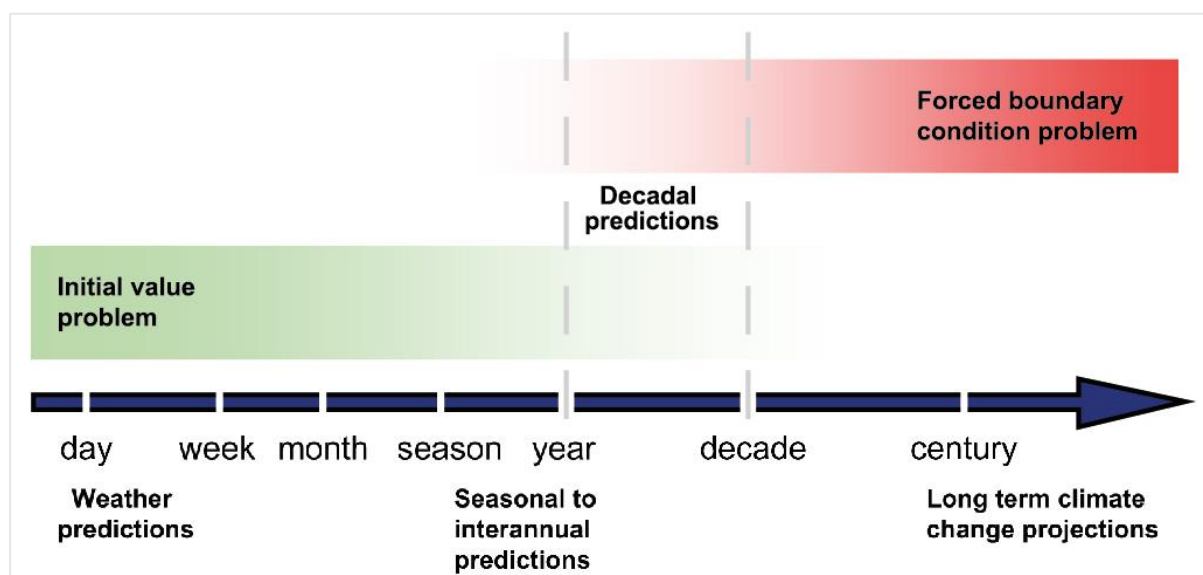


Figure 1.4: Schematic representing the timescales of weather prediction, seasonal to interannual predictions and climate projections; extracted from Kirtman et al. (2013)

Seasonal to decadal sea ice predictability has been evaluated in the Antarctic Seas (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021; Morioka et al., 2021; Libera et al., 2022). Antarctic sea ice predictability studies largely diverge in their method of assessing predictability mainly into two categories: potential prognostic predictability (hereafter prognostic predictability) and diagnostic predictability. The prognostic predictability, (also known as the perfect model experiment) introduced by Pohlmann et al. (2004), calculates sea ice predictability using an ensemble of model runs with slightly different initial conditions, and analyses how the ensemble members diverge over time (Holland et al., 2013). In the prognostic predictability method, the variance of the ensemble predictions are compared to the variance of some reference prediction (Marchi et al., 2019). When the ensemble spread is indistinguishable from the simulated natural variability, there is no predictive capability from the initial state (e.g., Figure 1.5 a and c). Diagnostic predictability (anomaly correlation coefficient (ACC)) can be calculated from a single model run (or from observations) and is based on lagged correlation or anomaly correlation calculation (e.g., Figure 1.5 d). Prognostic predictability analysis is computationally more expensive and difficult to compare with other data than diagnostic predictability analysis (Ordoñez et al., 2018).

The terms “prediction”, “predictability”, “prediction skill”, etc, can be confusing due to their literal usage as well as the various definitions and interpretations used in literature. Diagnostic predictability is used to assess sea ice and upper ocean predictability similar to previous studies (Blanchard-Wrigglesworth et al., 2011; Cheng et al., 2016; Ordoñez et al., 2018; Giesse et al., 2021). Diagnostic predictability (where the emphasis is on using lagged correlations) is commonly applied to sea ice and ocean variables in the Arctic (Blanchard-Wrigglesworth et al., 2011; Day et al., 2014; Cheng et al., 2016; Guemas et al., 2016; Giesse et al., 2021) and Antarctic (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021; Morioka et al., 2021) sea ice predictability studies, either as the only approach or in combination with other methods. Diagnostic predictability is used to assess the retention of anomalies in the ice-ocean system, and the ability of the system to retain anomalies is what gives rise to predictability. Therefore “predictability” in this thesis refers to the capacity of the ice-ocean system to retain memory over multiple months to seasons (which is assessed from persistence in the lagged correlations).

Sea ice predictability studies also vary based on the objective of the study. Many studies investigate ways to improve sea ice prediction skills, such as differences in initialization

techniques and the use of different variables. Other studies focus on understanding the sources of sea ice predictability and the mechanisms underlying it. Studies that investigate the impact of initialization of ocean-sea ice variables, atmospheric boundary conditions, etc., on sea ice predictability have identified that the initialization of sea ice conditions (concentration, thickness) is important for the skilful prediction of Antarctic sea ice at seasonal-to-decadal timescales (Morioka et al., 2019; Marchi et al., 2020; Bushuk et al., 2021; Kearney et al., 2021; Morioka et al., 2022)s. While the ocean, atmosphere, and sea ice itself are potential sources of memory for sea ice predictability, this thesis focuses on the oceanic source of memory underlying sea ice predictability. This thesis builds on the concepts from Holland et al. (2013) and Marchi et al. (2019) that have linked the upper ocean heat content anomalies to Antarctic sea ice predictability.

Holland et al. (2013) investigated Antarctic sea ice (ice-edge location) predictability using prognostic method initialized in January and found ice-edge location to be predictable up to two years in advance. They found that during the sea ice advance season (April to September) significant predictability is retained in some locations, with an eastward propagating signal. The retained predictability signal is lost in the sea ice retreat season (after October), however, in some locations predictability re-emerged in the following sea ice advance season. They associated sea ice predictability with ocean heat content anomalies that are retained at depth. They also explained that the loss of predictability in the sea ice retreat period occurs when these heat anomalies are isolated from the surface due to the summer stratification (sea ice meltwater capping). Predictability reemerges when the heat anomalies stored at depth resurface in the following sea ice advance season. Zunz et al. (2015) evaluated the impact of initialization on Antarctic sea ice predictability at interannual to multi-decadal timescales. They applied both prognostic and diagnostic methods. Their results were in good agreement with those of Holland et al. (2013), in finding multi-year ice edge predictability, eastward propagation of the signal and re-emergence of predictability. Zunz et al. (2015) calculated the correlation between ice edge location and the upper ocean heat content (OHC; computed south of 60° S, and 0-100 m depth along longitudes), and found a significant negative correlation between ice edge location and the OHC below the sea ice. Their findings provided further evidence that multi-year sea ice predictability is linked to the upper OHC achieved from the strong thermal inertia of the ocean.

In the first multi-model study of Antarctic sea ice predictability, Marchi et al. (2019) found high predictability in winter, low predictability in summer and predictability re-emerging from winter to winter in all of the models used (for up to three years). They applied a similar approach to Zunz et al. (2015) and evaluated the ice edge predictability against the predictability of the upper OHC. They found that in all cases, the locations with significant ice-edge predictability matched with those of high upper OHC predictability. The findings in Marchi et al. (2019) were consistent with those of Holland et al. (2013) and Zunz et al. (2015), and they explored the mechanism of sea ice predictability re-emergence. Marchi et al. (2019) developed on the re-emergence theory described by Holland et al. (2013) and tested the mechanism by computing the Pearson correlation between winter sea surface temperature (SST) and potential temperature in the upper ocean for different lags. This analysis revealed evidence for a thermal memory in the ocean that responds to the seasonal evolution of the upper ocean, consistent with the theory in Holland et al. (2013). Marchi et al. (2019) found that SST anomalies were closely tied to the seasonal cycle of the mixed layer. Incorporating their findings, Marchi et al. (2019) described the mechanism of sea ice predictability re-emergence: Sea ice growth and associated brine rejection causes mixed layer to deepen. Mixed layer deepening fosters interaction between the surface (to the mixed layer) and the interior ocean (below the mixed layer), producing strong positive correlations through the mixed layer, which persist until the end of winter (Figure 1.5b). When sea ice retreats, mixed layer shoals and the anomalies at depth are isolated from the surface in summer. When the mixed layer deepens again during the following winter, the temperature anomalies resurface and influence the sea ice advance.

Marchi et al. (2019) then focused on the large longitudinal variation (spatial variability) in their sea ice predictability results, i.e., evaluating why significant predictability and re-emergence of predictability were produced along certain longitudes and not others. They noticed that predictability re-emergence is limited to longitudes (and models) that hosted deep winter mixed layers, where winter SST anomalies were more effectively preserved. They stated that the effectiveness of the predictability re-emergence mechanism relied on sufficiently large variations of the mixed layer seasonal cycle. On encountering similar longitudinal variation in the sea ice predictability, Holland et al. (2013) also suggested that sufficiently deep mixed layer was needed for retaining thermal anomalies at depth. Regional differences in Antarctic sea ice predictability was evaluated in Ordoñez et al. (2018) and Bushuk et al. (2021) in five regions (with similar regional definition) around Antarctica. Ordoñez et al. (2018) used long control simulations of the Community Earth System Model (in a full depth dynamical and slab ocean

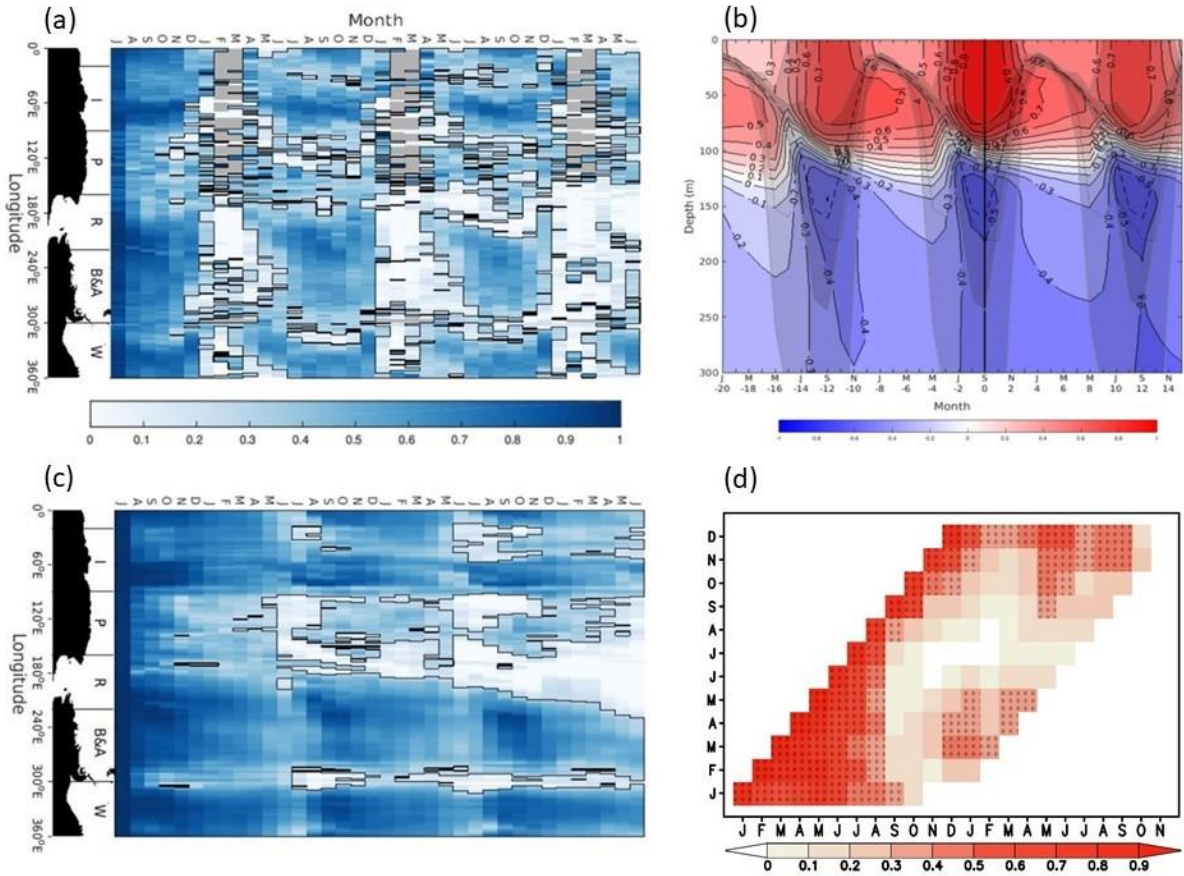


Figure: 1.5 (a) Prognostic predictability of ice-edge locations as a function of longitude and lead time. (b) Correlation between sea surface temperature in September and lagged upper ocean temperature. (c) prognostic predictability of the upper ocean heat content (0-100 m) for the same model used for (a). (d) diagnostic sea ice predictability of observed sea ice in the Weddell Sea region. (a-c) extracted from Marchi et al. (2019) and (d) from Morioka et al. (2021).

model configurations), while Bushuk et al. (2021) used three dynamical seasonal prediction systems (that differed in their coupled model components, initialization techniques, atmospheric resolution, and model biases). The sea ice predictability across regions showed high similarity in the analysis by Ordoñez et al. (2018), and were also comparable between the dynamical and slab ocean models. Whereas, in the study by Bushuk et al. (2021), which used similar regional definitions as in Ordoñez et al. (2018), considerable regional variability was found in the results.

Antarctic sea ice predictability is still an emerging field and it is of relevance, given the recent changes in observed Antarctic sea ice cover. Following years of record highs (the latest in September 2014), precipitous reduction in Antarctic sea ice extent started in 2016 (Eayrs et al.,

2021), it sank to a record low in 2017, and the decline that occurred in the Antarctic in three years (2014 to 2017) exceeded thirty years of sea ice loss in the Arctic. In February 2022, Antarctic sea ice reached a new record low approximately 32% below climatological values and studies suggest these changes might indicate a regime shift of more extreme annual fluctuations (Parkinson, 2019; Fogt et al., 2022; Raphael and Handcock, 2022). To improve our skills for predicting Antarctic sea ice and crucially for understanding the drivers of its changes, more studies probing deeper into the physical mechanisms underlying Antarctic sea ice predictability are required.

1.3 Thesis aim and objectives

Antarctic sea ice predictability studies agree that the ocean is the source of memory underlying sea ice predictability (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021). This memory is attributed to the large thermal inertia of the ocean (Marchi et al., 2019), that extends through the entire water column kilometres deep, but sea ice-related anomalies may only extend a couple of hundred meters into the ocean (except in regions of deep convection). *To understand the oceanic source of memory one needs to differentiate the memory in the ocean related to sea ice processes (which is likely present in the upper 100-200 m) from the inherent ocean memory (memory in the ocean due to its large thermal inertia). This is important to answer questions such as (1) To what depth does this memory extend? (2) Does it have regional variability, (3) Does this memory evolve with time?*

Some of the studies that evaluated the predictability of the ocean integrated the OHC within an arbitrary depth of 100 m (Zunz et al., 2015; Marchi et al., 2019). When integrating the OHC as done by Marchi et al. (2019), information about the evolution of OHC anomalies in the vertical oceanic layers is lost and limits our capacity to observe the physical process occurring within the ocean. *To understand the physical processes in the upper ocean that drive Antarctic sea ice predictability, it is essential to consider the vertical dimension (i.e., along depth).*

Previous studies align in finding that sufficiently deep mixed layers are required to sustain ocean memory and sea ice predictability over long periods. *However, is the presence of deep mixed layers, the only necessary condition for sea ice predictability? Do all regions with deep mixed layer produce long persistence in sea ice predictability?*

If sea ice predictability arises from sustained thermal anomalies in the upper ocean, what are the processes or conditions that can impact these thermal anomalies, other than the depth of the mixed layer? Holland et al. (2013) suggested that processes such as vertical mixing and ocean advection can modify the thermal anomalies in the upper ocean and thereby impact sea ice predictability. The upper ocean layer (WW) that is in thermal equilibrium with the sea ice lies above substantially warmer CDW (Gordon and Huber, 1984). The vertical mixing induced by sea ice growth and the entrainment of CDW into the WW can modify the thermal signature within the WW (Martinson, 1990; Martinson and Iannuzzi, 1998). Variability arises in regional ice-ocean interaction depending on the upper ocean condition, especially the differences in stratification strength (Martinson, 1990; Martinson and Iannuzzi, 1998; Stammerjohn et al., 2012; Lecomte et al., 2017; Wilson et al., 2019). Therefore, the local ice-ocean interaction can influence how anomalies are preserved or modified within the WW, and it can lead to regional differences in sea ice predictability. Assessment of sea ice predictability from the framework of sea ice-ocean interactions can be a new avenue for exploring the physical mechanisms that drive the variability in Antarctic sea ice predictability.

Objectives:

- 1:** *Using a model, develop a new physical framework based on sea ice-ocean interaction for evaluating sea ice predictability.*
- 2:** *Evaluate the drivers of regional differences in Antarctic sea ice predictability based on the findings from the first objective using the same model.*
- 3:** *Compare regional Antarctic sea ice predictability from observed sea ice against the model to evaluate how the representation of sea ice and upper ocean conditions in the model leads to biases in sea ice predictability.*

CHAPTER 2

Chapter 2: Ocean-sea ice processes and their role on multi-month predictability of Antarctic sea ice

Chapter overview

In this chapter, sea ice predictability is used as a tool to investigate ocean-sea ice processes underlying sea ice predictability in the Weddell Sea region in Antarctica. In the Southern Ocean, the Weddell Sea is one of the dominant regions for sea ice production and a large contributor to the overall Antarctic sea ice variability. Its geographical settings limit dynamical influence from advected oceanic and sea ice properties, making it ideal for studying local ice-ocean interaction.

Three sets of memory in the ice and ocean system: (i) sea ice predictability (sustained memory of sea ice), (ii) ocean predictability (inherent memory in the ocean due to its large thermal inertia), and (iii) ice-ocean predictability (memory in the ocean related to sea ice processes, separated from the inherent ocean memory) are investigated in this study. After evaluating the three sets of memory in the ice-ocean system, the memory in the upper ocean underlying sea ice predictability is differentiated from the background ocean memory. All three sets of memory in the ice-ocean system were then evaluated against the seasonal sea ice and ocean processes to identify the physical processes in the ocean influencing sea ice predictability at interseasonal timescale.

The main text of this section is a paper published in the *Geophysical Research Letters*:

Libera, S., Hobbs, W., Klocker, A., Meyer, A., & Matear, R. (2022). Ocean-Sea Ice Processes and Their Role in Multi-Month Predictability of Antarctic Sea Ice. *Geophysical Research Letters*, 49(8), e2021GL097047. doi:<https://doi.org/10.1029/2021GL097047>

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Candidates contribution to the manuscript:

The Candidate and the supervisory team (Dr Will Hobbs, Dr Andreas Klocker, Dr Amelie Meyer, Dr Richard Matear) jointly conceived and designed the study. The candidate undertook the data analysis and results interpretation with guidance from the supervisory team. The candidate wrote the manuscript with the guidance and critical revision provided by the supervisory team. Two anonymous reviewers also provided useful and constructive feedbacks in the peer review process which improved the structure and flow of the manuscript.

Key Points:

- Antarctic sea ice predictability is strongly determined by temperature and salinity profiles of the underlying upper ocean water column
- Every winter, the timing of the loss of sea ice predictability is defined by when deep water is entrained into the mixed layer
- Sea ice predictability depends not only on the depth of the Winter Water layer, but also on how strongly stratified its base is

Abstract

Antarctic sea ice is a critical component of the climate system and a vital habitat for Southern Ocean ecosystems. Understanding the underlying physical processes and improving Antarctic sea ice prediction is of broad interest. Using model outputs, we investigate sea ice and upper ocean predictability at interannual timescales in the Weddell Sea region. We find that oceanic predictability is largely confined to the Winter Water layer and responds to seasonal modifications of the water column, mainly driven by sea ice processes. Predictability depends not only on the depth of the Winter Water layer but also on how strongly stratified its base is. Predictability is lost when warm Circumpolar Deep Water with no sea ice-related memory entrains into the mixed layer. We show the strong dependence of sea ice predictability on the local upper ocean vertical structure, which suggests that both are likely to change in a warming climate.

2.1 Introduction

The growth and melt of Antarctic sea ice, arguably the strongest seasonal cycle on the planet (Handcock and Raphael, 2020), affects global climate through its interplay with planetary albedo, atmospheric circulation, thermohaline circulation, and ocean productivity (Brandt et al., 2005; Massom and Stammerjohn, 2010; Raphael and Hobbs, 2014; Abernathey et al., 2016; Hobbs et al., 2016). The close interaction of Antarctic sea ice with the ocean and atmosphere has been linked to

interannual variability and trends in sea ice (Martinson, 1990; Holland, 2014; Hobbs et al., 2016; Lecomte et al., 2017). Antarctic sea ice predictability studies have identified the strong dependence of sea ice predictability on oceanic processes, pointing towards sea ice-ocean interactions (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019). This study aims to better understand the physical processes in the ocean associated with sea ice predictability.

Sea ice predictability studies are diverse, with predictions ranging from seasonal to decadal timescales, using statistical or dynamical approaches, and based on observations or climate model data. They have a variety of applications ranging from planning operational activities (scientific research, tourism, shipping, fisheries management, and conservation) to evaluating climate projections, and policy decision making (Chen and Yuan, 2004; Blanchard-Wrigglesworth et al., 2011; Holland et al., 2013; Juricke et al., 2014; Zunz et al., 2015; Guemas et al., 2016; Yang et al., 2016; Ordoñez et al., 2018; Marchi et al., 2019; Massonnet et al., 2019; Zampieri et al., 2019; Marchi et al., 2020; Bushuk et al., 2021; Giesse et al., 2021; Kearney et al., 2021). In this study, we evaluate sea ice and ocean predictability at seasonal to interannual timescales in the Weddell sector (Figure 2.1a). We use lagged correlations to evaluate the inherent memory of the ice-ocean system in a forward-looking perspective of predictability. That is, indicating how a given initial state is retained in the system and continues to influence future states. There may be external sources of predictability from atmospheric forcings, e.g., the El Nino Southern Oscillation (Simpkins et al., 2012), and the Southern Annular Mode (Murphy et al., 2014) that we do not consider in this study. In the Southern Ocean, the Weddell Sea is one of the dominant regions of sea ice production. Its geographical setting limits dynamical influence from advected oceanic and sea ice properties into this region, making it ideal for studying local ice-ocean interaction.

Previous studies have established the link between the upper ocean heat content (OHC) and sea ice predictability (Holland et al., 2013; Marchi et al., 2019). These studies found that sea ice predictability can persist for some months but is then generally lost during the ice-retreat season before reemerging in the following ice-growth season. Marchi et al. (2019) calculated the predictability of integrated OHC in the upper 100 m and showed strong correspondence between regions of high sea ice predictability and oceanic predictability. When integrating the OHC as done by Marchi et al. (2019), information about the evolution of OHC anomalies in the vertical oceanic layers is lost, this limits our capacity to observe the physical process occurring within the ocean.

In this study, we retain the vertical dimension for oceanic predictability results and compare the evolution of predictability of sea ice and ocean simultaneously. We find the loss of predictability in summer followed by the reemergence of predictability in autumn consistent with Holland et al. (2013) and Marchi et al. (2019). We also find a sudden loss of predictability in mid-winter when warm Circumpolar Deep Water is entrained into the mixed layer, connecting the influence of local vertical ocean structure and sea ice processes. These findings not only give insights into the physical processes in the upper ocean underlying sea ice predictability but also directs towards hydrographic features that are valuable for understanding the regional differences in Antarctic sea ice trends and variability.

2.2 Methods

2.2.1 Data

We use the outputs from a global coupled ocean-sea ice model, the Australian Community Climate and Earth System Simulator (ACCESS-OM2), (ACCESS-OM2; Kiss et al., 2020). ACCESS-OM2 is based on the ocean MOM5.1 and ice CICE5.1 models coupled with the OASIS-MCT coupler. The model experiment analyzed in this study was forced using JRA55-do v1.4.0 (Tsujino et al., 2018). The high horizontal resolution of 0.1 ° in ACCESS-OM2-01 produces a good representation of Southern Ocean dynamics, and adequate simulations of the Antarctic sea ice extent and concentration: The mean annual cycle of Antarctic sea ice extent from ACCESS-OM2-01 closely matches observations and the historical sea ice trends are also well represented (Kiss et al., 2020).

To calculate the observed sea ice area (SIA) used in this study, we use sea ice concentration (SIC) derived from satellite passive microwave data (a product based on the NASA Goddard-merged parameter in the NOAA/NSIDC Climate Data Record (CDR)) (Meier et al., 2013). We use SIA rather than sea ice extent (SIE) because SIE excludes information on how consolidated the ice pack is.

2.2.2 Correlation Analysis and Statistical methods

In our diagnostic predictability analysis (i.e., using lagged correlations), we calculate the correlation between a given initial month and the twelve future months using monthly data from 1985 to 2015. For the sea ice area, we calculate the total SIA in the Weddell sector, create the monthly time series and detrend it by subtracting the linear least-squares fit, then we apply the correlation analysis. The correlation analysis on total SIA reveals persistent anomalies in the sea ice (hereafter referred to as ‘sea ice predictability’ i.e., the predictability arising from sea ice persistence). To evaluate the predictability of the ocean from its initial state, we calculate spatial averages within the oceanic area between 300-345°E and between 90°S and winter sea ice extent maximum (from the model) for the Weddell Sea region approximated at 58°S. Spatial averaging for conservative temperature (T) produces timeseries of vertical profiles in the upper 200 m. These timeseries are detrended before applying the correlation analysis, we correlate the temperature at every depth in an initial month with the temperature at the same depth with the future months (T from the initial month correlated with future T at the same depth in future months), hereafter referred as ‘ocean-ocean correlations.’ Then using the detrended monthly timeseries of total SIA and T in the upper 200 m, we calculate correlations between given initial SIA with future T at depth, to investigate the signature of ice-ocean interactions (hereafter referred to as ‘ice-ocean correlation’). We define statistically significant values as p-values greater than 95% in the two-tailed Student’s t-test.

2.2.3 Climatology

To understand the general ice and ocean seasonal evolution in the Weddell sector, we create the monthly climatologies of temperature (T) and salinity (S) and calculated climatological density (potential density referenced to the surface) profiles from the averaged T and S climatologies using TEOS equation (McDougall and Barker, 2011). From these climatologies, the vertical gradient of temperature (dT/dz), density (dp/dz) and mixed layer depth (MLD; defined by an increase in density by 0.03 kgm^{-3} from surface ocean density) are calculated.

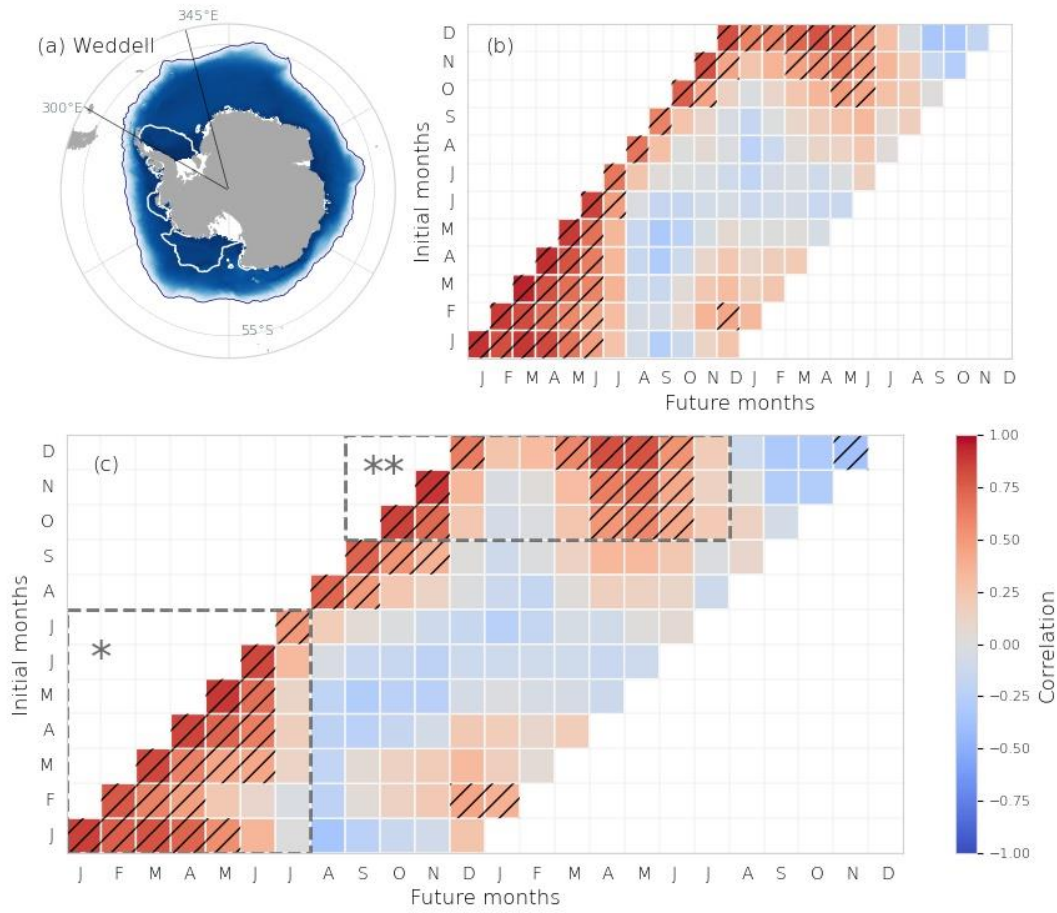


Figure 2.1: (a) Map defining the Weddell sector in Antarctica along with the model climatological winter maximum (September) of sea ice extent (in shading) and summer minimum (February, white contour). (b) Sea ice predictability: autocorrelation of sea ice area from observation, and (c) Sea ice predictability from model output. In (b) and (c), SIA from initial months (or lead) along the y-axis are correlated against the SIA in the future months (or lags) along the x-axis and statistically significant values (>95%) are hatched. Summer persistence (*) and spring reemergence (**) patterns are marked in (c).

2.3. Results

2.3.1 Sea ice predictability: Summer persistence and spring reemergence

Our analysis of sea ice and ocean predictability is in the forward-looking perspective, that is indicating how a given initial state relates to the future states. Sea ice area predictability results from observations and model simulations are similar (Figure 2.1 b, c), and show two predictability patterns: ‘persistence’ from summer initial months with correlations lasting till June, shown by a sustained significant autocorrelation (Figure 2.1c *); and ‘reemergence’ from spring initial months to the following autumn months (Figure 2.1c **), shown by the loss of correlation in summer months which ‘reemerges’ in April. These patterns of persistence in summer and reemergence from spring to autumn (hereafter ‘spring reemergence’) were identified in a similar diagnostic study by Ordoñez et al (2018). Prognostic studies have identified the reemergence of Antarctic sea ice predictability during ice-growth season (Holland et al., 2013; Zunz et al., 2015; Marchi et al., 2019). Our sea ice predictability results show the termination of both summer persistence and spring reemergence consistently occurring in July.

2.3.2 Ocean-ocean and ice-ocean correlations

The physical processes in the ocean underlying sea ice predictability patterns are investigated using the evolution of upper ocean predictability (Figure 2.2). We choose SIA correlations starting from January and October to represent summer persistence and spring reemergence patterns respectively (Figure 2.2 a and b). We correlate January and October SIA with lagged ocean temperature to explore how sea ice anomalies are related to upper ocean temperature (ice-ocean correlation) (Figure 2.2 c and d); the ocean’s internal predictability is characterised by correlating the January and October temperature with future temperature at the same depth (ocean-ocean correlation) (Figure 2.2 e and f). We find a consistent evolution of predictability between the three sets of correlations. Sea ice and ocean autocorrelations (Figure 2.2 a,b, e, f), are positive, while ice-ocean correlations are negative, since cooler temperature implies more ice. Although sea ice and ocean predictability (ice-ocean and ocean-ocean) show consistent evolution, small mismatches (in

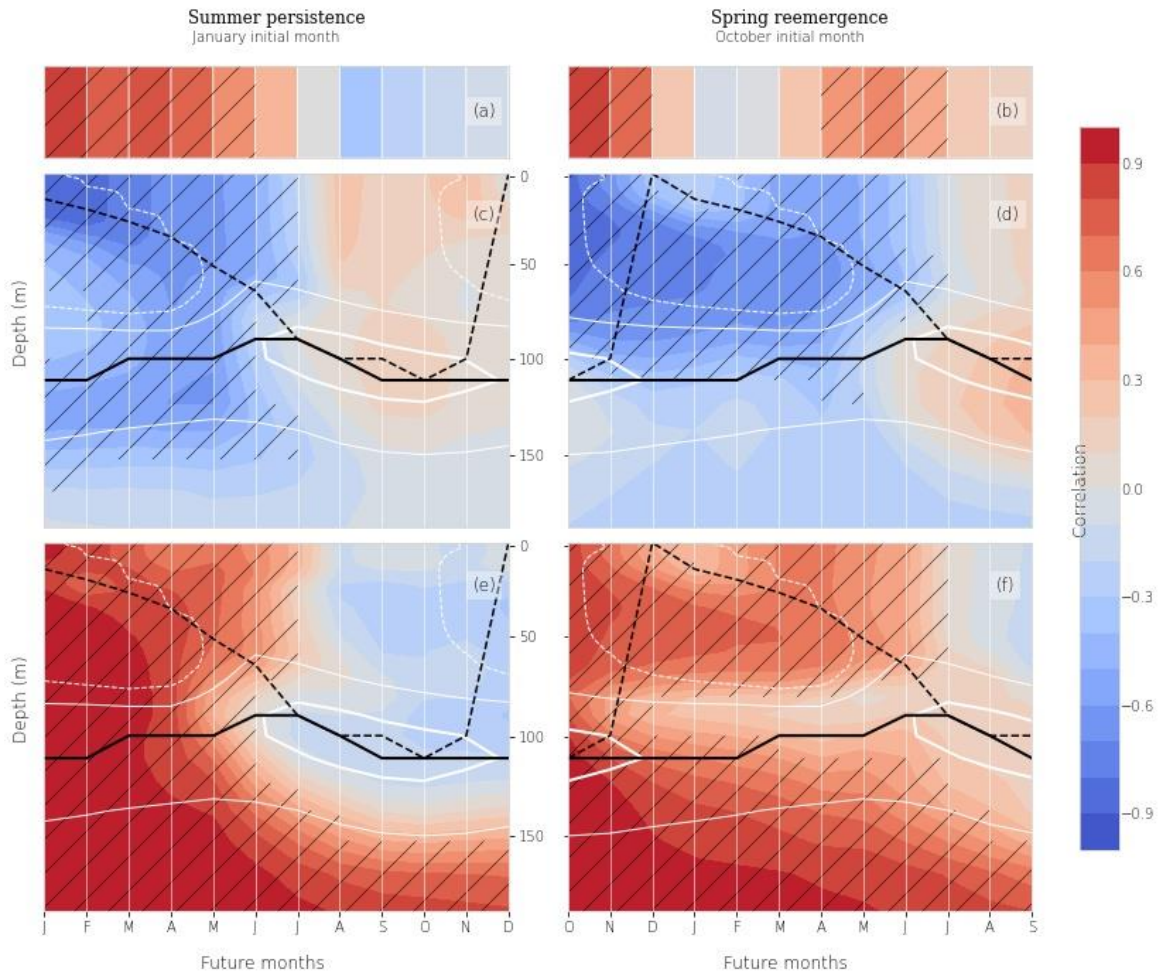


Figure 2.2: Comparison of summer persistence (plots on the left) and spring reemergence patterns (plots on the right) evident from sea ice (top row) and upper ocean predictability (0-200 m) (middle and bottom row). Sea ice predictability: correlation of SIA from January (a) and October (b) initial months with future months; Correlation between SIA in (c) January and (d) October with ocean temperature in future months; Correlation of ocean temperature in (e) January and (f) October with future months. Statistically significant values (>95%) are hatched in all panels. In the oceanic predictability results (c-f), the thick black line is the vertical temperature gradient (dT/dz) maximum, dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification.

timing and magnitude) are expected since we carry out the calculations with total SIA and averaged T from a large domain. For example, statistically significant sea ice predictability in summer declines one month earlier than in the ocean (Figure 2.2 a and c). Similarly, the sea ice predictability in spring sustains a month longer than in the ocean (Figure 2.2 b and d). This mismatch is likely a consequence of bulk analysis, and if a smaller spatial domain were considered, the correlations between the sea ice and ocean predictability are expected to be more consistent.

2.3.2.1 Seasonal evolution of the upper ocean

We overlay the climatological vertical thermal and density gradient contours to follow the seasonal evolution of the water column. The climatological dT/dz maximum (black line) represents the permanent pycnocline (PP) that separates the Winter Water (WW, i.e., cold water formed during sea ice production and its summer remnant) from slowly modifying Circumpolar Deep Water (CDW). During winter, the WW extends through the entire water column above the PP; during summer the seasonal stratification forms at the surface and the WW is sandwiched between the SP above and PP below. The dp/dz maximum (dashed black line) represents the evolution of seasonal pycnocline, which forms the base of the seasonally evolving mixed layer that is in direct exchange with the surface. Seasonal pycnocline acts as the base of the mixed layer in summer (January-March) and autumn (April-June), before it merges with PP in winter (more information in supporting information (TextS1, Figure S2)).

During the ice growth season, the PP coincides with the maximum depth of statistically significant correlation (Figure 2.2d). There is a layer of weak correlations at the base of the WW (Figure 2.2 f), which we interpret as noise, due to the high variability from mixing processes at the interface of the upper ocean and ocean interior. This separation (weak correlations) at the base of WW is expected, since WW is modified by sea ice production, implying that the entire WW is a source of memory for the ice-ocean system, while the ocean interior (below the PP) has no sea ice-related memory. However, the differentiation identified by the layer of weak correlations between the memory in the WW and the ocean interior is not present in the ocean-ocean correlations for summer persistence (Figure 2.2e). This is because the ice-ocean correlations from summer extend below the PP (Figure 2.2c), reducing the difference between the WW and the ocean interior when

the stratification strength at the PP weakens. How changes in the strength of the PP influence the anomalies within the WW is further discussed in section 2.3.2.3.

The high correlations below the PP in the ocean-ocean correlations (below ~150 m depth, Figure 2.2 e, f) are linked to the longer variance timescale of deep waters compared to surface waters but are not associated with sea ice variability. The ocean-ocean correlations illustrate the inherent capacity of the ocean to sustain thermal anomalies, also known as the thermal inertia of the ocean; it is not specific to sea ice-related thermal anomalies and can extend through the entire water column. Therefore, the stronger ocean-ocean correlations in the ocean interior (e.g., at 1000 m, not shown) below the seasonally evolving upper ocean layer (below the PP, e.g., at ~200 m in Figure 2.2 e and f) illustrate the inherent ocean memory and are not linked with seasonal sea ice processes. In contrast, sea ice-related memory is confined largely within the seasonally evolving upper ocean (above the PP, ~0-100 m; Figure 2.2 c, d) (Holland et al., 2013; Marchi et al., 2019). To evaluate the physical processes in the ocean that influence sea ice predictability, it is important to differentiate the memory in the upper ocean related to sea ice predictability from the background inherent oceanic memory. Hence, this paper focuses on predictability evolving in the upper ocean (largely within the WW layer; ~0 - 100 m, Figure 2.2 c-f).

2.3.2.2 Freeze and melt as limits to predictability

The correlations emerging from October encounter a loss of predictability during summer lag months, and present predictability reemergence (Figure 2.2 b,d,f). Consistent with Holland et al. (2013) and Marchi et al. (2019), the oceanic predictability shows the weakening or loss of correlations during summer near the surface, while strong correlations are retained below this surface layer and above the PP. Freshwater and surface ocean warming during the ice-melt season (December-February) produce a thin and highly-stratified surface layer that becomes the summer mixed layer (dashed black line) in Figure 2.2c-f). This summer layer separates the thermal anomalies in the WW layer from the surface, causing the loss of predictability between December and March (Figure 2.2 b, d, f).

In March, the regime shifts from sea ice melt (and a well-stratified summer mixed layer) to sea ice production (and destratification at the surface). Brine rejection from sea ice growth induces vertical mixing, resulting in entrainment across the seasonal mixed layer. Initially, this entrainment

reconnects the relatively cold remnant WW layer with the surface layer, leading to the re-emergence of both sea ice and ocean predictability (Holland et al., 2013; Marchi et al., 2019). After entraining through the WW layer, the mixed layer continues to deepen, eventually reaching the PP (merging of dashed black line with black line in Figure 2.2 c-f). Further entrainment causes the loss of predictability (Figure 2.1c and all panels of Figure 2.2) as it entrains water that has no sea ice process-related memory. The entrainment of CDW into the WW causes modification or loss of sea ice-related anomalies (persisting from the previous winter) and causes the loss of sea ice predictability terminating the spring re-emergence. We call this loss in predictability the ‘predictability barrier’, which is discussed in section (2.4.1).

2.3.2.3 Sensitivity of sea ice predictability to the stratification strength at the base of WW

The main distinction between ice-ocean correlations and ocean-ocean correlations is that ice-ocean correlations are largely bounded by the PP (upper 100 m), while the ocean-ocean correlations produce significant correlations below the PP (below 100 m). As discussed in section 3.2.1, ice-ocean correlations emerging from October are bounded by the PP, which we attribute to the sea ice memory being confined to the WW. However, the January SIA is correlated with ocean temperatures below the PP (up to 150 m; Figure 2.2c). Here we put forward the hypothesis that this is due to changes in the strength of the stratification at the base of the WW (or at the PP) (Figure 2.3e).

Winter cooling and sea ice production create a WW layer that is very distinct from CDW, which maintains a strong PP; therefore, the sea ice memory is confined above PP. Stratification strength at the PP reaches its maximum immediately before the occurrence of the predictability barrier in July, following which the PP stratification weakens due to entrainment-driven mixing across the PP, reaching its minimum in spring (by October, Figure 2.3e). Ice-ocean correlations for all 12 initial months (Supplementary figure 1 (S1)) show how the correlations respond to changes in stratification strength. The correlations initialised from November to May (i.e. when the PP is weakest) extend below the PP, but otherwise, correlations are bounded by the PP. This suggests that the strength of the PP, which itself is modified by sea ice growth/brine rejection (and related

mixing of the CDW and WW), is an important indicator of how well heat anomalies are retained in the sea ice-mixed layer system.

We have not quantified the various processes that affect the vertical mixing of sea ice-related anomalies, e.g., buoyancy-driven mixing, turbulent mixing, upwelling of CDW. Buoyancy-driven vertical mixing (due to brine rejection from sea ice formation) is expected to dominate in winter; while other modes of mixing may become more significant in summer, when heat and freshwater fluxes increase stratification and impede buoyancy-driven mixing. Doddridge et al. (2021) demonstrated that during the ice melt season, turbulent mixing can move heat anomalies downwards across the summer mixed layer and into the remnant WW layer; here we posit a similar process happening before the development of the summer mixed layer, so that temperature anomalies penetrate below the PP. In this case, the memory from those thermal anomalies is lost to the CDW (which acts as a thermal sink).

Importantly, this variability in the depth of significant correlation (Figure 2.2 c,d) demonstrates that the strength of stratification, as well as the depth, of the PP, is important for the regime of sea ice predictability in a given sector. We further discuss the dependence of sea ice predictability on the hydrographic profile in section 4.2.

2.4 Discussions

2.4.1 Predictability barrier and predictability suppression

Our study is consistent with findings from existing literature connecting upper OHC (oceanic thermal memory) with sea ice predictability (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019). In the Weddell Sea, sea ice anomalies persisting from spring, are lost temporarily in summer (December-May), and then re-emerge in May before they are lost permanently in mid-winter (July) (Figure 2.2, all panels). Seasonal loss of sea ice predictability (in summer) is associated with the development of a highly stratified summer mixed layer due to sea ice melt that separates the surface ocean and sea ice from the heat content anomalies below the summer mixed layer. Below the summer mixed layer, OHC anomalies are retained and re-emerge when the summer mixed layer erodes (i.e., weakens in the stratification strength) and deepens in autumn. This is consistent with the re-emergence mechanism explained by Holland et al. (2013) and Marchi et al. (2019).

After reemerging, predictability is suddenly lost in mid-winter (in July). The loss in predictability is consistent among all three sets of correlation analysis. We call this loss in predictability the “predictability barrier.” In our analysis, the predictability barrier is a clear, sharp loss of correlations in July (regardless of the lead month) and not the gradual decline we might expect from statistical red noise. This implies there is a change in the physical system in July. Previous studies (Blanchard-Wrigglesworth et al., 2011; Ordoñez et al., 2018; Giese et al., 2021) also show the permanent loss of predictability on a specific month (or a set of months, in the same season) but they do not explain the physical mechanism of this barrier.

Since predictability arises from OHC in the mixed layer, the loss of predictability suggests modification of OHC within the mixed layer. The changes in OHC causing loss of predictability in mid-winter can be caused by the mixed layer losing heat to the atmosphere as it cools the upper ocean and produces sea ice, or as the mixed layer gains heat from the ocean interior. In winter, since sea ice cover insulates the ocean from direct heat flux exchange with the atmosphere, heat loss from the ocean to the atmosphere should be reflected as a change in sea ice volume. The climatological sea ice freezing rate (Figure 2.3a) shows no sudden increase (unlike the very sudden loss of predictability), suggesting there are no sudden changes in the ocean-atmosphere fluxes that could explain the predictability barrier. In section 3.2, we have shown that the predictability barrier coincides with the time at which the seasonal pycnocline merges with PP.

During the ice growth season, the atmosphere cools the upper ocean inducing sea ice growth and deepening the mixed layer through enhanced vertical mixing from the brine released. Initially, this entrains remnant WW containing sea ice memory into the mixed layer, which explains the reemergence of predictability. Once the mixed layer deepens to reach the PP (after entraining all of the WW that contained sea ice-related memory), further sea ice growth entrains heat from the ocean interior (CDW) into the mixed layer (Gordon and Huber, 1984; Gordon and Huber, 1990; Martinson, 1990; Wilson et al., 2019), which has no sea ice-related memory, therefore terminating predictability associated with sea ice-related memory in the upper ocean.

Martinson (1990) and Wilson et al. (2019) showed that vertical heat flux driven by brine rejection placed a constraint on winter sea ice growth. Our analysis shows that this constraint is likely invoked in mid-winter in the Weddell sector when warm CDW is entrained into the mixed layer. We consider that the timing of predictability barrier signals when the negative ice-ocean feedback described by Martinson (1990) that limits sea ice growth rate due to entrainment is activated.

Goosse and Zunz (2014b) and Lecomte et al. (2017) showed how increased stratification in the upper ocean reduced vertical heat flux from CDW during ice growth, which enabled positive ice-ocean feedbacks. Lecomte et al. (2017) found the positive ice-ocean feedback, associated with a long-term trend in sea ice concentration only in the Ross sector. The fact that they do not find positive ice-ocean feedback in the Weddell Sea, is consistent with our finding of the existence of a predictability barrier that prevents the persistence of anomalies beyond 12 months. Therefore, the existence of this predictability barrier is a consequence of the absence of near-surface ice-ocean feedbacks that could potentially lead to long-term trends (in upper ocean properties and sea ice concentration). By investigating regional Antarctic sea ice predictability one can determine the presence or absence of predictability barriers that will provide valuable insights into long-term sea ice trends.

2.4.2 Dependence of sea ice predictability to mixed layer depth/ winter water depth

Mixed layer depth (MLD) has been given high relevance in previous studies of sea ice predictability. Holland et al. (2013) and Marchi et al. (2019) observed spatial variability in their prognostic sea ice predictability analysis and suggested that sufficiently deep mixed layers were required for retaining heat anomalies and hosting sea ice predictability. Our results align closely with findings from Marchi et al. (2019) in that the temperature anomalies relevant to sea ice predictability are stored at the depth range typical of WW. However as discussed in section 3.2.2, we find instances where the depth to which temperature anomalies extend, varies depending on the stratification strength at the PP. When the stratification is weak the T anomalies (memory) extends deeper than the WW. Also, Marchi et al. (2019) suggested that the effectiveness of the reemergence mechanism is associated with sufficiently large seasonal cycle of MLD (i.e., a transition from a shallow highly-stratified summer mixed layer to the deep WW). However, in finding comparable sea ice predictability from dynamic and slab ocean models, Ordoñez et al. (2018) suggested that variable mixed layer depth is less important to sea ice predictability than basic mixed layer temperature persistence, suggesting that the MLD is not the only important criterion for sea ice predictability during ice melt and growth season.

By using the maximum density gradient for evaluating the upper ocean evolution against MLD in our study, we were able to identify a physical constraint (predictability barrier) on sea ice predictability. This shows that it is not simply the depth to which the anomalies extend in the WW or ML that determines the potential for sea ice predictability, but also how well the thermal signatures linked to sea ice processes are retained in the upper ocean. The predictability barrier occurs with the transfer of anomalies (thermal signatures) across the PP, implying that the water across the PP is overcoming the ‘barrier’ of maximum stratification in the upper ocean. Our key finding is that sea ice and its underlying oceanic predictability follow changes in vertical ocean structure (dT/dz and $d\rho/dz$ gradients; Figure 2, all panels). Therefore, the vertical ocean structure in any region and its modification via sea ice processes determines its potential for retaining oceanic thermal memory, and by focusing only on the mixed layer depth, we lose other key features and processes related to sea ice predictability and its spatial variability.

Our sea ice predictability analysis has produced results consistent with previous studies, however, we take caution while comparing our results with the earlier studies: (1) we use a coupled ocean-sea ice model driven by reanalysis atmospheric fields, while previous Antarctic sea ice predictability studies (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019) used coupled atmosphere-ocean-sea ice models; (2) the regional diagnostic (using lagged correlations) sea ice predictability analysis in our study is similar to Ordonez et al (2018), while other studies use prognostic methods (using model ensemble spread). The coupled ocean-sea ice model provides a fair representation of the ice-ocean system required for our analysis at the spatial and temporal scales used in our study. The effect of atmospheric coupling would be important if the atmosphere responded significantly to sea ice variability, however previous work (Raphael et al., 2011; England et al., 2018) suggested this is not true at the spatial scales that we consider here. Our analysis looks at changes in total sea ice area and ocean properties averaged over a large area, in the Weddell Sea. Spatial plots of the variation of temperature, dT/dz and $d\rho/dz$ (Supplementary figures S3, S4) reveal two regimes, (1) the approximate ‘2-layer’ system that dominates our correlations, and (2) a narrow band of destratification (indicating Dense Shelf Water production) near the Ronne ice shelf. The destratified region does not show seasonal changes (especially below 50 m), whereas, in the 2-layer system, the vertical gradients evolve closely aligning with the averaged climatological profiles in Figure 2.3. Since the destratified region lacks seasonal

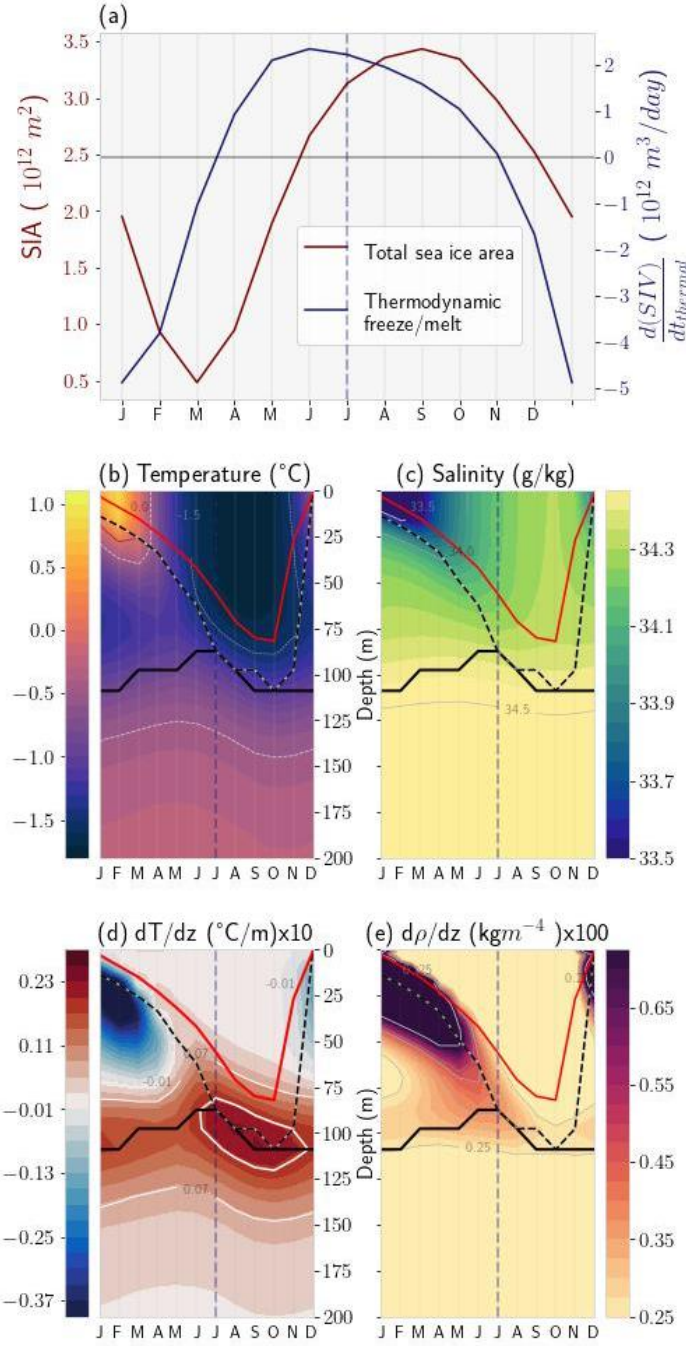


Figure 2.3: Annual evolution of climatological (a) Sea ice area (red line) and thermodynamic freeze and melt (blue line), (b) Temperature, (c) Salinity, (d) vertical temperature gradient (dT/dz), and (e) vertical density gradient ($d\rho/dz$), spatially averaged over the sea ice-covered ocean zone of the Weddell sector in the upper 200 m. The vertical temperature gradient maximum (thick black line), vertical temperature gradient (white lines), and vertical density gradient maximum (dashed line) are the same as used in Figure 2, and mixed layer depth (red line) is marked on all panels.

variability, it is the seasonal changes in the 2-layer system that drive the correlations in the oceanic predictability results presented in the study. Figures S3, S4, and S5 reveal that the SIA is affected largely by sea ice variability in the 2-layer system (more information and discussion in supporting information (TextS2)). This paper attempts to understand the processes in the upper ocean that drive sea ice predictability at a regional spatial scale. However, evaluating sea ice and ocean predictability using smaller domains may provide stronger correspondence between oceanic and sea ice predictability, but it is beyond the scope of this study.

We also do not consider the variability within the region, such as the transport of sea ice, nor the advection of oceanic properties. Sea ice motion can transport ice in or out of the sector or move it within the sector. Our estimate from the model shows that ~92% of the sea ice that forms in the Weddell sector melts within it; hence on a bulk scale, the net dynamic term is minimal compared to thermodynamic freeze/melt (Figure S7). Sea ice motion driven by the atmosphere and the ocean (Weddell gyre) impacts predictability by reducing the skill. Divergence of sea ice (due to sea ice motion) changes ice-covered and open-ocean areas that can lead to the modification of OHC anomalies causing loss or modification of ocean memory. Ice motion dominated by synoptic variability is inherently ‘unpredictable’ at the monthly-to-seasonal timescales that we consider here; Oceanic advection, especially strong northward drift reduces predictability skill (Holland et al., 2013; Bushuk et al., 2021). There may also be some relationship between sea ice variability and the strength of Weddell gyre (Neme et al., 2021) potentially impacting sea ice predictability. However, that is a complex relationship beyond the focus of this analysis. By averaging over a larger area such that thermodynamic effects become more important, we can see the predictability that arises from upper ocean heat storage (ocean memory), and the problem becomes tractable.

We have relied solely on the correlations to draw our interpretations and use the climatological oceanic parameters to guide our arguments. Quantifying the seasonal exchanges and thermal modifications occurring in the upper ocean is a potential follow-up analysis.

2.5 Conclusions

Over the 40 years of satellite record of Antarctic sea ice, the last decade has seen particularly large fluctuations in sea ice extent, including a record high value in 2014, followed by a record low in 2016-17. These recent fluctuations and the uncertainties in sea ice variability and trends linked to climate change make the emerging field of sea ice prediction particularly relevant. In this study we have analyzed the predictability of sea ice and underlying ocean in the Weddell sector of the Southern Ocean, using lagged correlations. We find that 1) sea ice predictability emerging from summer months persists until mid-winter, and 2) sea ice predictability emerging from spring months has a temporary loss during summer months and reemerges in autumn months. We also find that the predictability in the ocean linked to sea ice predictability is largely confined to the Winter Water layer, and it is dependent not only on the depth of the Winter Water layer but also heavily controlled by changes in the strength of stratification at the base of the Winter Water layer. Therefore, both these hydrographic parameters may be valuable for understanding regional differences in Antarctic sea ice trends and variability.

Our results are consistent with Holland et al. (2013) and Marchi et al. (2019) in (1) connecting upper ocean heat content with sea ice predictability and (2) with their proposed mechanism of predictability reemergence. In addition to the temporary loss of predictability in summer lag months prior to predictability reemergence, we find a more permanent loss of predictability in mid-winter. In mid-winter when the seasonal pycnocline merges with the permanent pycnocline, the mixed layer entrains warm Circumpolar Deep Water with no sea ice-related memory and terminates the predictability. Key insights from our study are in finding that (1) regional sea ice predictability is tied to the vertical structure of its oceanic properties and how this structure evolves, especially when forced by sea ice processes. This implies that the spatial variability in sea ice predictability can now be addressed based on local upper ocean vertical structure and sea ice processes. We also find that (2) the strength of stratification at the base of the Winter Water layer is relevant in determining the potential for sea ice predictability.

Oceanic predictability can be summed up as thermal anomalies lingering in the ice-ocean system at interannual timescales. These thermal anomalies generate sea ice predictability, which implies that sea ice predictability is a signature of local ice-ocean interaction mediated by residual thermal

anomalies. Therefore, our analysis not only improves our knowledge and capacity for operational Antarctic sea ice forecast, but it presents a potential tool for evaluating the regional signature of ice-ocean interactions. The fact that sea ice predictability is strongly tied to the vertical structure of oceanic properties suggests that changes in the upper ocean in a warming climate are likely to alter Antarctic sea ice predictability patterns in the future.

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Open Research

All model outputs including the simulations presented in this manuscript are stored as part of the COSIMA data collection here: <https://doi.org/10.4225/41/5a2dc8543105a>. Passive microwave sea ice data are publicly available, hosted by the National Snow and Ice Data Center (<https://doi.org/10.7265/N59P2ZTG>).

CHAPTER 3

Chapter 3: Ocean-sea ice processes driving the variability in regional Antarctic sea ice predictability

**This chapter is written as a manuscript in preparation*

Abstract

Satellite observations of Antarctic sea ice over four decades show strong regional variability, indicating the importance for a regional investigation of sea ice processes and its drivers. This study investigates sea ice and upper ocean predictability in six sectors around Antarctica to understand the drivers of this regional variability. The dominant sea ice predictability regimes identified are summer persistence and spring re-emergence. Strong summer persistence stemming from deep reaching (~0-200 m) ocean memory is found in the western sectors (Amundsen, Bellingshausen, and Weddell Sea sectors) but is lacking in the eastern sectors (King Haakon Sea, East Antarctica, and Ross Sea). This lack of summer persistence in the eastern sectors is due to the ocean memory in the winter water layer in these regions dissipating early in spring. Weaker summer stratification in the ocean and the lack of summer sea ice cover are linked to this faster dissipation of ocean memory to the atmosphere. In all sectors, the memory in the upper ocean underlying sea ice predictability is largely confined to the winter water layer. The intensity and the depth to which ocean memory forms are controlled by the vertical structure of the upper ocean. Further, ocean memory responds to seasonal changes in the upper ocean, especially changes in stratification strength driven mainly by sea ice processes. Our results show that the regional differences in the upper ocean structure and local ice-ocean processes are the main drivers of the variability in regional sea ice predictability.

3.1 Introduction

Sea ice is a key modulator of the climate system, a vital habitat for high-latitude ecosystems, and a sensitive indicator of climate variability (Massom and Stammerjohn, 2010; Hobbs et al., 2016). A characteristic of Antarctic sea ice is its significant regional seasonal and interannual variability (Hobbs et al., 2016). This regional variability results in differences in the duration of sea ice cover and contrasting seasonality of sea ice trends in regions around Antarctica (Stammerjohn et al., 2012; Holland, 2014; Matear et al., 2015; Schroeter et al., 2018; Raphael et al., 2019). Local ice-ocean interactions are tied to a range of sea ice factors such as the freezing and melting of sea ice, duration of sea ice cover, and the heat and freshwater fluxes associated with sea ice processes (Martinson, 1990; Stammerjohn et al., 2012; Goosse and Zunz, 2014b; Lecomte et al., 2017; Wilson et al., 2019). With the strong regional variability of these key factors, ice-ocean interactions at regional scales also vary. Therefore, it is important to investigate the physical drivers influencing sea ice at these regional scales around Antarctica and how this influences sea ice behaviour and predictability.

Regional variability is also observed in sea ice predictability studies. Sea ice predictability studies that have assessed the potential prognostic predictability (studies using an ensemble of model runs and evaluating predictability based on the ensemble spread with respect to a reference or control run) of the sea ice edge location show meridional differences (Holland et al., 2013; Zunz et al., 2015; Marchi et al., 2019; Bushuk et al., 2021). That means that they find higher predictability skills in certain regions (along a range of longitudes) over others. Regional differences have been found in diagnostic sea ice predictability (studies using a single model, evaluating predictability based on lagged correlations) (Ordoñez et al., 2018; Bushuk et al., 2021). Antarctic sea ice predictability studies, having established the role of the ocean as the primary source of memory that underlies sea ice predictability, linked its regional variability to the variability in mixed layer depth (Holland et al., 2013; Marchi et al., 2019). Studies have shown that ocean heat content (or thermal) anomalies retained in the upper ocean give rise to sea ice predictability, hence the argument is that regions with deeper mixed layers can retain these thermal anomalies longer, resulting in long-sustained sea ice predictability (Holland et al., 2013; Marchi et al., 2019). Holland et al. (2013) also suggested that processes such as vertical mixing and ocean advection can modify the thermal anomalies in the upper ocean and thereby impact sea ice predictability.

Indeed, studies investigating Antarctic sea ice-ocean interaction and feedbacks emphasise the role of vertical mixing between the upper ocean [winter water layer (WW) that is in thermal equilibrium with the sea ice] and the ocean interior in determining the regional variability of Antarctic sea ice (Martinson, 1990; Martinson and Iannuzzi, 1998; Stammerjohn et al., 2012; Lecomte et al., 2017; Wilson et al., 2019). Sea ice forms and interacts with the WW (~100-200 m deep) at near freezing temperature in winter, below this layer is the ocean interior [in most instances Circumpolar Deep Water (CDW)], which is significantly warmer and deeper (Gordon and Huber, 1984). A relatively thin (permanent) pycnocline separates the cold WW from warmer CDW, and in regions where the stratification strength is weaker, vertical mixing induced by sea ice growth entrains warm CDW into the WW, modifying (warming) the WW and limiting further sea ice growth (Martinson, 1990). Whereas in regions where the stratification strength at the permanent pycnocline (PP) is stronger, the entrainment of heat from CDW into the WW is limited and can lead to increased sea ice growth (Lecomte et al., 2017). With the ocean interior acting as a large reservoir of heat, right below the layer retaining the thermal memory for sea ice predictability, vertical ocean heat flux can modify the thermal anomalies and cause changes and potential loss of sea ice predictability.

Libera et al. (2022) investigated sea ice and upper ocean predictability with a focus on physical processes in the ocean drive sea ice predictability and found that ocean memory responded to the seasonal evolution of the upper ocean water column, especially to the changes in the stratification. They found that sea ice predictability was lost in mid-winter when CDW is entrained into the WW, producing evidence linking sea ice predictability to the local ice-ocean interaction. They showed that sea ice predictability emerging from a region has a strong dependence on the local vertical ocean structure, and that sea ice predictability not only depended on the depth to which the ocean retains memory but how well it is preserved which was tied to the strength of stratification. However, Libera et al. (2022) conducted the analysis in only one region (the Weddell Sea), therefore, whether the regionally varying ice-ocean interactions (dependant on upper ocean vertical structure) contribute to regional variability of sea ice predictability is yet to be addressed.

This study investigates sea ice and upper ocean predictability in six regions around Antarctica, as a follow-up analysis to Libera et al. (2022). By investigating the regional variability in sea ice and upper ocean predictability, this study explores the contributions of local ice and ocean conditions, as well as ice-ocean interaction, to regional sea ice predictability.

3.2 Methods

3.2.1 Data

The data, methodology and terminology used in this study follow from Libera et al. (2022), in which the focus was on the Weddell Sea region, here extended to the entire Southern Ocean. Outputs from the high horizontal resolution (0.1°) version of a global coupled ocean-sea ice model, the Australian Community Climate and Earth System Simulator (ACCESS-OM2-01) are used. ACCESS-OM2 is based on the ocean MOM5.1 and ice CICE5.1 models coupled with the OASIS-MCT coupler (Kiss et al., 2020), and the experiment used in this study was forced using JRA55-do v1.4.0 atmospheric reanalysis (Tsujino et al., 2018). Adequate simulations of the Antarctic sea ice extent and concentration are produced in ACCESS-OM2-01 (Kiss et al., 2020). The historical sea ice trends are well represented and the mean annual cycle of Antarctic sea ice extent from ACCESS-OM2-01 closely matches observations (presented in Kiss et al., (2020), shown in Sx7 and Sx8). ACCESS-OM2-01, due to its higher horizontal resolution, produces a good representation of the Southern Ocean dynamics by simulating the mesoscale variability, which is critical in capturing the evolution of the system (Kiss et al., 2020).

3.2.2 Correlation analysis

The sector definition used in this study (Figure 3.1a) closely follows that described in Raphael and Hobbs (2014), based on spatial autocorrelation calculations and decorrelation analysis. The Ross-Amundsen and Amundsen-Bellingshausen regions from Raphael and Hobbs (2014) are differentiated into three regions: Ross Sea, Amundsen Sea, and Bellingshausen Sea to focus on the regional differences. In each of these sectors, the northern boundaries are approximated to the latitude of the average winter sea ice extent maximum over the 1985 to 2015 period. The regional boundaries for the six sectors and their abbreviations are as follows: King Haakon VII Sea (KH, $345\text{--}70^\circ\text{E}$, coast to 55°S), East Antarctica (EA, $70\text{--}150^\circ\text{E}$, coast to 57°S), Ross Sea (RS, $150\text{--}225^\circ\text{E}$, coast to 62°S), Amundsen Sea (AS, $225\text{--}270^\circ\text{E}$, coast to 63°S), Bellingshausen Sea (BS, $270\text{--}300^\circ\text{E}$, coast to 60°S), and Weddell Sea (WS, $300\text{--}345^\circ\text{E}$, coast to 58°S). The regional boundaries, including the northern limit, are constant throughout the analysis for all months.

Using 30 years of data from 1985 to 2015, the diagnostic predictability of Antarctic sea ice and upper ocean temperatures is analysed by calculating the correlation between an initial month and the eleven future (or lagging) months. The results presented in this study are limited to eleven lagged months because of the decorrelation time scale in most sectors. For sea ice

predictability, the total sea ice area (SIA) in each sector is calculated from sea ice concentration (SIC) data. The SIC is multiplied against the grid cell area and summed within each sector. The correlation analysis is then applied to the detrended total SIA timeseries. The correlation analysis on total SIA reveals persistent anomalies in the sea ice (hereafter referred to as ‘sea ice predictability’ i.e., the predictability arising from sea ice persistence). For oceanic predictability, attention is directed to the upper 200 m, assuming that most of the upper ocean processes associated with sea ice will be confined within this layer (except in regions of deep convection or where the upper ocean is significantly influenced by horizontal and vertical advection). Detrended timeseries of conservative temperature (T) in the upper ocean are created after spatially averaging within the respective sectors. Since this study aims to understand the ocean ‘memory’ linked to sea ice processes, the correlation analysis is applied between the sea ice and ocean fields. The correlation between the detrended SIA timeseries from an initial month against the monthly detrended upper ocean temperature timeseries for the same month (zero-lag) and with eleven future (or lagging) months (hereafter ‘ice-ocean correlation(s)’) reveals the sea ice-related memory in the upper ocean. Statistically significant values are defined as p-values greater than 95% in the two-tailed Student's t-test.

3.2.3 Oceanic climatologies and definitions of permanent and seasonal pycnocline

Monthly climatologies of oceanic variables are used to follow and understand the upper ocean’s seasonal changes. Monthly climatologies of oceanic variables are created from their spatially averaged vertical profiles within each sector. The potential density climatology (potential density referenced to the surface; ρ) is calculated from the temperature (T) and salinity (S) climatologies. Using (T) and (ρ), the climatologies for the vertical temperature gradients (dT/dz) and vertical density gradients ($d\rho/dz$) are then calculated. The maximum gradients of (dT/dz) and ($d\rho/dz$) are used as proxies for the permanent pycnocline (PP) and seasonal pycnocline (SP) respectively. Although buoyancy is haline-dominated in these regions, dT/dz maximum is used to define the PP, since it provides a clearer distinction between the surface and deep ocean layers.

The PP and SP are used to identify the boundary between the upper ocean's three main layers (or water masses) - the summer surface water, the WW, and the CDW. The PP separates the CDW in the ocean interior with little seasonal variation, from surface waters that are strongly modified by sea ice processes. Above the PP lies the WW, the cold water layer formed during

sea ice growth which retains most of the sea ice-related anomalies. In summer, the freshwater from sea-ice melt leads to high stratification at the ocean surface, generating the SP as an interface between this surface water (meltwater) and the WW below, creating the upper ocean's three main layers (i.e., summer surface water, WW, and CDW). The upper ocean layers change seasonally, there are three layers in summer, while in winter there are only two layers (WW and CDW) as the SP merges with PP. The summer surface layer (or seasonal stratification) forms in summer and erodes by early winter. When the summer surface layer forms at the surface, it sandwiches the WW layer between the SP above and PP below. As the upper ocean cools through autumn, leading to increased vertical mixing induced by sea ice growth, the WW entrains into the surface layer, the SP weakens and deepens until it eventually merges with the PP in winter (eroding the SP and decaying the summer surface layer).

3.3 Results

3.3.1 Sea ice predictability modes and features

Predictability is represented using correlations in a forward-looking perspective, which indicates how a given initial sea ice state relates to future ice and ocean states. Persistence and re-emergence are the two main modes of sea ice predictability identified in the literature (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021; Giesse et al., 2021). 'Persistence' in this context is the monotonic occurrence of high correlations originating from an initial month sustained over the first few future months. Persistence shows strong seasonality, i.e., sea ice anomalies in some months have longer persistence than in other months. Our results indicate that summer and winter anomalies have the longest persistence (or longer sustained memory) and are termed 'summer persistence' and 'winter persistence' (marked as 1 and 2 in Figure 3.1b; originating from summer and winter initial months). Similar seasonality of persistence and the occurrence of strong summer and winter persistence have been observed in other diagnostic sea ice predictability studies in both the Arctic and Antarctic (Blanchard-Wrigglesworth et al., 2011; Ordoñez et al., 2018; Bushuk et al., 2021; Giesse et al., 2021; Morioka et al., 2021).

Re-emergence is a special case of persistence that has received much attention, especially in the Antarctic (Holland et al., 2013; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al.,

2021; Libera et al., 2022). This study finds re-emergence for correlations from spring initial months and refers to it as the ‘Spring re-emergence’ (marked as 3 in Figure 3.1b; originating from spring initial months). In spring re-emergence, high correlations persisting in the spring months disappear in the summer months and re-emerge in the future autumn and winter months. Previous studies that explored the mechanism of re-emergence (Holland et al., 2013; Marchi et al., 2019; Libera et al., 2022) have established that the ocean acts as the primary source of memory for sea ice predictability and that re-emergence is associated with the seasonal evolution of the upper ocean. In winter when sea ice is in thermal equilibrium with the WW, thermal anomalies at the surface linked to sea ice extend deep into the ocean (depth of WW or winter mixed layer). In summer, when sea ice melts and associated freshwater stratifies the ocean near the surface, these thermal anomalies retained at depth are isolated from the surface causing the loss of predictability. But when the summer stratification erodes and the SP deepens it causes the thermal anomalies to resurface giving rise to the re-emergence of sea ice predictability.

Along with the main modes of sea ice predictability, two associated features are also identified. First, the ‘predictability suppression’, that is associated with the spring reemergence pattern (marked as 4 in Figure 3.1b). Predictability suppression is the intermediate loss of predictability in summer, prior to the re-emergence due to the seasonal stratification that forms near the surface (in autumn future months). The second feature is the ‘predictability barrier’; ‘predictability barrier’ or ‘barrier’ has been used in sea ice predictability studies in various contexts (Ordoñez et al., 2018; Marchi et al., 2020; Bushuk et al., 2021; Libera et al., 2022). The same description as in (Libera et al., 2022) is followed in this study, where the predictability barrier is described as the sharp loss of persistence (correlations) which mostly coincides with the timing of when the SP merges with the PP (marked as 5 in Figure 3.1b). Predictability barrier and predictability suppression represent two different processes in the ocean in two different seasons that impact sea ice predictability. Predictability suppression is the transient loss of sea ice predictability related to the formation of seasonal stratification that isolates the memory in the WW from the surface). The predictability barrier is linked to the entrainment of signals from the ocean interior into the WW and thereby modification (or a permanent loss) of sea ice-related memory from the WW.

Thus, predictability suppression and predictability barrier can also be perceived as a temporary versus permanent loss of correlation. Some exceptions are observed, e.g., in the Weddell and Ross Sea sectors, the memory from the summer initial months produces correlations in the

following summer future months (~10 months lag) beyond the predictability barrier. This signal is also different from the traditional mechanism of re-emergence. The source or mechanism of this summer-to-summer occurrence of predictability in the Ross and Weddell Sea sectors is not identified in this study. Overall, the analysis reveals summer persistence and spring re-emergence as the most prevalent predictability modes for all six sectors.

On examining the sea ice predictability from all six sectors (Figure 3.1 c-h), most sectors have multi-month persistence emerging from summer, winter, and spring sea ice anomalies. The clearest distinction in sea ice predictability between the sectors is their differences in the persistence of summer anomalies in east and west Antarctic sectors. Sectors in west Antarctica (Amundsen Sea, Bellingshausen Sea, and Weddell Sea; Figure 3.1 f-h) produce summer persistence lasting until winter (~6 months lag), while the sectors to the East (King Haakon and East Antarctica, Figures 1 c, d) lack such long sustained summer persistence and summer anomalies do not persist beyond the start of sea ice growth (February-March, i.e., ~3 months lag or less). The Ross Sea sector (located between the eastern and western sectors) produces summer persistence at an intermediate time scale. Such an east-west contrast is not observed for other modes of predictability.

Spring re-emergence can be defined as the memory from spring months persisting in the ice-ocean system until the future autumn or winter months, however, the memory is temporarily lost in summer future months (lags) within this persistence period. The loss of persistence in summer (predictability suppression) is associated with the formation of seasonal stratification and the memory re-emerges once the SP deepens and as the seasonal stratification decays. Spring re-emergence is found in the King Haakon Sea, Amundsen Sea, and Weddell Sea sectors (Figure 3.1 c, f, h). It is also present in the East Antarctic and Ross Sea sectors but with a weaker signal. Notably, the Bellingshausen Sea sector lacks spring re-emergence (Figure 3.1 g) and instead produces substantially longer persistence from winter and spring to the following autumn (up to 11 months lag). The King Haakon sector has a long period of predictability suppression (5 months lag) where the memory of sea ice anomalies from later winter-spring (Aug-Dec) is suppressed in January and does not re-emerge until the following April (Figure 3.1 c). In most other sectors that produce spring re-emergence (i.e., except the Bellingshausen Sea), predictability suppression is shorter, with the shortest being in the East Antarctic sector (one month lag).

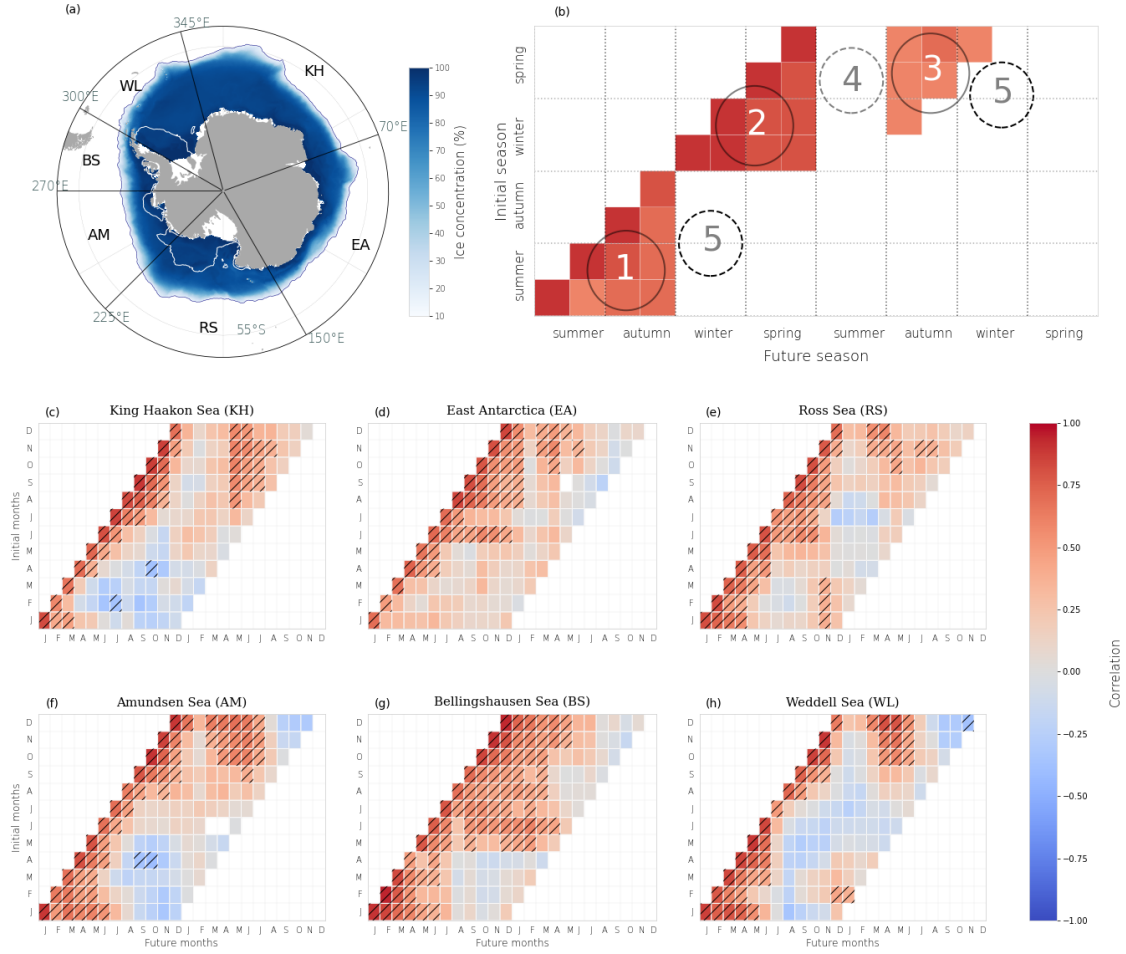


Figure 3.1: (a) Map of the six regions around Antarctica (a), also showing the climatological winter sea ice extent maximum (in September; shading) and minimum (in February; white contour). (b) Graphical representation of the main seasonal sea ice predictability regimes and associated features; (1) Summer persistence, (2) Winter persistence, (3) Spring re-emergence, (4) Predictability suppression, and (5) Predictability barrier. (c)-(h) Sea ice predictability results for the six regions, where the total sea ice areas from the initial months (along the y-axis) are correlated against the total sea ice areas in the future months (along the x-axis) and statistically significant values (>95 %) are hatched.

In some sectors, the high correlations from spring re-emergence as well as summer persistence encounter a predictability barrier, which is a sharp loss of persistence in winter. This predictability barrier is found in the Amundsen Sea and Weddell Sea sectors. Libera et al. (2022) found that the predictability barrier in the Weddell Sea is linked to the merging of the SP with the PP, where water from below the PP (CDW, with no sea ice-related memory) is entrained into the WW and dissipates the upper ocean memory.

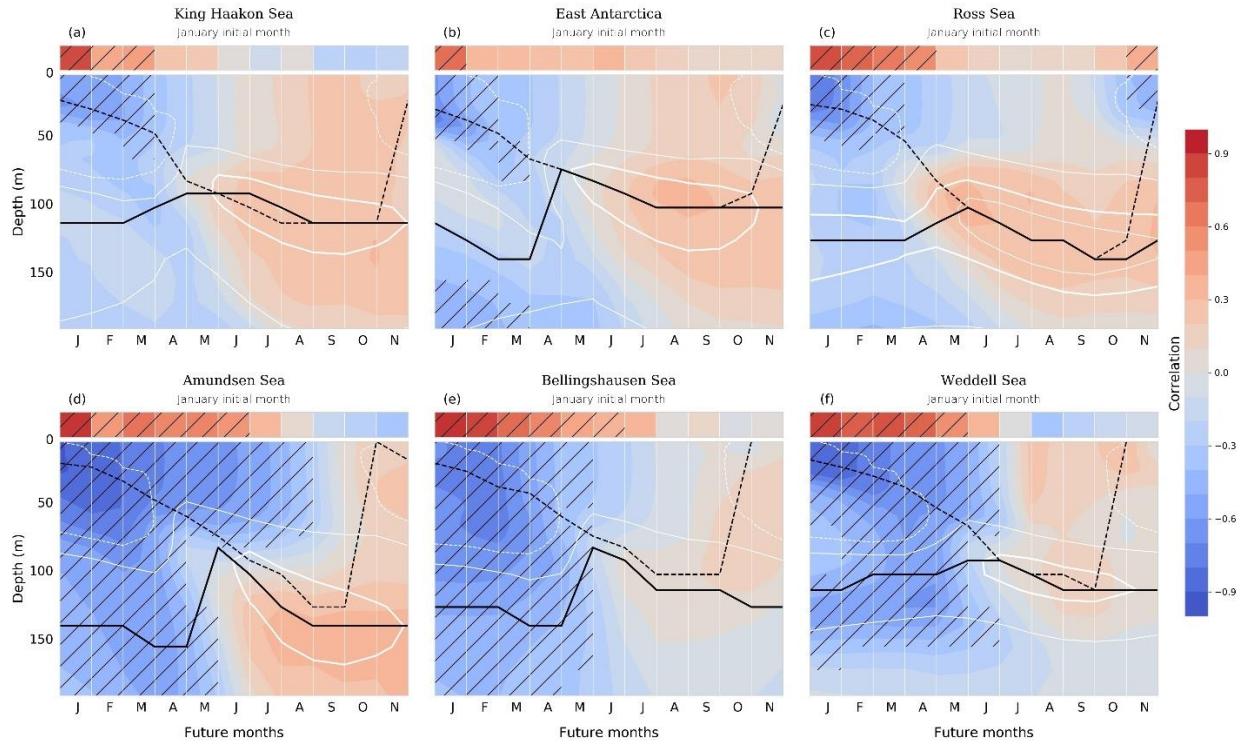


Figure 3.2: Ice-ocean correlations in the upper ocean (0-200 m) for the six sectors showing summer persistence from January initial month. Each of the panels also includes the sea ice predictability from January initial month for the respective sectors for comparison. The thick black line and the dashed black line over the ice-ocean correlations are the climatological vertical temperature gradient (dT/dz) maximum and vertical density gradient (dp/dz) maximum. The vertical temperature gradients are marked using white contours, white dashed contours bound the vertical temperature gradient values that are negative due to summer stratification. Statistically significant values (>95%) are hatched on all panels

3.3.2 Upper ocean predictability and drivers of its regional variability

To evaluate the oceanic source of memory underlying sea ice predictability and to explore the drivers of its regional variability, the predictability in the ocean associated with sea ice processes is assessed. Strong statistically significant negative ice-ocean correlations are found in the ocean (~upper 200 m, e.g., Figure 3.2, all panels), where there are strong positive statistically significant correlations in the sea ice predictability (in each sector at the respective initial and future months). The negative correlation in the ocean occurs due to the cold (warm) surface ocean conditions that are associated with sea ice formation (melt). These negative ice-ocean correlations are found largely in the upper 50 to 100 m in most sectors, but in some sectors, these correlations extend deeper during certain seasonal ranges (Figure 3.2, 3.4, and

Sx1 – Sx6). Statistically significant positive ice-ocean correlations are also found in some sectors (mostly in the Ross Sea and the Amundsen Sea) near the PP (~100 m depth) and below (discussed in section 4). Since summer persistence and spring re-emergence are the most prevalent sea ice predictability modes for all sectors, the upper ocean predictability focusing on these two modes is presented below.

3.3.2.1 Summer persistence from oceanic predictability

The upper ocean predictability initialized in January is used to evaluate the summer persistence for all sectors. The eastern and western sectors contrast in the oceanic predictability consistent with the findings with sea ice predictability in section 3.1.1. Summer persistence in the eastern sectors (King Haakon Sea, East Antarctica) and in the Ross Sea sector (Figure 3.2 a-c) occurs in a shallower layer of the upper ocean (above the SP; ~50 m) and lasts shorter (2-3 lag months); summer persistence in the western sectors (Amundsen, Bellingshausen, and Weddell Sea sectors; Figure 3.2 d-f) extends deeper into the ocean (even below the PP, deeper than 150 m) and lasts longer (5-8 lag months). Our oceanic predictability results (Figure 3.2) indicate that the depth to which thermal anomalies penetrate the ocean can determine a region's capacity to produce long-sustained persistence.

Potential causes for the contrast in summer persistence between the eastern and western sectors are evaluated to understand (1) why does ice-ocean correlations penetrate deeper in the western sectors than the eastern sectors? (2) Are the deep ice-ocean correlations in the western sectors formed during summer initial months or are they retained from those formed during winter and spring initial months? If they are retained from the winter or spring initial months, then (3) what processes contribute to the rapid dissipation of memory in the eastern sectors? The climatological sea ice freeze calculated in December and January (not shown) confirms that there is no sea ice growth in summer. Hence, there is no sea ice growth-related vertical mixing bringing ice-related ocean memory into the deeper layer (WW) in summer. That is, the ice-ocean memory in the deeper layers is the remnant from the previous ice growth season. Now the question is why does ice-ocean memory persist longer in the western sectors, and why do they dissipate earlier in the eastern sectors?

The differences in the upper ocean and sea ice climatology between the sectors are evaluated to identify the processes or conditions that support the retention of winter and spring ice-ocean anomalies into summer in the western sectors. Comparing the stratification strength (dT/dz , and dp/dz) at the SP and PP, the eastern sectors (KH, EA) have weaker stratification at the SP

in summer (Figure 3.3d). Also, the eastern sectors lose sea ice cover earlier and become ice-free for most of the summer, whereas the western sectors lose sea ice cover gradually and retain some fraction of sea ice throughout summer (white line in Figure 3.3e). The eastern sectors have low summer sea ice cover but are not completely ice-free for all summer months and therefore have variability (and can produce correlations). The differences in the upper ocean and sea ice conditions between the eastern and western sectors result in the hypothesis that the combination of weaker summer stratification and near-absence of sea ice cover in summer plays a role in the early dissipation of winter and spring ice-ocean anomalies (from within the WW) in the eastern sectors.

Sea ice cover insulates the upper ocean, limiting high rates of heat exchange with the atmosphere, and preserving the ice-ocean memory in the upper ocean. Similarly, strong stratification over the WW (i.e., the SP) acts as a barrier preventing the dissipation of ice-ocean memory within the WW to the atmosphere. Therefore, the near absence of sea ice cover in summer (Figure 3.3e) and weaker summer stratification (Figure 3.3d) contribute to increased surface heat flux in the eastern sectors. This in turn causes the dissipation of memory from WW and the lack of long persisting summer anomalies in the eastern sectors. The climatological surface heat flux in late spring and early summer months is lower for the western sectors (under the climatological sea ice cover) than the eastern sectors (Figure 3.3e). Weaker stratification and the lack of summer sea ice cover therefore facilitate increased surface heat flux exchange in the eastern sectors, resulting in the earlier dissipation of the ice-ocean memory from the WW (from the previous sea ice season). However, memory forming in summer is present in the shallow mixed layer (above the SP) and this memory persists for a shorter duration.

To summarise, the east-west contrast in the sea ice and ocean predictability for summer persistence is explained as follows: Eastern sectors host ice-ocean memory from summer only in the mixed layer above the SP and have shorter persistence, while the western sectors retain ice-ocean memory from previous winter and spring deeper into the ocean below the PP and also host memory from summer in the mixed layer above the SP. This enhances the overall ice-ocean memory in the western sectors leading to longer summer persistence.

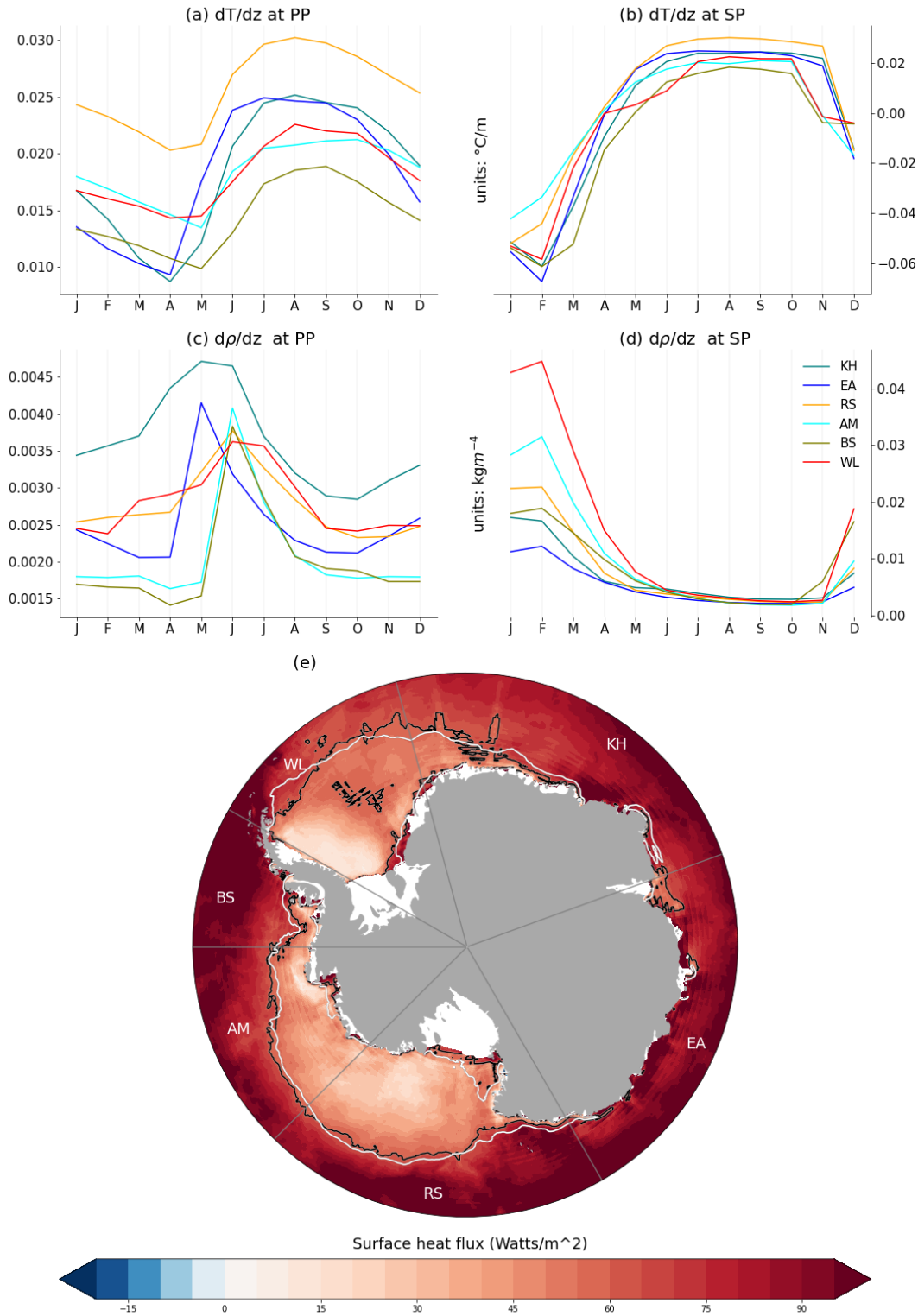


Figure 3.3: Stratification strength at the permanent pycnocline (PP) and seasonal pycnocline (SP) for all six sectors. Annual evolution of the vertical temperature gradient at (a) the PP and (b) at the seasonal pycnocline. And annual evolution of vertical density gradient at (c) the PP and (d) the SP. (e) Map of climatological Surface Heat Flux (SHF) in January, showing the six sectors used in this study and highlighting SHF values equalling 60 Wm^{-2} (black line) and climatological sea ice extent (white line).

3.3.2.2 Spring Re-emergence from Oceanic Predictability

Examining each sector for the spring re-emergence regime reveals variation in the timing when individual sectors produce a pronounced signature. Hence, instead of using a single representative initial month (as used for summer persistence in Figure 3.2), ocean predictability initialised in different months (ranging from August to November) that represents best the signature of spring re-emergence in each sector are presented (Figure 3.4). The results of ice-ocean correlations from all twelve initial months are provided in the supplementary Figures Sx1-Sx6. Spring re-emergence consists of two features: the predictability suppression (i.e., the temporary loss of ocean memory at the surface in summer) followed by the re-emergence of ocean memory at the surface. As established in previous studies, the formation of seasonal stratification in summer leads to the occurrence of predictability suppression (See Figure 3.4; shallow SP). In most cases, the loss of ocean memory occurs around the depth of SP in summer. When the SP deepens in autumn and winter, and as the stratification strength weakens (see Figure 3.3d), the ocean memory that was isolated from the surface by the summer stratification resurfaces.

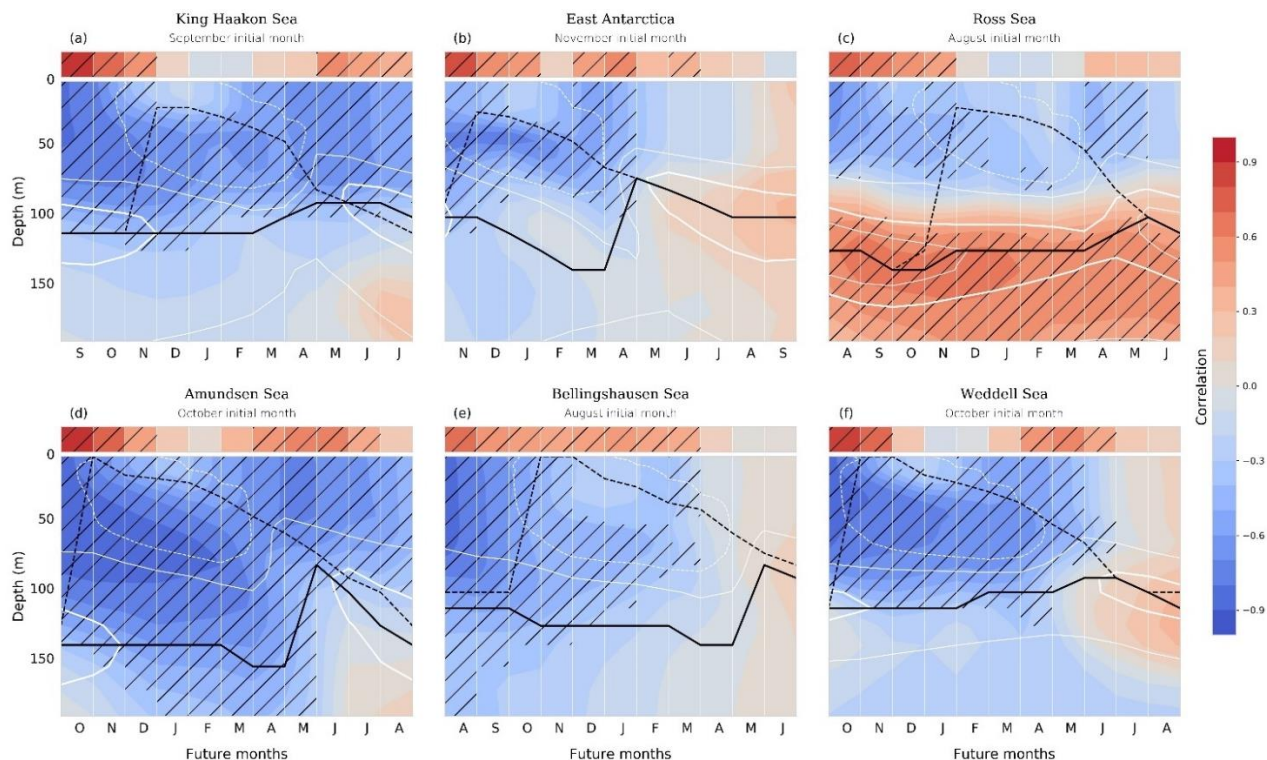


Figure 3.4: Ice-ocean correlations in the upper ocean (0-200 m) for the six sectors showing the spring re-emergence pattern occurring in the six sectors from various initial months. Each of the panels includes corresponding sea ice predictability for respective sectors and initial months. Hatching and all contours used in the panels are described in Figure 2.

All sectors produce a signature of spring re-emergence between mid-winter and spring, of which the earliest occurrence is in the East Antarctic and Ross Sea sectors in June and the latest in the Weddell Sea sector in October (Figures Sx2, Sx3, and Sx6). This variability in timing among sectors may be attributed to the regional differences in the seasonality of sea ice (Holland, 2014). Further, the differences in how well the ice-ocean correlation is formed and preserved in respective sectors at the different stages of the sea ice seasonal cycle (winter and spring initial months) also contribute to the regional variability in the timing of spring re-emergence. Based on Antarctic sea ice predictability studies that show the ocean as the primary source of memory underlying sea ice predictability (Holland et al., 2013; Marchi et al., 2019; Libera et al., 2022), strong correspondence is expected between sea ice and ocean predictability (i.e. the timing of predictability suppression and the subsequent re-emergence in ice and ocean predictability occur simultaneously). But in most instances, there is a slight mismatch in this timing (by about one to two months).

Since the analysis assesses changes in the upper ocean conditions and sea ice area over large spatial scales (spatially averaged) and at monthly timescales (which is averaged output from the model), it is not surprising that there is some mismatch. If the analysis was instead conducted using an idealised 1D sea ice-ocean model, the timing in the ice and ocean predictability responses are expected to match closely. However, instances where major discrepancies in the timing between ocean and sea ice predictability may indicate that drivers other than the ocean dominate in those instances. Such a major discrepancy is found in the Bellingshausen Sea sector for spring re-emergence (Figure 3.4e), where the ocean has a clear loss of memory at the surface, but predictability in the sea ice persists with no hindrance. This result directly contradicts the predictability re-emergence mechanism explained in previous studies (Holland et al., 2013; Libera et al., 2022; Marchi et al., 2019) and suggests that other sources of memory are sustaining or driving the persistence in sea ice predictability in the Bellingshausen Sea region (Figure 3.4e and Figure Sx5).

Besides the ocean, the atmosphere and sea ice pack can act as potential sources of predictability. The atmosphere is considered to hold memory only at daily to weekly timescales and in most cases is inherently unpredictable at the monthly to seasonal timescales used in this study. However, studies have shown a strong El Niño-Southern Oscillation (ENSO) teleconnection with the Amundsen-Bellingshausen Sea region (Harangozo, 2000; Simmonds et al., 2003; Raphael and Hobbs, 2014; Purich et al., 2016), which is a longer acting atmospheric mode, potentially providing a source of predictability in this region. The Amundsen Sea Low (ASL),

with its climatological low-pressure centre mostly located in the Amundsen-Bellinghshausen Sea region, also exerts a strong influence on the regional sea ice concentration and thickness (Raphael et al., 2016; Turner et al., 2016; Raphael et al., 2019). Holland et al. (2018) found that the ASL exerts a lagged influence on sea ice for up to eight months. They pointed out that this lagged influence of the ASL on sea ice could have implications for Antarctic sea ice predictability.

Sea ice thickness was found to be a source of predictability in regions that host thick multi-year ice (Bushuk et al., 2021; Morioka et al., 2021). Bushuk et al. (2021) describe that thick ice requires additional energy to melt, hence producing positive sea ice anomalies during the melt season and providing a source of summer sea ice prediction skill. Bushuk et al. (2021) and Morioka et al. (2021) focused on the Weddell Sea region (that hosts thick multi-year sea ice) and found an additional source of sea ice predictability from sea ice thickness. Studies show that the Bellingshausen Sea region also hosts thick multi-year ice (Holland et al., 2010; Xie et al., 2011), hence it is likely that sea ice thickness provides the additional source of sea ice predictability in the Bellingshausen Sea sector during the melt season. Analysing these potential sources of predictability (other than the ocean) is beyond the scope of this paper but presents opportunity for further research.

3.3.2.3 Predictability barrier

The predictability barrier identified by Libera et al. (2022) is a sharp (permanent) loss of ice and ocean memory, found to occur when the SP merges with the PP. The predictability barrier (marked as 5 in Figure 3.1 b) is linked to the entrainment of CDW into the WW when the mixed layer deepens to merge with the permanent pycnocline and usually occurs in late autumn or winter future (lag) months. Unlike the predictability suppression (which is associated with the spring re-emergence regime), the predictability barrier is not linked to a single sea ice predictability regime. The predictability barrier occurs along different predictability regimes (mainly following summer persistence and spring re-emergence). For example, in the Weddell Sea sector, summer persistence terminates after ~6 months lag and spring re-emergence terminates after ~9 months lag in July due to the predictability barrier (Figures 3.2f and 3.4f). Libera et al. (2022) described the mechanism of the predictability barrier as follows: The memory in the ice-ocean system is largely confined within the WW layer. In summer this memory is isolated below the SP (and above the PP) when sea ice melt produces strong

stratification near the surface. Surface ocean cooling and sea ice growth cause the SP to deepen in autumn, which initially entrains water from WW containing ice-ocean memory leading to re-emergence of sea ice predictability (consistent with the literature). But once the SP deepens beyond the WW, it entrains the warmer water from CDW below, which has no sea-ice related memory. This alters the thermal signature within the WW and results in the loss of predictability or 'predictability barrier'.

When examining all sectors, the predictability barrier is found to occur for predictability emerging from summer (e.g., Weddell Sea Figures 3.1h, 3.2f), winter (e.g., East Antarctica and Ross Sea; Figure Sx2, Sx3), and spring. However, as described in earlier sections, summer persistence has an east-west contrast and winter persistence is not a dominant mode of predictability in all sectors. Hence, regional variability of the predictability barrier is evaluated from the spring re-emergence regime (as it is present for all sectors). A clear signature of the predictability barrier (loss of ice-ocean correlations when SP merges with PP) is found in the East Antarctic, Ross Sea, and Weddell Sea sectors. Merging of SP with PP occurs in winter: April-May in East Antarctica, May-June in the Ross Sea, and June-July in the Weddell Sea sectors.

Without a predictability barrier, the ocean memory is expected to persist longer and dissipate gradually, given there is no other mechanism for dissipating the memory. Indeed, amongst the King Haakon VII Sea and Amundsen Sea sectors (where the predictability barrier is not identified), ice-ocean correlations persist all 12 lag months in some initial months. The predictability barrier is interpreted as the stage where water from the ocean interior (i.e., CDW) enters the WW (aligning with the merging of SP and PP). However, in the Amundsen and Bellingshausen Sea regions, the SP and PP do not merge (Figure 3.4 d, and e). Instead, after the SP approaches the PP in winter, it follows the PP closely in spring. This slight difference in depth between the SP and PP is due to the vertical discretisation of the model levels which are about 10-15 metres thick at this depth. Even though the SP and PP do not merge, the point where the SP, after being shallow in summer approaches PP in winter is considered as the time when the surface layer (WW) with sea ice memory mixes or interacts with the ocean interior.

It is important to note that the sectors that lacked the predictability barrier (King Haakon, Amundsen, and Bellingshausen Sea sectors; Figure 3.4 a, d, and e) lie adjacent to large sea ice producing sectors. To test if the import or export of sea ice played any role in the presence or absence of the predictability barrier, the relative contribution of dynamic sea ice volume was

evaluated in all sectors. Table 3.1 presents the results of calculating the net dynamic sea ice growth term integrated over a year and dividing it by the integrated annual sea ice volume for each sector. The results show in fraction (ranging from -1 to 1) how much of the total ice is advected in (positive) or out (negative). As expected, King Haakon and Amundsen Sea regions have a net gain of sea ice from adjacent sectors and they act as net melting regions. Excess freshwater from sea ice melt stratifies the upper ocean and acts to inhibit the negative ice-ocean feedback (Martinson and Iannuzzi, 1998; Wilson et al., 2019). It is possible that the advected ice and ocean anomalies may be modifying ice-ocean feedbacks in the regions downstream.

Sector	Net dynamic sea ice growth ÷ Net sea ice volume (year ⁻¹)
King Haakon Sea	0.164
East Antarctica	0.027
Ross Sea	-0.15
Amundsen Sea	0.11
Bellingshausen Sea	-2.15
Weddell Sea	-0.05

Table 3.1: Dynamic contribution of sea ice growth in the six sectors showing the net dynamic sea ice growth term integrated over a year and divided with the integrated annual sea ice volume.

Sea ice and ocean predictability (based on correlations) and climatological upper ocean conditions are used in this study; however, entrainment calculations are not performed. Therefore, to explain the regional variability in the occurrence of predictability barrier, especially, the absence of predictability in the three sectors, the findings from ice-ocean interaction studies are relied on. Since the predictability barrier represents the entrainment of CDW into WW exerting negative ice-ocean feedback to sea ice growth, it is expected that these regions have winter-time heat ventilation. Wilson et al. (2019) investigated wintertime upper ocean stability and heat availability in sea ice-covered Southern Ocean using observations and found that the thermodynamic coupling between winter sea ice growth and ocean ventilation has significant regional variability. They found that a combination of weak stratification in the upper ocean and a large temperature gradient below the mixed layer in winter favoured high

rates of wintertime heat ventilation and strong negative ice-ocean feedback. Also, regions with low rates of sea ice growth may never attain the amount of brine needed to erode the PP.

Wilson et al. (2019) found high ventilation efficiency in the regions comparable to the Weddell Sea and East Antarctic sector in this study. They also found the sea ice growth that occurs in a single winter season in the Amundsen and Bellingshausen Sea regions, was several times lower than what is needed to erode the PP. This reason may be valid in our study. In the diagnostic developed by Martinson (1990), they evaluate the amount of sea ice growth required to remove the excess freshwater from the surface layer to initiate convection. Both Amundsen and Bellingshausen Sea sectors in this study are considerably fresher than other sectors (potentially due to the melting of advected sea ice from the Ross Sea sector) and have relatively weaker gradients at the thermocline (not shown), they are also not the largest sea ice producers. Hence aligning with the findings from Wilson et al. (2019), the lack of predictability barrier in the Amundsen and Bellingshausen Sea regions is likely linked to the excess freshwater at the surface and lower sea ice production rates. King Haakon sector maintains a strong density gradient at the PP throughout the year, which is stronger than the rest of the sectors. It is possible that for the King Haakon sector, the strong stratification inhibits the negative ice-ocean feedback.

3.4 Discussion

Consistent with the findings from Libera et al. (2022), this study shows that in all sectors the upper ocean memory (residual thermal anomalies associated with sea ice processes) acts as a primary source of memory underlying sea ice predictability. This study finds the regional variability in the vertical structure of the upper ocean and its seasonal evolution drives the regional differences in the sea ice predictability in these regions. Studies by Bushuk et al. (2021) and Ordoñez et al. (2018) using a similar method and regional definitions, also find the occurrence of long persisting predictability in summer and winter. Sea ice predictability results in Ordoñez et al. (2018) show summer persistence, winter persistence, spring re-emergence, and weak correlations in autumn.

The main regional variability in sea ice predictability and underlying ocean predictability found in this study is the east-west contrast in the occurrence of summer persistence. Western sectors retain ice-ocean memory from the previous winter and spring deep (~0-200m) in the ocean into

summer and produce long-sustained summer persistence (~6 months lag). Eastern sectors, on the other hand, lack ice-ocean memory from the previous winter in the WW layer, as they dissipate the memory earlier. While regional variability in Antarctic sea ice predictability has been found in previous studies (Ordoñez et al., 2018; Bushuk et al., 2021), such an east-west contrast has not been reported. However, the work by Stammerjohn et al. (2012), has shown an east-west contrast in memory sustained over summer.

Stammerjohn et al. (2012) described this signal of summer memory as an ocean thermal feedback in autumn in response to changes in ocean temperature from the previous spring. Summer persistence in our study, as well as the signature in Stammerjohn et al. (2012), represents memory in the ocean linked to sea ice that originates in the previous sea ice growth season and persists through the summer into the following sea ice season. This study finds in both cases, that regions with sea ice cover preserve the memory better than regions that become largely ice-free in summer. Therefore, the insulating effect of sea ice cover preserves the memory in the upper ocean by suppressing surface heat flux exchanges with the atmosphere. Further sea ice thickness is also identified as a source of sea ice predictability (Bushuk et al., 2021; Morioka et al., 2021), hence the presence of sea ice itself may provide a source of predictability in the western sectors compared to the eastern sectors. With the recent changes in observed Antarctic sea ice cover reaching a new record low in February 2022, future projections of low summer sea ice covers will change these relationships.

Lecomte et al. (2017) investigated the mechanism behind the observed sea ice expansion in the Ross Sea sector. They identified a local ice-ocean feedback mechanism, where increased sea ice coverage leads to more stratification, trapping heat at the base of the mixed layer (or below the PP) leading to more sea ice growth. In this study, the occurrence of positive ice-ocean correlations (indicative of warmer thermal anomalies associated with sea ice growth) is found in the Ross Sea and Amundsen Sea sectors near the PP (at ~100-200 m). In the Ross Sea sector, these positive (warm) ice-ocean correlations persist through all 12 lag months during the sea ice growth period (May to October initial months; Figure Sx3). Consistent with the concept in Lecomte et al. (2017), the Ross Sea sector produces the strongest vertical temperature gradient at the PP of all sectors. The positive ice-ocean correlations that form in the Ross Sea sector are considered a signature of similar ice-ocean feedback as described by Lecomte et al. (2017), with an extension of this signature also being found in the adjacent Amundsen Sea sector.

In this study, the Bellingshausen Sea sector (Figure 3.1g) stands out as the region with high winter predictability and the lack of predictability suppression in sea ice predictability although it was present in the ocean predictability (Figure 3.4e, and Sx5). Bushuk et al. (2021) also described the Bellingshausen Sea sector as a region of high winter prediction skill. They found in the Bellingshausen Sea sector sea ice extent initial conditions provide a source of memory for short timescales while the key source of memory is still provided by advected ocean heat anomalies. The sea ice predictability that persists while the ocean loses memory from predictability suppression is likely provided by the sea ice pack. The Bellingshausen Sea sector is noted in other predictability studies to stand out for similar situations, Ordoñez et al. (2018) also find a lack of predictability suppression only in the Bellingshausen Sea region amongst all of the regions considered in their study. Marchi et al. (2019) were not able to clearly associate the predictability in the Bellingshausen Sea sector to the local ocean, and they remarked that this region is characterised by an eastward shift of ice predictability originating from the Ross Sea and that the predictability in Bellingshausen may be controlled by horizontally advected ocean temperature anomalies or by the atmosphere. Further, advected sea ice (and associated sea ice anomalies) are found to be important for decadal sea ice predictability in the Bellingshausen Sea sector (Morioka et al., 2022). Hence advected sea ice and ocean anomalies can potentially modify sea ice predictability in regions downstream.

The investigation of regional variability in sea ice predictability reveals that factors such as the depth of the WW, the strength of upper ocean stratification, and the presence or absence of sea ice cover can influence a region's ability to retain memory for sea ice predictability. However, this study does not evaluate the relative contributions of each of these factors to sea ice predictability. An investigation that includes model experiments to assess how sea ice predictability (using ice-ocean correlations) responds to prescribed mixed layer depth, stratification, and sea ice cover will complement the findings of this study. While model studies have been crucial for developing our understanding of the physical mechanisms responsible for Antarctic sea ice predictability, the findings in each of these studies are determined by the representation of the upper ocean and sea ice processes in these models. It is encouraging to see that key findings from most of the model studies overlap and suggest that there is potential for some of the mechanisms described in these studies to be identified in observations. It is possible to evaluate sea ice predictability from observations spanning over four decades. While continuous and long-term upper ocean observations with good spatial coverage in the Southern Ocean are limited, especially under sea ice in winter, it may still be possible to compute

climatological upper ocean conditions in some regions. Applying the insights from model studies to the observed sea ice predictability and climatological upper ocean conditions can help evaluate the predictability of the ice-ocean system in the real world and will be a follow-up to this study.

3.5 Conclusion

The high regional and seasonal variability observed in Antarctic sea ice is closely linked to local ice-ocean interactions and associated feedbacks. These local processes are also crucial for understanding regional sea ice predictability (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021). Building on the work by Libera et al. (2022), this study investigates the drivers of regional variability in Antarctic sea ice predictability with a focus on local sea ice and ocean processes. The results of this study present further evidence of the role of the ocean as the source of sea ice predictability. Consistent with findings from Libera et al. (2022), this study also finds that thermal anomalies associated with sea ice processes linger in the upper ocean largely within the Winter Water layer (WW) and give rise to sea ice memory in all regions (except for spring to summer memory in the Bellingshausen Sea sector). The findings of this study also show that sea ice predictability is strongly tied to the local vertical ocean structure and its seasonal evolution in all sectors. The variability in the local vertical ocean structure and ice-ocean processes drives the regional variability of sea ice predictability. The main regional variability found in this study is an east-west contrast in summer persistence. Eastern sectors lose memory stored in the WW earlier in the year due to higher ocean surface heat flux exchange and modification of the upper ocean from weaker seasonal stratification and lack of summer sea ice cover. The predictability barrier as described in Libera et al. (2022) is found in half of the sectors. Sectors that lack a clear and consistent signature of the predictability barrier are located adjacent to large sea ice-producing regions, indicating that advected ice and ocean anomalies may be modifying ice-ocean feedbacks in the regions downstream.

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Open Research

All model outputs including the simulations presented in this manuscript are stored as part of the COSIMA data collection here: <https://doi.org/10.4225/41/5a2dc8543105a>. Passive microwave sea ice data are publicly available, hosted by the National Snow and Ice Data Center (<https://doi.org/10.7265/N59P2ZTG>).

CHAPTER 4

Chapter 4: Stratification dominates accurate simulation of sea ice predictability

**This chapter is written as a manuscript in preparation*

Abstract

This study evaluates regional Antarctic sea ice predictability from observations and a global coupled ocean-sea ice model. Modelled sea ice predictability has considerable biases. In the observations, winter persistence is the prominent predictability regime, while summer persistence and spring re-emergence are prevalent in the model. Overall, sea ice predictability from the model is more persistent than those from observations and the bias in model sea ice predictability is traced back to the systematic bias in the representation of upper ocean vertical structure, specifically the stratification in the model.

An overly stratified upper ocean in the model preserves upper ocean memory over longer timescales and produce longer predictability skill. This overly stratified upper ocean is a consequence of the model ocean being too fresh above the permanent pycnocline, and too warm below. Our study shows that a better representation of the upper ocean in the sea ice zone of the Southern Ocean can produce sea ice predictability of better agreement with the observations. The findings of this study also suggest that in the warming climate, increased stratification can enhance the ocean's capacity to retain memory, which can potentially increase sea ice predictability in the future.

4.1 Introduction.

Antarctic sea ice plays an important role in the regional and global climate through its influence on local and global oceanic and atmospheric circulations, planetary radiative balance, and the crucial support it provides to Southern Ocean ecosystems and biogeochemical cycles. Following the record-high Antarctic sea ice extent (SIE) in 2014, extreme annual fluctuations

have occurred with a sharp decline in 2016, a record low in 2017, and a new record low in 2022 (Fogt et al., 2022; Raphael and Handcock, 2022). It is of high relevance to understand the processes influencing Antarctic sea ice in this period of pronounced fluctuations. This study uses sea ice predictability analysis as a tool to gain insights into the physical drivers in the ice-ocean system that produce memory at seasonal-to-interannual timescales. This predictability analysis can help in understanding the persisting signature of low or high sea ice anomalies over multiple seasons.

Antarctic sea ice predictability studies at seasonal-to-interannual timescales have found predictability for two to three years (Turner et al., 2013; Hobbs et al., 2016; Meredith et al., 2019; Roach et al., 2020). Two main modes of predictability have been identified, namely persistence and re-emergence. Persistence is the continuous retention of memory from the initialization. In the case of correlations, this is shown as sustained high values of correlations starting from the initial state. Re-emergence, on the other hand, occurs when a persistent signature is lost temporarily, and predictability (correlations) recovers or re-establishes at a later period. Sea ice persistence is generally lost in summer and re-emerges in the following autumn. The primary memory source for this predictability is the upper ocean, with the seasonal evolution of the upper ocean being closely linked to the re-emergence of predictability (Holland et al., 2013; Zunz et al., 2015; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021; Libera et al., 2022).

Surface heat anomalies linked to sea ice give rise to ocean memory. During winter, these heat anomalies extend deep into the ocean as the mixed layer (or seasonal pycnocline) deepens. Sea ice melt and the radiative warming of the surface ocean layers in summer leads to the formation of seasonal stratification that isolates the ocean heat anomalies at the surface, causing the loss of predictability in summer, termed as predictability suppression (Libera et al., 2022). When the mixed layer deepens in the following sea ice advance season, the ocean heat anomalies resurface and produce re-emergence of sea ice memory (Holland et al., 2013; Ordoñez et al., 2018; Marchi et al., 2019; Bushuk et al., 2021; Libera et al., 2022). Marchi et al. (2019) found that the effectiveness of re-emergence depended upon the depth of the mixed layer, and that regions with deep mixed layer could sustain sea ice memory over multiple years re-emerging from one winter to the next. Libera et al. (2022) (presented here as Chapter 2) found that sea ice predictability is strongly tied to the vertical ocean structure and its seasonal evolution. They showed that the memory in the ocean is largely confined to the winter water layer (WW: upper

ocean layer at thermal equilibrium with sea ice, containing sea ice memory), and that the stratification above and below the WW help preserve the memory in the upper ocean.

Sea ice predictability does not only depend on the depth to which the ocean memory is stored but also on how well this memory is preserved. Chapter 2 (Libera et al. (2022)) found a sharp loss of predictability when the seasonal pycnocline (SP) merges with the permanent pycnocline (PP) in winter, termed as predictability barrier. This predictability barrier is the consequence of warm Circumpolar Deep Water (CDW), without sea-ice related memory, entraining into the WW of the ocean. The follow-up analysis in Chapter 3 showed that regional variability of Antarctic sea ice predictability is driven by the differences in local upper ocean structure and sea ice processes. They found that regions with stronger stratification as well as summer sea ice cover preserved the memory deep into the ocean and produced predictability (emerging from summer) that persisted over a longer period. Multi-model assessment of Antarctic sea ice predictability has shown that the presence of deep mixed layers enhances sea ice predictability (Marchi et al., 2019). However, an evaluation on the influence of stratification strength in determining sea ice predictability is yet to be done.

In this Chapter, regional Antarctic sea ice predictability from observations is compared with that from a global coupled ocean-sea ice model and the results reveal that the model overestimates sea ice predictability. Then, the hypothesis is tested to determine if stratification biases in the model impact sea ice predictability, resulting in longer retention of predictability in over-stratified models compared to observations. Although sea ice predictability from the model and the observations are evaluated in this Chapter, it is not intended as model evaluation. Instead, in this Chapter the insights from model-based Antarctic sea ice predictability studies (especially those from Chapters 2 and 3) are applied to the observed data.

4.2 Methods

The methodology and terminology used in this study follow from Chapters 2 and 3 (Libera et al. (2022)). Outputs from the global coupled ocean-sea ice model, the Australian Community Climate and Earth System Simulator (ACCESS-OM2-01), run with a grid spacing of 0.1° are used. ACCESS-OM2-01 is based on the ocean MOM5.1 and ice CICE5.1 models coupled with the OASIS-MCT coupler. The model experiment analysed in this study was forced using JRA55-do v1.4.0 (Tsujino et al., 2018). The ACCESS-OM2-01 experiment ran for 60 years

from 1 Jan 1958 to 31 Dec 2018. It was started from a 40-year spin-up under repeated 1 May 1984 – 30 April 1985 JRA55-do forcing (Kiss et al., 2020). The observed sea ice concentration (SIC) is derived from satellite passive microwave data, a product based on the NASA Goddard-merged parameter in the NOAA/NSIDC Climate Data Record (Meier et al., 2013). Observed ocean profiles are sourced from the World Ocean Database. Thirty years of data from 1985 to 2015 is used for both the observations and model.

Sea ice predictability is investigated using correlation analysis in six sectors around Antarctica (Fig. 1). The regional boundaries for the six sectors used for regional sea ice predictability analysis are as follows, King Haakon Sea (15°W-70°E), East Antarctica (70-150°E), Ross Sea (150°E-135°W), Amundsen Sea (135°E-90°W), Bellingshausen Sea (90-60°W), and Weddell Sea (60-15°W). The sea ice area (SIA) is calculated by multiplying the sea ice concentration (SIC) by the grid cell area and then summing it within each sector to calculate the total SIA. Calculating the total SIA results in the loss of spatial information (i.e., grid cell by grid cell changes), but enables understanding of how sea ice behaves at a regional scale. Sea ice predictability (due to the persistence of sea ice anomalies) is evaluated by correlating the detrended SIA in an initial month to the detrended SIA in the eleven subsequent (or lagging) months. In most sectors statistically significant correlations do not extend beyond the initial eleven lag months, hence the predictability results presented in this study are limited to eleven lagged months. Statistically significant values are defined as p-values greater than 95% in the two-tailed Student's t-test.

Monthly climatologies of the upper 200 m ocean conditions are calculated using observed and model data. A binned dataset with 2° horizontal resolution is created using the available observed ocean profiles. Observed ocean profiles are limited in the high-latitude Southern Ocean and particularly scarce under the sea ice in winter (Figure 4.1b). Due to the selected period of study (1985 to 2015), some of the recent observed data could not be used in the analysis. The period post-2015 is excluded due to the large fluctuations in Antarctic sea ice cover that have occurred since 2015. This study aims to understand the physical processes in the ice-ocean system at the mean or climatological state, including recent data (2015 to present) would introduce large anomalies and hinder our ability to understand the processes occurring in the mean state. The available ocean profiles in the high-latitude Southern Ocean are also non-uniform in their temporal and spatial distribution (i.e., more data in some years or decades than others and observations concentrated in some regions, etc). The availability of observed ocean profiles for individual decades is shown in Figure Sz2. Observed ocean profiles contain

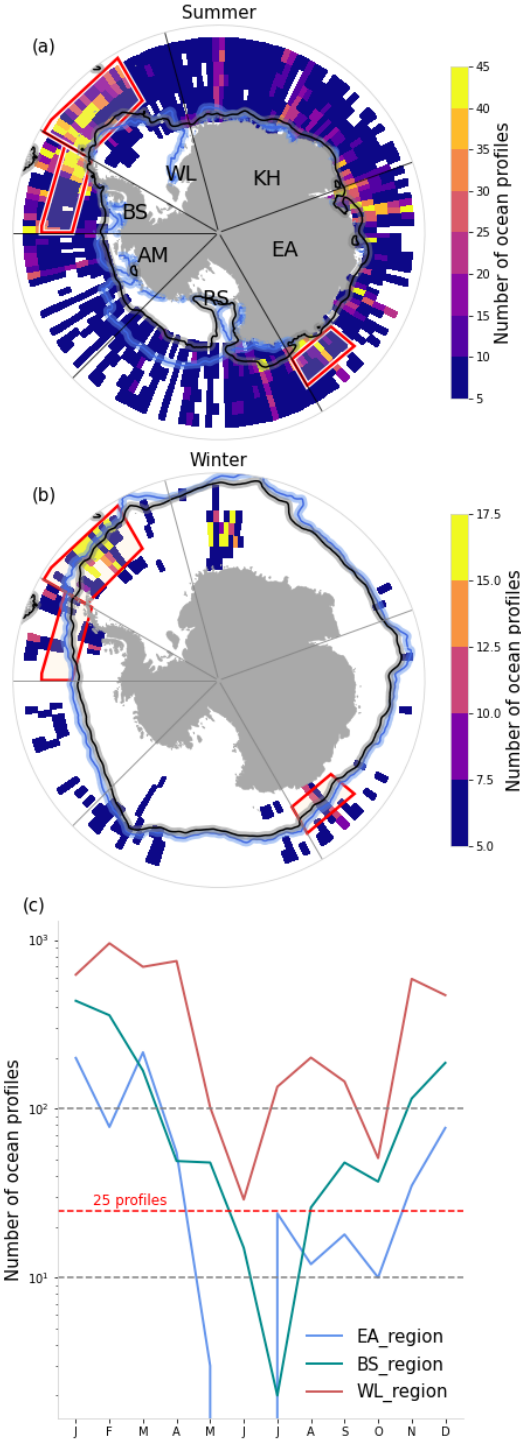


Figure 4.1: Map of Antarctica showing the six regions used in the study King Haakon Sea (KH), East Antarctica (EA), Ross Sea (RS), Amundsen Sea (AS), Bellingshausen Sea (BS) and Weddell Sea (WS). (a) the number of available observed ocean profiles (for the period of study 1985-2015) in summer (January-March) and (b) in winter (July-September). The selected regions used for calculating upper ocean climatologies are marked using red polygons. (c) Total number of profiles per month for the three selected regions (for 1985-2015). The months where the climatological ocean observations have less than 25 profiles in (c) are hatched to indicate less confidence in Figures 4.3, 4.4, and 4.5

ocean observations from various platforms, most of the profiles are obtained from Argo floats, seal data (Autonomous Pinniped Bathythermographs), and CTD profiles (Figure Sz1).

Considering the spatial distribution of ocean profiles in summer and winter from this binned data (Figure 4.1 a, b) three regions are selected with an adequate number of profiles for this study. These regions lie within the East Antarctic, Bellingshausen Sea, and Weddell Sea sectors according to our sector definitions. In the selected regions, coastal/shelf areas are excluded as they are destratified and dominated by Dense Shelf Water formation and other coastal processes that can skew the ocean climatology. Of these regions, only the Weddell Sea sector contains a substantial amount of ocean profiles in summer and winter. In the two remaining regions, the ocean profiles in winter are very scarce and lack good spatial coverage. Considering the lack of ocean profiles in winter, an arbitrary quality check is placed: if the monthly climatology is made up of less than 25 ocean profiles, the data is marked to be of low confidence. The corresponding (geographical) region in the model is selected to calculate the spatially averaged monthly climatology for comparison.

4.3 Results

4.3.1 Regional sea ice predictability from modelled and observed sea ice area.

Sea ice predictability is calculated in a forward looking perspective, showing how memory emerging from an initial month (on the y-axis) is retained into the future months/ lags (along the x-axis). There are considerable differences in regional sea ice predictability between the model and observations (Figure 4.2). Winter persistence (i.e., significant correlations emerging from winter persisting for ~ 5 month lag) is the most common predictability regime in the observation. All sectors (Figure 4.2 a-2, b-2, c-2, d-2, and e-2) (except Weddell Sea sector; Figure 4.2 f-2) produce winter persistence in the observation. Summer persistence (significant correlations emerging from summer persisting for ~ 6 lags) and spring re-emergence (significant correlations emerging from spring after persisting in the initial few lags are lost in summer lags, significant correlation re-emerge in subsequent autumn and winter lags) are more prevalent in the model.

Overall, in most cases, the model shows higher sea ice predictability than the observations, with well-established predictability regimes that often persist longer. There are only a few instances when the observed sea ice predictability is similar or higher than in the model. The

differences in winter persistence between the model and observation can be taken as an example, in the model, the winter persistence is in most cases followed by spring reemergence. Indicating that the memory in the ice-ocean system persists in the winter and spring months. Whereas in the observation, in most cases the winter persistence tapers off by spring, and the spring reemergence (which is a prominent regime, with memory persisting for ~6 to 10 months lag) is mostly absent. In the observations, winter persistence terminates in spring (October or November) in most sectors and spring re-emergence is not found in most sectors except in the Amundsen and Weddell Sea sectors. In the model, in most sectors winter persistence is followed by spring re-emergence, indicating that the upper ocean in the model is hosting memory over a longer duration. The dissipation of memory after winter persistence (tapering of winter persistence by spring initial months, Figure 4.2 a-2, b-2, c-2, and e-2) without the development of spring re-emergence in the observation could be indicative of either the memory forming in the upper ocean is weaker and does not persist long enough, or that the memory is being lost to the ocean interior or the atmosphere.

The East Antarctic sector produces winter persistence and a summer-to-spring re-emergence (correlation between the initial summer and early autumn months (January to May) with the future spring months (October and November); which is not seen in any other sector) in the observations (Figure 4.2 b-2). The model produces winter persistence and spring re-emergence and has almost no memory in summer (Figure 4.2b-1). The mechanism for a summer-to-spring re-emergence (memory sustained over 11 months lag; Figure 4.2b-2) has not been explored in the Antarctic sea ice predictability literature to the best of the author's knowledge. The closest to such a regime (summer-to-spring re-emergence) in the model is found in the Ross Sea sector. Investigating the predictability signature in the ocean corresponding to the summer-to-spring re-emergence in the Ross Sea sector (Sx3), no continuous memory in the ocean between the summer persistence and the signature of re-emergence in spring was found. This is in contrast to the established mechanism of spring re-emergence and indicates that there are other pathways (beyond the ocean) through which predictability at near-multiyear timescale can occur. Investigating the mechanism for the summer-to-spring re-emergence of sea ice predictability is however beyond the scope of this paper.

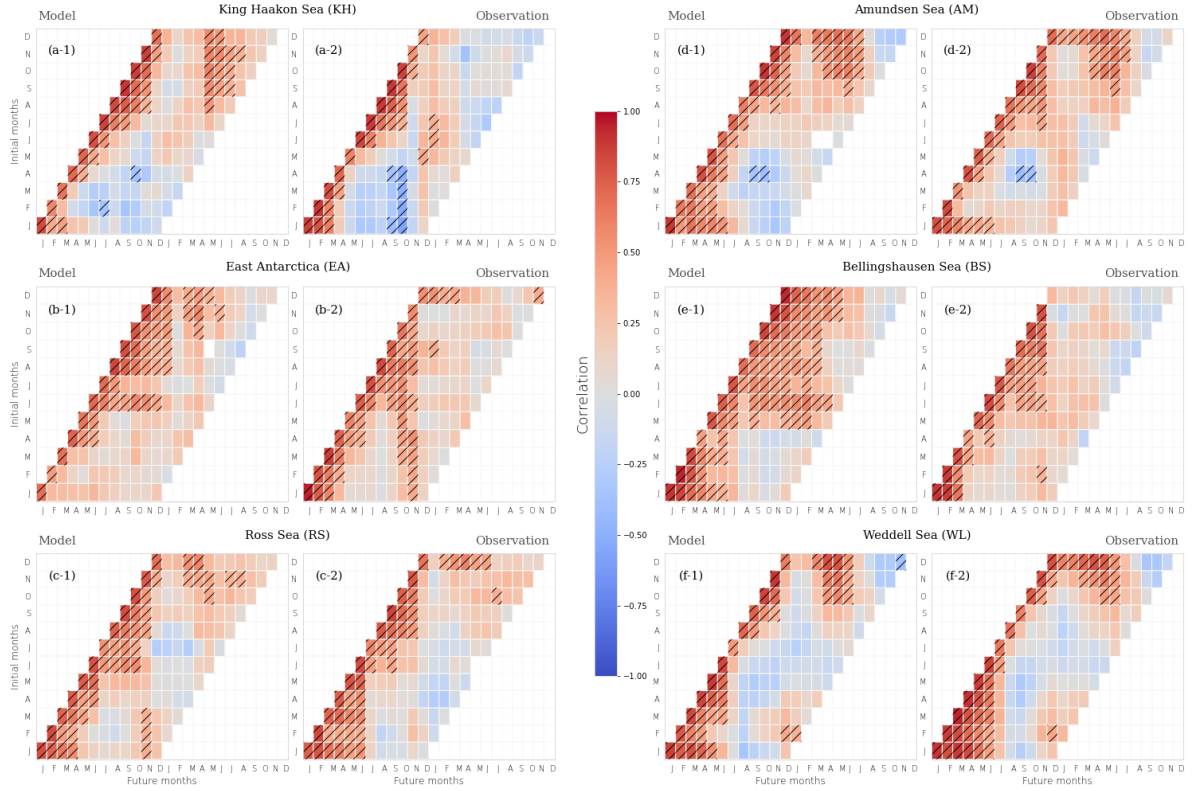


Figure 4.2: Regional Sea ice predictability (autocorrelation of sea ice area) for the six sectors around Antarctica. Model sea ice predictability (LHS) and observed sea ice predictability (RHS) are presented side-by-side. In each panel, the sea ice area from the initial month (along the y-axis) is correlated against the future months or lags (along the x-axis) and statistically significant values (>95%) are hatched. The model sea ice predictability for the six sectors (a-1, b-1, c-1, d-1, e-1, and f-1) are the same as shown in Figure 3.1.

Assessing the predictability of the upper ocean can show whether the ocean memory in the observation is weaker, how it is evolving, or if it is being dissipated. However, continuous observations (over a few decades) with high spatial and temporal resolution are required for assessing ocean predictability. Upper ocean observations are limited in the Southern Ocean, especially in winter. Therefore, it is not possible to evaluate upper ocean predictability from observations. However, existing ocean observations can be used to evaluate the representation of climatological vertical ocean structure and its seasonal evolution that model studies identify to have a strong influence on the persistence of sea ice predictability. Yet the comparison of the upper ocean conditions in the model and observations has caveats due to the non-uniformity (uneven spatial and temporal distribution) of the observed ocean data.

4.3.2 Modelled upper ocean below the ice: Consequences to sea ice predictability.

The disparity in upper ocean conditions between the model and observation and its impact on emerging sea ice predictability is assessed in this section. The climatological temperature and salinity in the model and observations, and the difference (model-observation) in vertical temperature gradients and vertical density gradients for the three regions are presented in Figures 4.3, 4.4, and 4.5. The vertical temperature and density gradient maximum are used as proxy for the permanent pycnocline (PP) and seasonal pycnocline (SP) respectively and they are overlaid on the climatology plots. The PP acts as the base for the Winter Water layer (where sea ice-related ocean memory is largely stored) and it separates the Circumpolar Deep Water (CDW, water from ocean interior with little seasonal variation) from surface waters that are strongly modified by sea ice processes. The SP mostly acts as the base of the mixed layer; it is shallow in summer when the seasonal stratification forms near the surface and deepens in winter as sea ice growth induces vertical mixing. In many cases, the SP deepens and merges with the PP in winter. The seasonal and permanent pycnocline help to follow the seasonal modification of the upper ocean.

Comparing the upper ocean climatology between the observation and model, the model is stratified more than observations in all three regions (panels e and f in Figures 4.3, 4.4, and 4.5). The strong distinction between the fresher and cooler surface (above the PP) and warmer and saltier deep layers (below the PP) makes the stratification at the permanent pycnocline stronger in the model compared to the observed climatologies (panel e for Figure 4.3, 4.4, and 4.5). The upper ocean is much fresher in the model (panels c and d in Figures 4.3, 4.4, and 4.5), and freshwater in the Bellingshausen Sea region extends deeper (>200 m) than other regions (in the case of model and observations). The summer (thermal) stratification (i.e., surface warming) is stronger in the model for all three regions. It is beyond the scope of this study to investigate why the model is more stratified than observed ocean conditions, however, the potential causes for the biases in the representation of the upper ocean in the model are discussed in section 4.4.

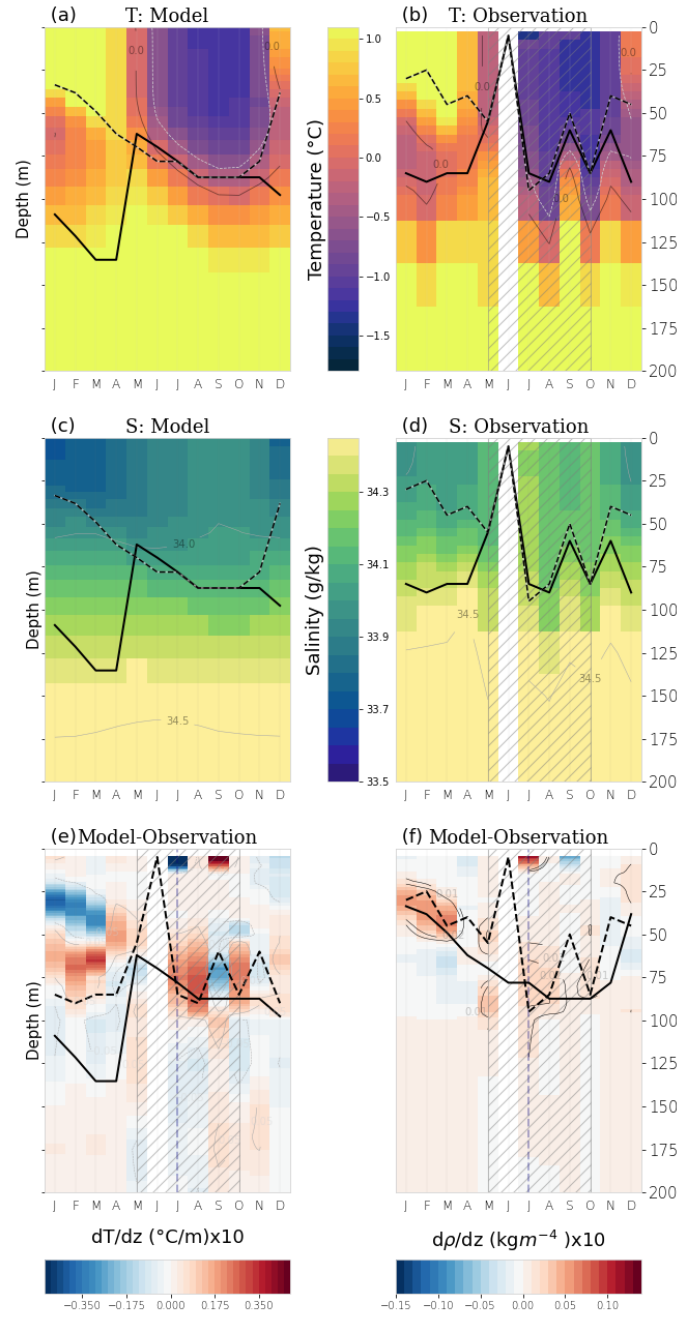


Figure 4.3: Ocean climatology of the upper ocean in a region within the East Antarctic sector for (a) temperature from the model and (b) observation. (c) Salinity from the model and (d) observation, (e) the difference in vertical temperature gradient between the model and observation and (f) the difference in vertical density gradient between the model and observation. In panels (a-e), the solid black line is the climatological vertical temperature gradient maximum and the dashed black line is the climatological vertical density gradient maximum. In panels (e) solid and dashed black lines are the climatological vertical temperature gradient maximum in the model and observation respectively. In panel (f) solid and dashed black lines are climatological vertical density gradient maximum in the model and observation respectively.

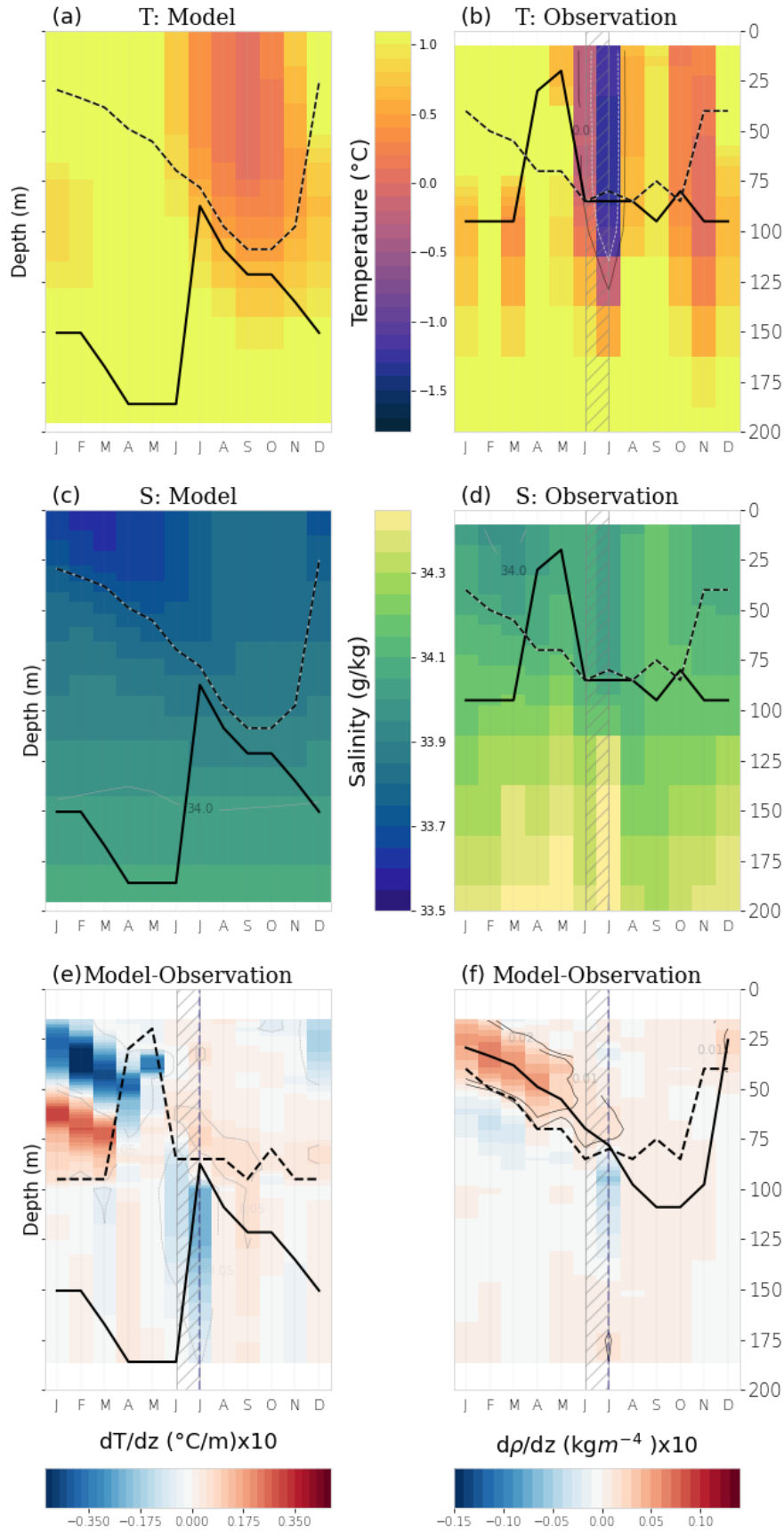


Figure 4.4: Same as Figure 4.3 but for the region in the Bellingshausen Sea sector.

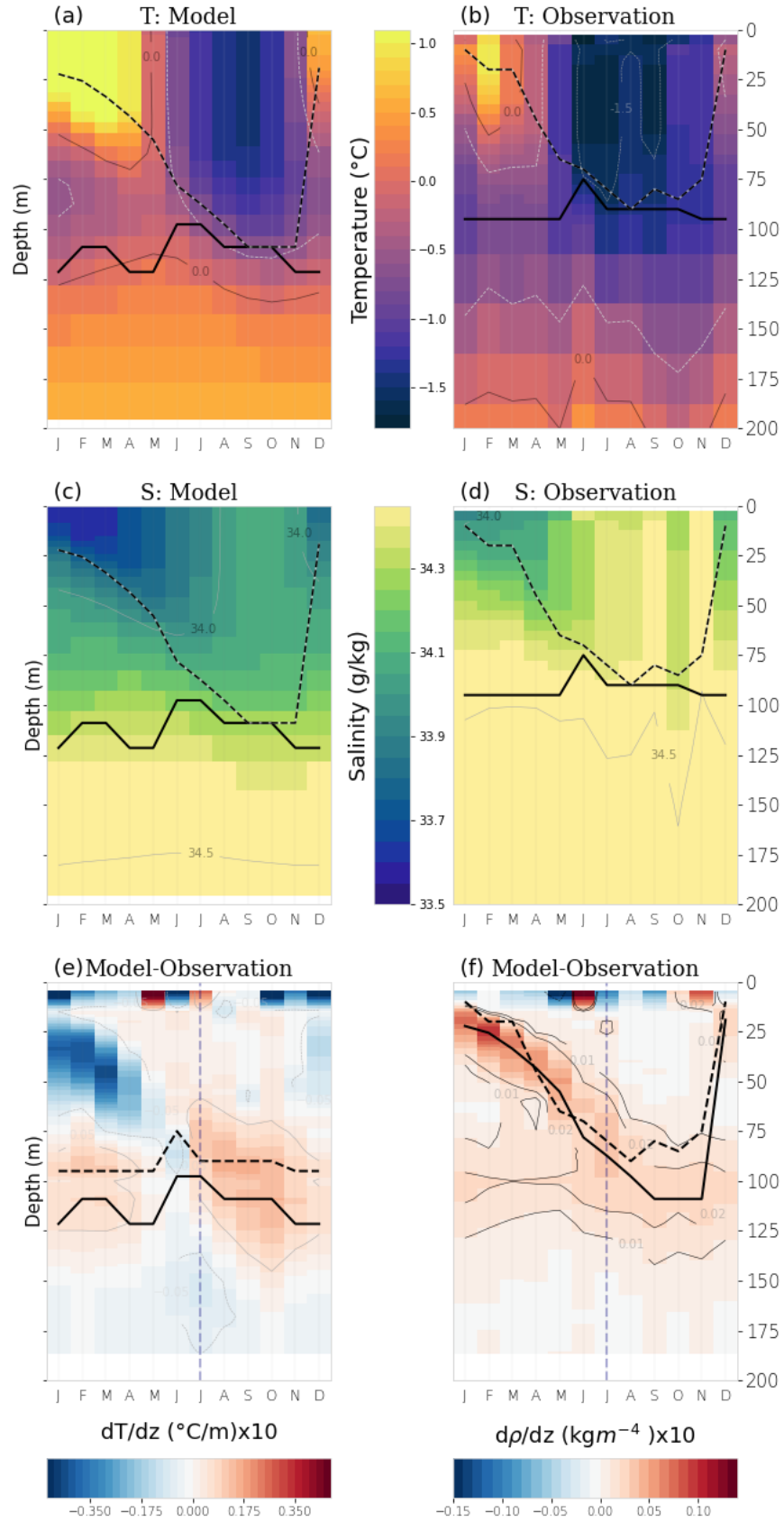


Figure 4.5: Same as Figure 4.3 but for the region in the Weddell Sea sector

Based on findings from Libera et al. (2022) and Chapter 3, the memory in the upper ocean is largely confined within the Winter Water layer. Highly stratified seasonal pycnocline and permanent pycnocline at the top and base of the Winter Water layer help preserve the upper ocean memory longer in the model. In the observations, on the other hand, where stratification is weaker, the Winter Water is more prone to modifications by the ocean interior or by the atmosphere. Hence, the ocean memory underlying the winter persistence in the observations dissipates in the spring before it can re-emerge in the subsequent autumn and winter.

Among the three regions considered, the Bellingshausen Sea sector shows a fresher upper ocean, and it lacks a distinctive transition between the surface and subsurface ocean (i.e., has weaker vertical gradients) in the model (Figure 4.4 a, c, e). The fresh near surface and weaker stratification may be due to the excessive melt of sea ice exported from the Ross Sea sector into the Bellingshausen Sea sector but could also be due to a freshwater bias in the model (discussed in section 4.4.1). In the absence of sea ice volume observations, it is impossible to ascertain whether the fresher upper ocean is a model bias or an accurate representation of sea ice transport. However, the upper ocean anomalies (mainly deeper WW) may cause the model's long persistence of sea ice predictability in the Bellingshausen Sea sector. Marchi et al. (2019) found deeper mixed layer or deep convection produces predictability more than what is achievable from the observed conditions. Similar to the findings in Marchi et al. (2019), the Bellingshausen Sea sector in this study also produces an anomalously long winter persistence (and spring persistence; Figure 4.2 e-1, e-2) in the model, while the observed winter (and spring) sea ice persistence is much shorter. Freshwater extending deeper into the ocean also implies that the ice-ocean anomalies extend deep into the ocean and are potentially stored longer, which would contribute to the anomalously long persistence of sea ice predictability in the Bellingshausen Sea sector in the model.

Linking the concepts from Marchi et al. (2019), Libera et al. (2022) and Chapter 3, it further shows how the vertical ocean structure plays a driving role in determining the sea ice predictability that can emerge from a region. (1) If the region has a considerable distinction between surface and deeper layers at the permanent pycnocline but relatively weak stratification then the memory forming in the Winter Water layer in that region will be more prone to being modified by entrainment from the deeper layer (as small volume of entrainment can make a large difference to the heat anomalies in the Winter Water layer). But (2) if the stratification is stronger (for case 1), such that memory in the Winter Water layer is preserved against modifications, it can lead to sustained predictability. And (3) if the distinction (between

surface and subsurface layer) is almost non-existent, either due to deep convection or due to the presence of large bias in mixed layer depth, it can also lead to long-sustained predictability. In this case (3), sea ice-related anomalies can extend a lot deeper into the ocean and will require extensive rates of heat flux exchange at the surface to modify the memory in such a large volume of the upper ocean. Also, the near-accessible heat reserve (warmer Deep ocean) that would otherwise modify the heat anomalies in the Winter Water layer (predictability barrier) would be at a much deeper level that seasonal sea ice processes cannot drive entrainment.

Sea ice predictability in the Weddell Sea sector has arguably the closest model-observation agreement (Figure 4.2 f-1, f-2). Summer persistence and spring re-emergence are produced in both the model and the observations. Although the model ocean climatology is fresher in the surface layer and warmer in the subsurface layer, which leads to a slight increase in the stratification at the permanent and seasonal pycnoclines, the seasonal evolution represented in the model and observation have better agreement than in the other two ocean regions considered (Figure 4.5). The seasonal stratification is stronger in the model, especially in summer (vertical density gradient, Figure 4.5f), potentially from the higher amount of freshwater in the surface compared to observations. The stronger seasonal stratification in the model produces well defined and more pronounced predictability suppression in the model compared to observation, as predictability suppression is linked to the seasonal stratification. Even with strong stratification and the deep mixed layer, both the model and observation lose predictability in mid-winter for the Weddell Sea (for both the summer persistence and spring re-emergence regimes). This is due to the winter predictability barrier, which Libera et al. (2022) show is due to the entrainment of warm Circumpolar Deep Water (with no sea ice-related memory) into the Winter Water layer (containing sea ice memory), resulting in the modification of the memory in the Winter Water layer and dissipating predictability in the ice-ocean system. The observed ocean climatology and sea ice predictability indicate that the predictability barrier occurs in the Weddell Sea region in the real world.

Overall, in the three ocean regions assessed, the differences in the sea ice predictability between the model and the observation could be associated with the differences in the representation of the upper ocean conditions between the model and observation. (1) In the Weddell Sea region: the ocean climatology and sea ice predictability have agreement between model and observation. (2) The Bellingshausen Sea region: the model produces anomalously high predictability and the model ocean climatology had considerably deep permanent pycnocline (i.e., deep winter mixed layer) lacking strong gradients between the surface and subsurface

ocean layer (above and below the permanent pycnocline). Bellingshausen Sea region in the model, therefore, serves as an example of how excessively deep mixed layers in the model can generate unrealistic levels of predictability. (3) In the East Antarctic region: the model produced sea ice predictability more persistent in winter and spring and the model ocean is more stratified than observation. (1) (2) and (3) show that stronger stratification in the model above and below the Winter Water layer preserves the memory longer producing predictability that persists longer.

4.4 Discussion

4.4.1 Over-stratified subpolar Southern Ocean in the model

The Southern Ocean has large climatical significance as it acts as a major gateway carrying anthropogenic carbon and heat into the ocean interior (Sallée et al., 2012; Frölicher et al., 2015). It is therefore of relevance that climate models represent well the complex dynamics, physical conditions, and processes of the Southern Ocean for reducing the uncertainty of the climate projections. Studies have shown that most climate models have disparities in their representation of the Southern Ocean properties, including the depth of the mixed layer and the strength of stratification at the base of the mixed layer (Sallée et al., 2013; Meijers, 2014; Downes et al., 2015). Many climate models represent shallower mixed layer depths than in observation due to excessive overlying freshwater input (Sallée et al., 2013). The positive haline (fresh and stable) components also act to intensify the stratification strength at the base of the mixed layer (Meijers, 2014). This study identified differences between the model ocean and the observation, such as considerably deeper PP in the Bellingshausen Sea sector and warmer CDW in the Weddell Sea sector. However, a consistent bias found in all three regions is the freshwater bias in the upper ocean (~0-100 m), which is linked to the increased stratification at the base of the mixed layer. In this section, the potential causes that drive the freshwater bias in the model are discussed.

The model (ACCESS-OM2-01) uses salinity restoration to fix large uncertainties in the surface freshwater flux. Salinity restoration is applied to get the sea surface salinity in the model closer to the observation which is referenced to the interpolated 0.25° World Ocean Atlas 2013 v2 (WOA13) monthly climatology (Kiss et al., 2020). However, due to the sparsity of observations in the Southern Ocean, especially under winter sea ice cover, the WOA13 reference itself can introduce a bias relative to the real world through salinity restoration. Additional freshwater

from excessive sea ice melt can also contribute to the freshwater bias in the model. Note that the observations used in this study are made from binned ocean profiles from the World Ocean Database and it is not the same as WOA13. The WOA13 interpolates data to 1° horizontal grids, within the data-sparse sea ice zone, which may produce an inaccurate representation of the ocean in high latitude Southern Ocean. Applying salinity restoration to the inaccurate representation of surface salinity can amplify freshwater bias in the model.

The model produces winter sea ice area (SIA) with a strong agreement to the observation but has lower summer SIA than observation, indicating excessive sea ice melt (which may develop from a warm ocean bias) that adds freshwater at the surface. There is also potential for a higher sea ice volume bias in the model that can produce more freshwater in the upper ocean. Besides these potential causes of how more freshwater is introduced at the surface; the reasons why not enough salt from deeper ocean enters the upper ocean also needs to be considered. Such as reduced upwelling rates of Circumpolar Deep Water (i.e., biases in the rate of overturning circulation) and systematic errors in the convective parameterization, sub-grid-scale turbulence, enhanced or dampened rates of vertical mixing, friction schemes, etc. There are many possible reasons for the freshwater bias in the model and even if many targeted simulations are run, it will be hard to disentangle the factor(s) responsible as there are so many interconnected processes.

This study finds that stratification in the upper ocean plays an important role in maintaining ice-ocean memory and generating sea ice predictability. The depth to which ice-ocean memory is stored also plays an important role as described in previous studies (Holland et al., 2013; Marchi et al., 2019). However, the correlation analysis cannot assess the relative contribution of these two factors. Process-based model experiments to evaluate how sea ice predictability responds to varying depths of WW and changes in the strength of stratification would be complementary to the findings in this study.

4.5 Conclusion

This study evaluated regional sea ice predictability from observed and modelled sea ice data in six sectors around Antarctica. Sea ice predictability has a strong regional variability in both the observations and the model, and the analysis revealed key differences between the observations and the model. Winter persistence (sea ice predictability or memory emerging in winter and persisting until spring) is the most prevalent predictability regime in the observations. The two

predictability regimes that prevail in the model are however summer persistence (memory emerging from summer persisting until mid-winter) and spring re-emergence (memory from spring persisting until the following autumn or winter, with a temporary loss in summer). In most sectors, the model produces higher sea ice predictability than the observations do. Winter persistence in the observation is lost early, whereas in the model after losing the memory in spring and summer lag months (due to predictability suppression) it re-emerges in subsequent autumn and persists till mid-winter in the model. Further, the model produces two or more predictability regimes spanning over multiple seasons in most sectors, which is not seen in the observations.

In parallel to comparing the sea ice predictability in the observations and model, this study also investigated what differences in the climatological upper ocean structure lead to the disparity in sea ice predictability between the observations and the model. The model has a fresher upper ocean (above the permanent pycnocline), a warmer subsurface ocean (below the permanent pycnocline) and stronger stratification both above and below the Winter Water layer. Previous studies have shown that sea ice-related memory in the ocean is largely confined within the Winter Water layer ($\sim$ upper 150 m). The stronger stratification above and below the Winter Water layer in the model is therefore preserving the ocean memory over longer timescales producing higher sea ice predictability in the model. In summary, the model overestimates sea ice predictability, beyond what is achievable from the observation. This study finds that the formation of deeper mixed layers and stronger stratification in the model leads to an overestimation of sea ice predictability.

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Open Research

All model outputs including the simulations presented in this manuscript is stored as part of the COSIMA data collection (<https://doi.org/10.4225/41/5a2dc8543105a>). Passive microwave sea ice data are publicly available, hosted by the National Snow and Ice Data Center (<https://doi.org/10.7265/N59P2ZTG>). Ocean observations are available at the world ocean database (<https://www.ncei.noaa.gov/products/world-ocean-database>)

CHAPTER 5

Chapter 5: Discussion and Conclusions

Summary

Antarctic sea ice affects global climate dynamics through its interplay with the planetary albedo, ocean, atmosphere, adjacent ice shelves, and biogeochemical cycles in the Southern Ocean. Observed Antarctic sea ice concentration exhibits large regional, seasonal to decadal variability. Many processes influencing Antarctic sea ice are lacking sufficient understanding, which results in large uncertainties in future projections of Antarctic sea ice. Biases in the representation of sea ice in models reduce confidence in a range of associated fields, ultimately reducing confidence in the future projections of the overall Antarctic climate system, from ocean circulation to the melting of the Antarctic ice sheet. Understanding the complex interactions between Antarctic sea ice and the components of the climate system, and how these interactions relate to broader climate change issues is an important scientific challenge. This thesis contributes to this scientific challenge by investigating the complex interactions between Antarctic sea ice and the underlying ocean.

A new approach assessing sea ice predictability based on local ice-ocean interactions was developed and applied in Chapter 2 using a global coupled ocean sea ice model. To evaluate ice-ocean interactions in a region, continuous sea ice and upper ocean data over a few decades with good spatial coverage are required. However continuous observations are very limited in the Southern Ocean, especially under the seasonal sea ice cover. Therefore, outputs from a global coupled ocean sea ice model with high-resolution monthly upper ocean and sea ice fields are used.

Previous studies attributed sea ice predictability to the inherent ocean memory, i.e., the large thermal inertia of the ocean (Holland et al., 2013; Marchi et al., 2019). However, inherent ocean memory can be affected by many factors. In Chapter 2, (1) Sea ice predictability (based on persistence of sea ice anomalies from sea ice area autocorrelation), (2) upper ocean predictability (autocorrelation of upper ocean temperature), and (3) ocean memory linked to sea ice processes (ice-ocean memory; sea ice area correlated with lagged upper ocean temperature) were evaluated in the Weddell Sea region of the Southern Ocean. This analysis

separated the memory in the ocean linked to sea ice processes (ice-ocean memory) from the inherent upper ocean memory in the background.

The analysis in Chapter 2 showed that the ice-ocean memory is largely confined within the winter water layer (WW: cold water formed during sea ice production and its summer remnant, approximately in 0–100 m), it is strongly dependant on the vertical structure of the upper ocean (temperature, salinity and therefore stratification) and it responds closely to the seasonal evolution of the upper ocean. The inherent ocean memory, on the other hand, extends through the entire water column and is more persistent, especially below the permanent pycnocline. The ice-ocean memory in the Weddell Sea region produces sea ice predictability re-emergence consistent with the mechanism described in previous studies, where sea ice predictability emerging from spring months is lost in summer and they re-emerge in the following sea ice advance period (autumn) (Holland et al., 2013; Marchi et al., 2019).

A new mechanism that applies a limit on sea ice predictability linked to local ice-ocean feedback was identified in Chapter 2. Besides the temporary loss of sea predictability in summer (predictability suppression) associated with the re-emergence signal, a more permanent loss of sea ice predictability (a predictability barrier) was identified, occurring in mid-winter, when the seasonal pycnocline merges with the permanent pycnocline. The predictability barrier was linked to the entrainment of warm Circumpolar Deep Water (CDW) into the WW, from below the permanent pycnocline. In autumn, the seasonal pycnocline deepens due to brine rejection and vertical mixing linked to sea ice production, initially entraining water from the WW (containing memory from the previous sea ice season) into the mixed layer. Once the seasonal pycnocline deepens beyond the WW, it starts to entrain water from the CDW, which in most cases is significantly warmer and does not contain sea ice-related memory. Entrainment of CDW changes the thermal signature in the WW and dissipates the ice-ocean memory causing the loss of sea ice predictability.

The ice-ocean memory (correlation between sea ice area and lagged upper ocean temperatures) underlying sea ice predictability can be interpreted as residual thermal anomalies from sea ice processes lingering in the ice-ocean system at inter-seasonal timescales. This implies that sea ice predictability presents a signature of local ice-ocean interactions. The predictability barrier identified in Chapter 2 is linked to the entrainment of CDW, and the entrainment of CDW applying a thermal limit to sea ice growth through negative ice-ocean feedback is well established in the literature (Martinson, 1990; Martinson and Iannuzzi, 1998). Hence, the

predictability barrier identified in Chapter 2 shows that local ice-ocean feedbacks is reflected in sea ice predictability, which further confirms the ice-ocean interactions nature of sea ice predictability. Therefore, sea ice predictability can be used as a tool to assess ice-ocean interactions (and feedbacks) in effect in a region.

Key findings from Chapter 2, are that (1) sea ice predictability presents a signature of local ice-ocean interaction which is tied to the local vertical ocean structure, (2) and that sea ice predictability is dependant not only on the depth of the Winter Water layer but also controlled by changes in the strength of stratification at the base of the Winter Water layer, are applied and tested in Chapter 3 for six sectors around Antarctica using the same ocean-sea ice model. The results in Chapter 3 are consistent with those in Chapter 2. In most sectors, the ice-ocean memory underlying sea ice predictability was present within the WW. Sea ice predictability and ice-ocean memory results for all six sectors revealed that variability in local vertical ocean structure and ice-ocean processes drives the regional differences in sea ice predictability. The main regional difference found in Chapter 3 is an east-west contrast in the persistence of summer sea ice anomalies, with this regional contrast linked to differences in regional upper ocean and sea ice conditions.

Western sectors (i.e., Amundsen, Bellingshausen, and Weddell Seas) sustained the persistence of summer sea ice anomalies over a longer period (~6 months lag) than the sectors in the east (~3 months lag, or less). The east-west contrast was evident with considerable contrast in the ice-ocean memory results, where ice-ocean memory emerging from summer months in the western sectors extended deeper into the ocean (below the permanent pycnocline, deeper than 150 m) and persisted longer, while those in the eastern sectors occurred in a shallower layer (above the seasonal pycnocline, ~50 m) and persisted less. This means that the ice-ocean memory within the WW in the eastern sectors is lost by summer, while it is retained in the western sectors. Evaluation of upper ocean and sea ice conditions revealed that weaker stratification at the seasonal pycnocline (above WW) and lack of summer sea ice cover resulted in higher rates of surface heat flux exchange in the eastern sectors. Higher surface heat flux in summer – which is enabled by the lack of the sea ice barrier - results in the loss of ice-ocean memory to the atmosphere.

Previous Antarctic sea ice predictability studies also encountered regional variability, which was largely attributed to the variability in mixed layer depth. The analysis in Chapter 3 is the first to explore the physical explanation underlying the regional variability in sea ice

predictability, from the perspective of ice-ocean processes. The results in Chapter 3 present further evidence on how local sea ice and ocean processes determine sea ice predictability in a region. Our findings revealed that sea ice predictability is not simply an outcome of passive retention of thermal anomalies (from large thermal inertia of the ocean), but rather a product of ice-ocean interactions, which are heavily controlled by the local vertical ocean structure and sea ice processes. Results from Chapter 3 were consistent with previous studies in finding that the persistence of sea ice predictability depends on the depth to which ice-ocean memory is stored. In addition, they showed the important role of stratification in preserving ice-ocean memory.

A predictability barrier was found in three of the six sectors (East Antarctic, Ross Sea, and Weddell Sea sectors). Of the three sectors that do not have a predictability barrier, the King Haakon and the Amundsen Sea sectors are net sea ice melting regions (i.e., they receive a net import of sea ice from adjacent sectors). Excess freshwater from sea ice melt stratifies the upper ocean and acts to inhibit the negative ice-ocean feedback (Martinson and Iannuzzi, 1998; Wilson et al., 2019). This result suggests that advected ice and ocean anomalies from major sea ice producers may be modifying ice-ocean feedbacks in regions downstream. Finding a predictability barrier in half of the sectors considered in this study demonstrates that the physical mechanism of predictability barrier driven by sea ice-ocean interaction is consistent in many regions around Antarctica.

The insights from Chapters 2 and 3 are applied in Chapter 4, comparing sea ice predictability in observations with predictability in the model. Chapter 4 assesses how realistic the model results from Chapters 2 and 3 are compared to observations. Overall, the sea ice predictability in the model is more persistent than that in observations. The overestimation of sea ice predictability in the model is traced back to systematic bias in the representation of the upper ocean vertical structure, specifically to the stratification in the model. An overly stratified upper ocean in the model is found to preserve ice-ocean memory over longer timescales and produce longer persistence of sea ice predictability. The overly stratified upper ocean is traced back to the model ocean being too fresh above the permanent pycnocline, and too warm below. The findings from this thesis further advance our knowledge of the complex interactions between Antarctic sea ice and the ocean below. The analysis used in this thesis presents a novel approach of using sea ice predictability as a tool to investigate regional ice-ocean interactions and biases in the representation of these interactions in fully coupled models.

Implications

- Our work shows that a better representation of the upper ocean in the sea ice zone of the Southern Ocean in coupled models will produce more accurate sea ice predictability through improved representation of local ice-ocean interactions.
- Based on our results, in a warming climate, strengthened stratification (from increased glacial meltwater, changes in precipitation, and sea ice production) will likely support the retention of memory in the upper ocean for a longer period and thereby increase the potential for sea ice predictability. Conversely, projected low summer sea ice cover will act to decrease sea ice predictability from higher rates of thermal memory loss to the atmosphere in ice-free ocean areas. Hence the findings from this thesis suggest these changes will likely create competing effects on sea ice predictability, which are at present unclear.

Future research directions

- In this thesis, the findings and interpretations are based solely on correlation analysis, and the arguments made are guided by climatological ocean and sea ice parameters. Quantifying the seasonal exchanges and thermal modifications occurring in the upper ocean identified in this thesis relevant to sea ice predictability is a potential follow-up analysis.
- Further, this thesis focuses primarily on the role of the ocean in driving Antarctic sea ice predictability. Antarctic sea ice has strong interaction with other components of the climate system than the ocean, which may also influence predictability (either by being a source of predictability or by reducing predictability). This thesis does not examine the influence of other components of the climate system on sea ice predictability. In this thesis, the synoptic atmospheric variability is assumed to reduce sea ice predictability (at the seasonal-to-interannual timescale). However, long-lived modes of atmospheric variability that affect sea ice at longer timescales (such as ENSO) may act as a source of sea ice predictability. Therefore, research focusing on the role of the

atmosphere in driving Antarctic sea ice predictability will be complementary to this study and can help to build a more complete picture of the drivers of Antarctic sea ice predictability.

- On the basis of strong regional variability identified in various Antarctic sea ice metrics, the analysis in this thesis was conducted by splitting the circumpolar Southern Ocean into six sectors. However, variability can arise in ice-ocean interaction within the sectors, especially in contrasting geographic areas within the sectors, such as the coastal or shelf regions (which are destratified) and the open ocean regions (which have warm CDW below WW). An analysis applying the method used in this thesis to evaluate the sea ice predictability and ice-ocean interactions in these contrasting regions is another potential follow-up study.
- The findings from this thesis suggest that increased stratification, reduced sea ice production, and lower summer sea ice cover can have competing effects on sea ice predictability. Therefore, a study to evaluate the impacts of projected Antarctic and Southern Ocean changes on sea ice predictability would be particularly pertinent.

Appendices

Appendix A: Supplementary information for chapter 2

Introduction

This supporting information contains the description of the methodological choice of selecting hydrographical profiles to denote the permanent pycnocline in the sea ice-covered Weddell Sea region (Text S1). Figure S1 shows ice-ocean correlations in the Weddell Sea region for all 12 initial months and Figure S2 shows the climatological hydrological profiles for temperature, salinity, and their vertical gradients for the Weddell Sea region.

Text S1. Defining the permanent pycnocline and seasonal pycnocline

In our analysis, we want to identify the boundary between three water masses: Circumpolar Deep Water (CDW), which has little seasonal variation; Winter Water (WW, the cold layer formed during sea ice formation); and the mixed layer (i.e., the surface layer at any month that is in close thermal equilibrium with sea ice). In polar waters, the traditional methods for identifying mixed layer depth may not be always as relevant as it is in much of the open-ocean regions, because polar waters are typically weakly stratified and are dominated by haline rather than thermal buoyancy fluxes. In this analysis, we have used a few simple proxies for our discrimination of these oceanic boundaries, mainly for computational simplicity.

The permanent pycnocline (PP) that bounds the CDW from the seasonally evolving layer above remains at approximately constant depth throughout the year. For the Weddell Sea region, the PP is encountered at about 100m, where all hydrographic profiles (Figure S2, a-d) converge from their highly fluctuating seasonal values to the nonvarying properties of CDW. To identify this point from the hydrographic profiles, we find that the vertical temperature gradient (dT/dz) maximum and the secondary maximum of vertical salinity gradients are located at the depth range of PP. It is computationally easier to pick out dT/dz maximum values than finding the secondary maximum values of dS/dz and both of these values are occurring at more or less the

same depth. This is because, due to the presence of WW, temperature inversions are produced, generating minima for vertical thermal gradient (negative values, instead of positive ones in dS/dz) due to the warm summer surface ocean layer over the cold WW in summer. Whereas with dS/dz with the sea ice meltwater, strong vertical salinity gradients are produced, and compared to these strong gradients near the surface, the secondary vertical salinity gradients are hard to tease out. Hence, we find using dT/dz maximum values (Figure S2, all panels black circles) as an effective and easy way of identifying the depth of PP.

For the seasonal pycnocline, which acts mostly as the base of the mixed layer, we find using the maximum vertical density gradient (Figure S2, all panels red crosses) as the effective method. Although not shown in Figure S2, the profiles of vertical density gradient closely follow those of vertical salinity gradient, once again to say that the density changes are haline-dominated. Freshwater and surface ocean warming during the ice-melt season (December-February) produce a thin and highly stratified surface layer that becomes the summer mixed layer and the maximum density gradient in the upper ocean (0-100m) helps to follow the seasonal evolution of this stratification that forms the base of the mixed layer.

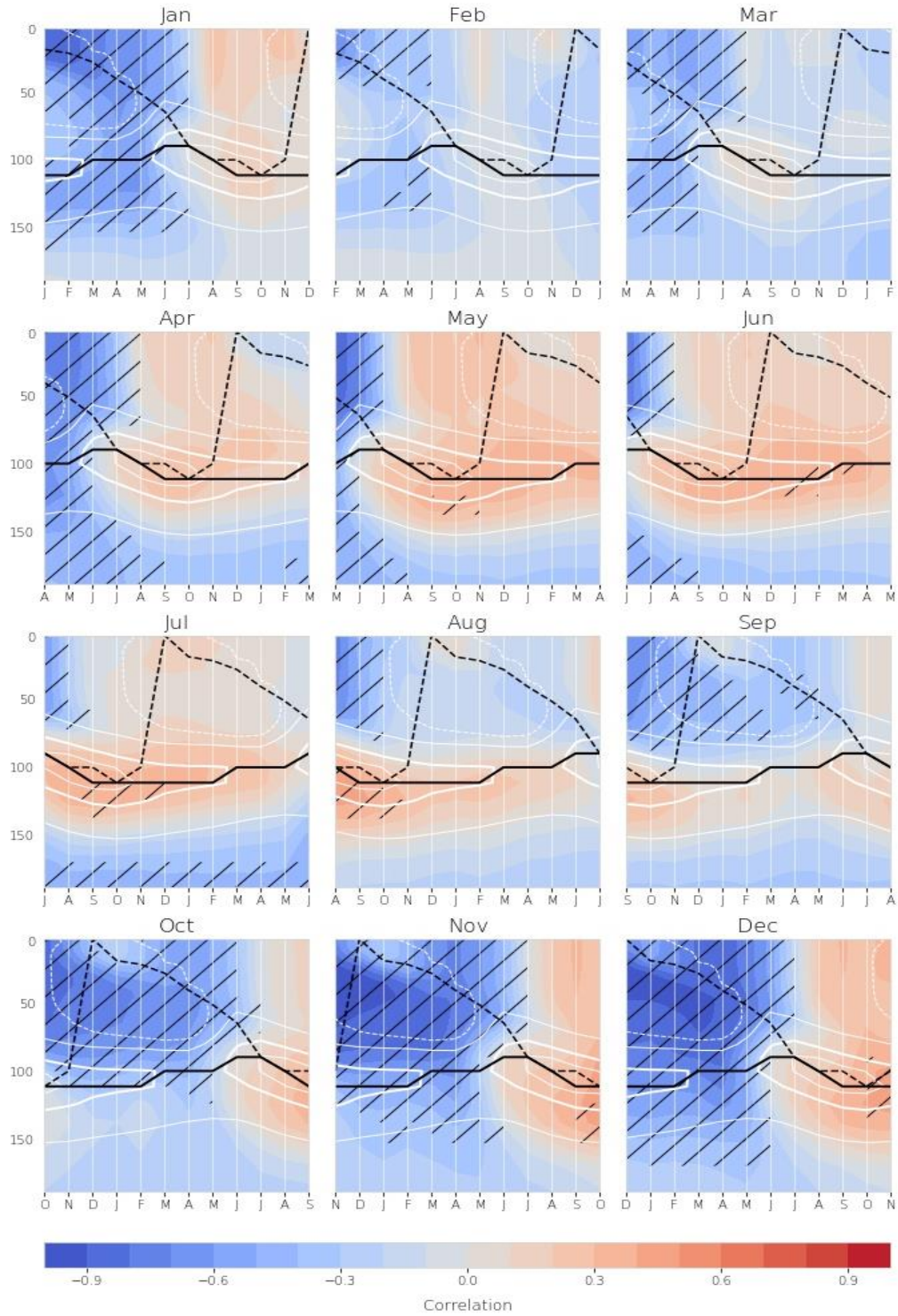


Figure S1: Correlation between sea ice area (SIA) and ocean temperature in future months (ice-ocean correlations) for all 12 initial months in the Weddell Sea sector. The black line is the vertical temperature gradient (dT/dz) maximum, the dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification. Statistically significant values (>95%) are hatched in all panels.

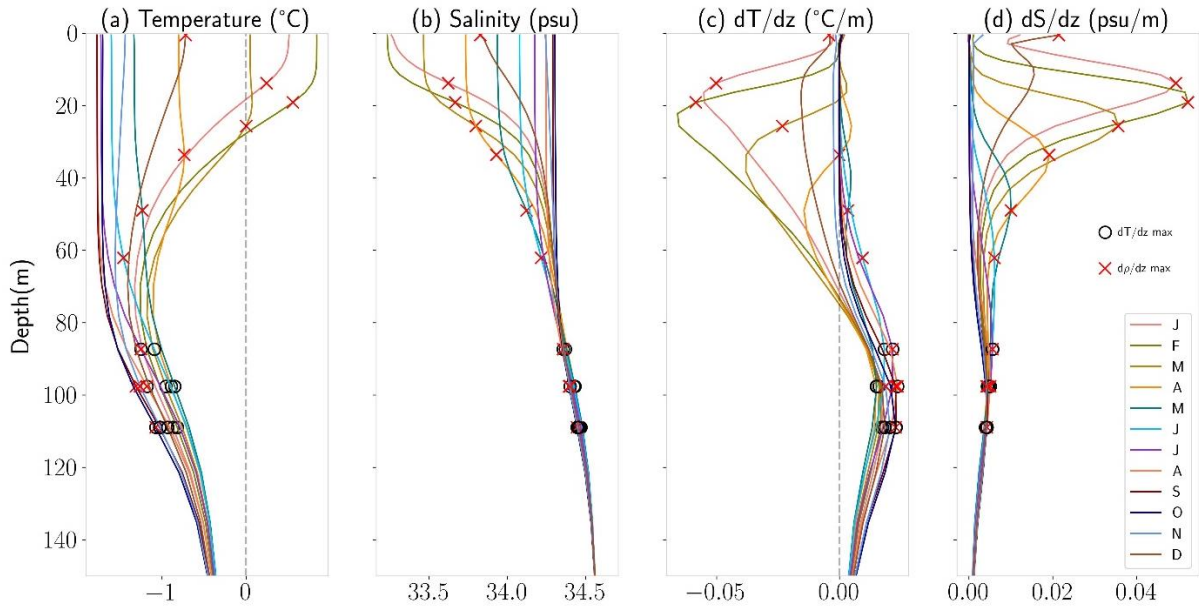


Figure S2: Hydrological profiles of spatially averaged monthly climatological (a) temperature, (b) salinity, (c) vertical temperature gradient, (d) vertical salinity gradient with markers of vertical temperature gradient maximum (black circles) and vertical density gradient maximum (red circles) in the Weddell Sea sector.

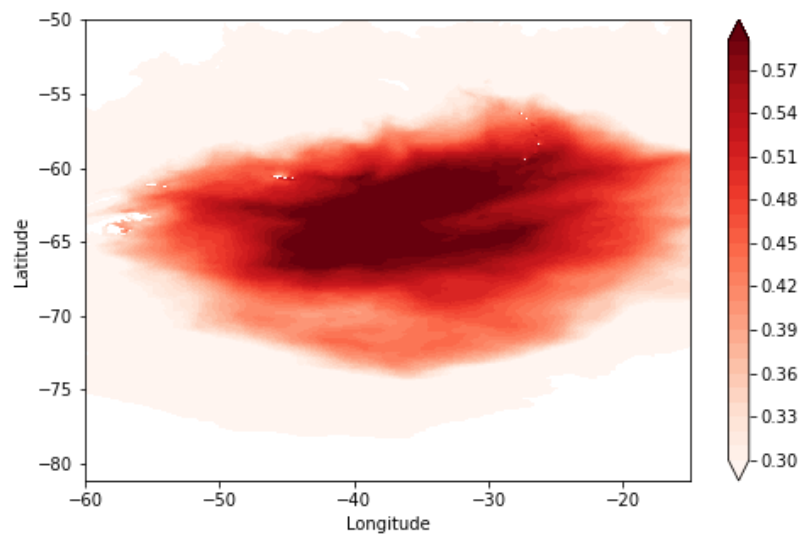


Figure S3: Correlation between the total sea ice area and sea ice concentration in the Weddell Sea region. Longitude along the x-axis and latitude along the y-axis.

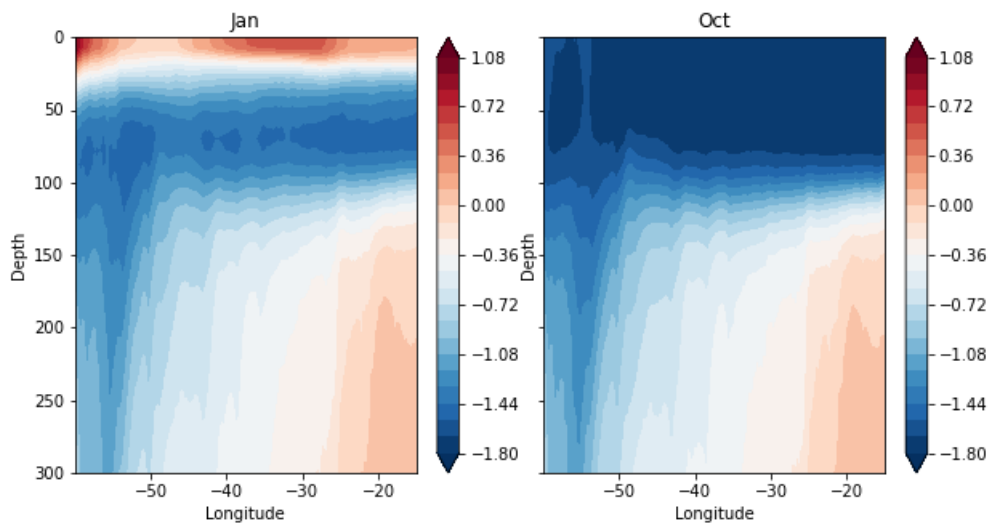


Figure S4: Longitude-depth section plot of Weddell Sea ocean temperatures in January (representing summer) and October (representing winter).

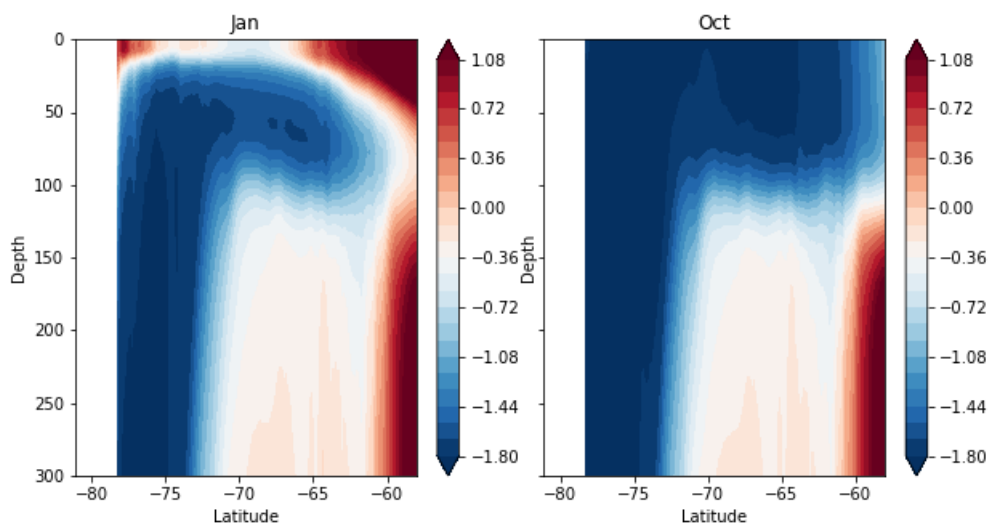


Figure S5: Latitude-depth section plot of Weddell Sea ocean temperatures in January (representing summer) and October (representing winter).

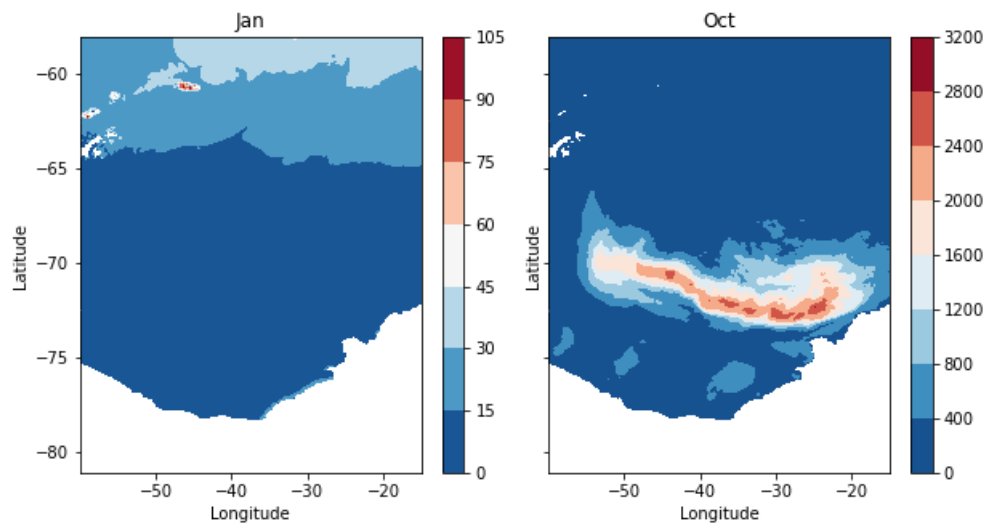


Figure S6: Regional plot of Weddell Sea mixed layer depth distribution in January (representing summer) and October (representing winter). The winter mixed layer depths show the model producing considerably deeper mixed layers near the continental shelf edge (compared to the observed depth ranges, not shown).

Weddell (300-345°E), 1985-2018

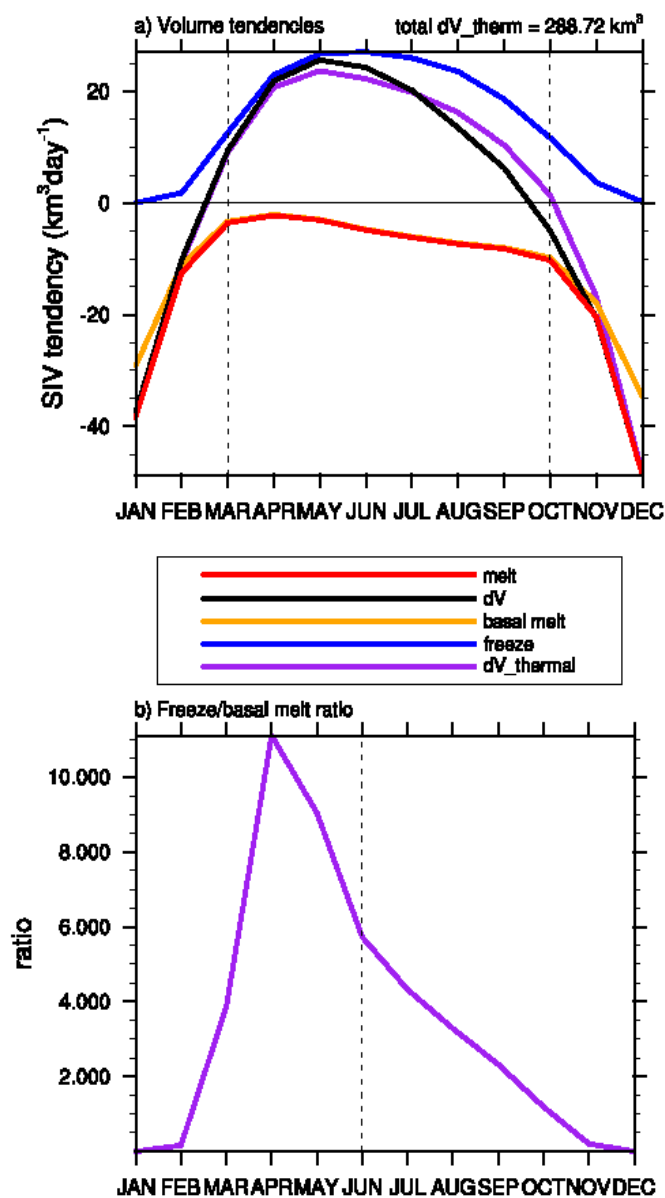


Figure S7: (a) Sea ice volume tendencies in the Weddell Sea sector, showing the contribution from various freeze and melt components; (b) the net freeze by net melt ratio for the Weddell Sea sector.

Appendix B: Supplementary information for chapter 3

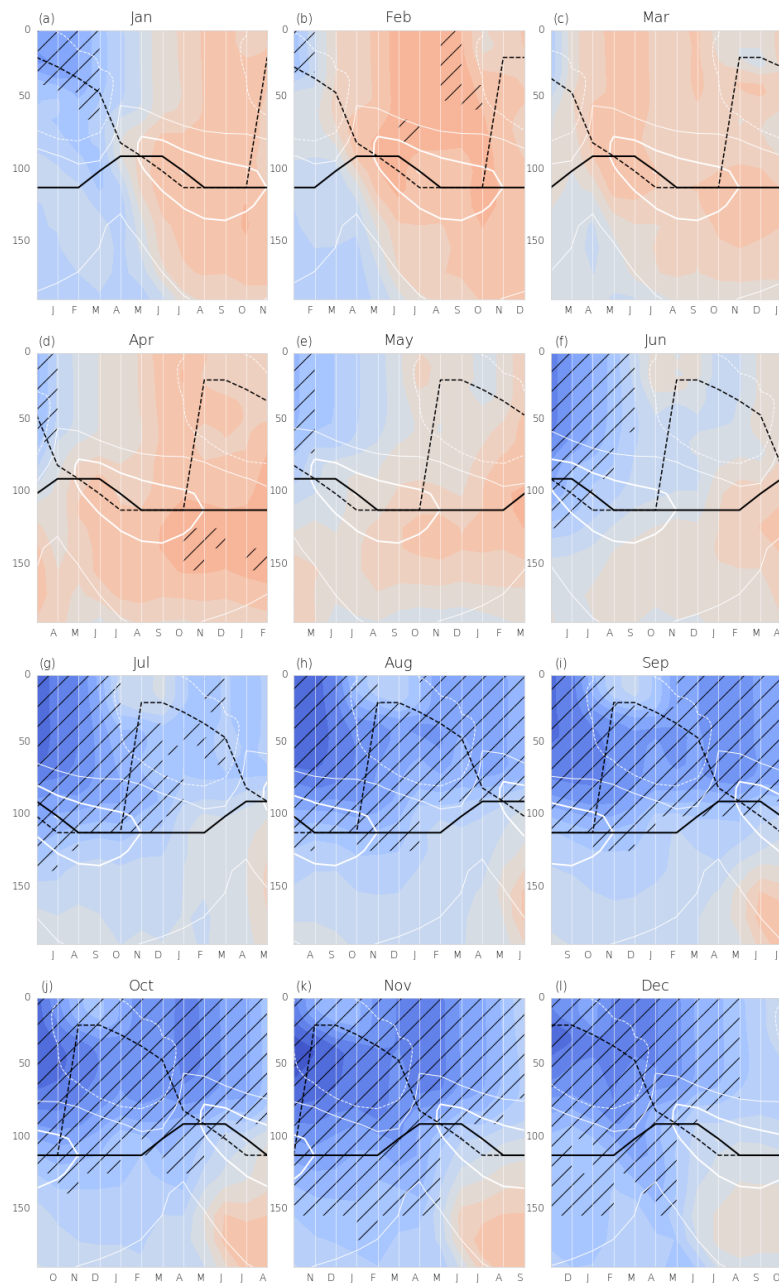


Figure Sx1: Correlation between initial sea ice area and ocean temperature in future months (ice-ocean correlations) for all 12 initial months for the King Haakon VII sea region. The black line is the vertical temperature gradient (dT/dz) maximum, the dashed black line is the vertical density gradient (dp/dz) maximum, white contours are dT/dz contours, and white dashed contours bound the dT/dz values that are negative during summer stratification. Statistically significant values (>95%) are hatched in all panels. The initial months for the sea ice area are marked as the title on each subplot)

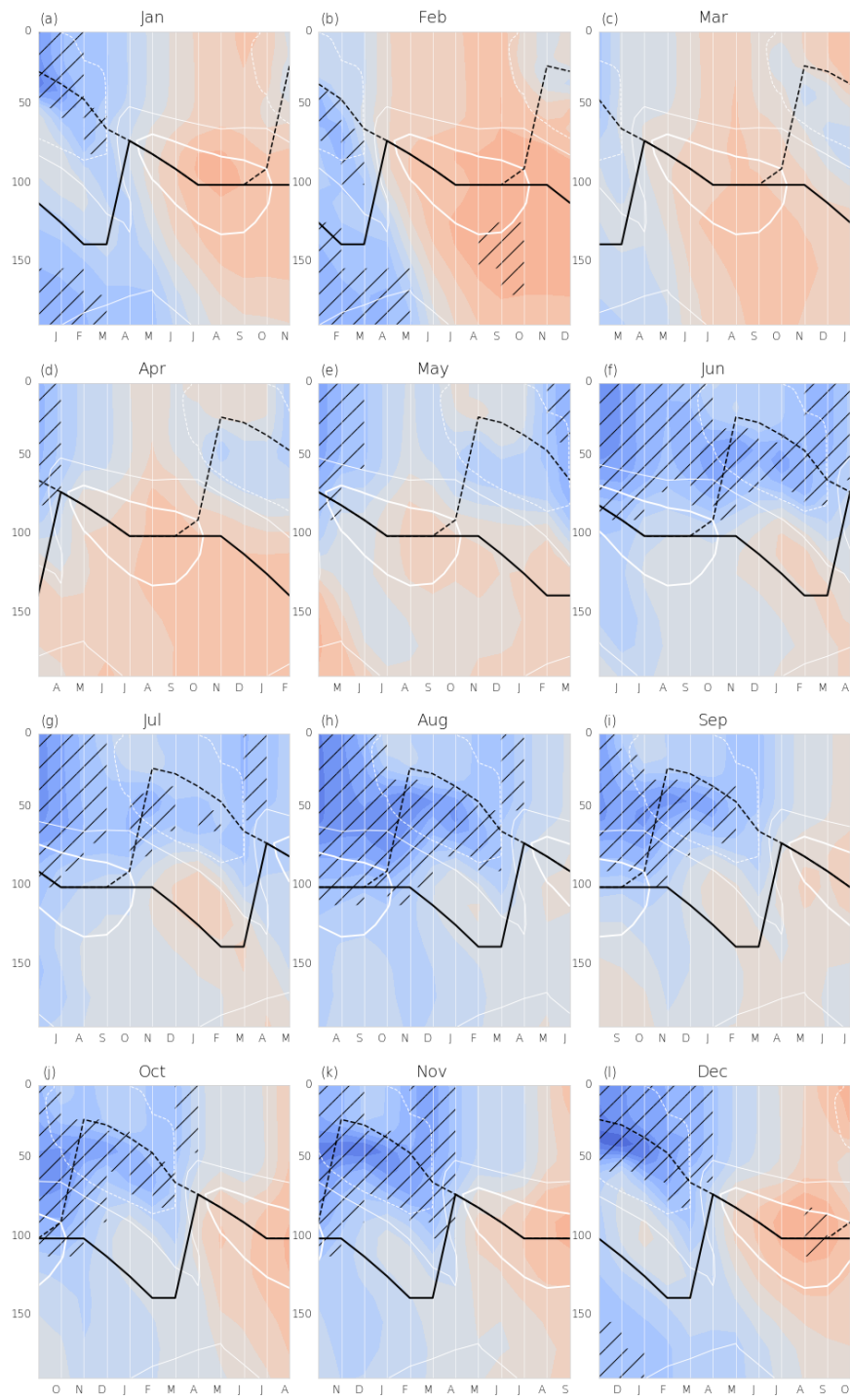


Figure Sx2: Same as Sx1, but for the East Antarctic sector.

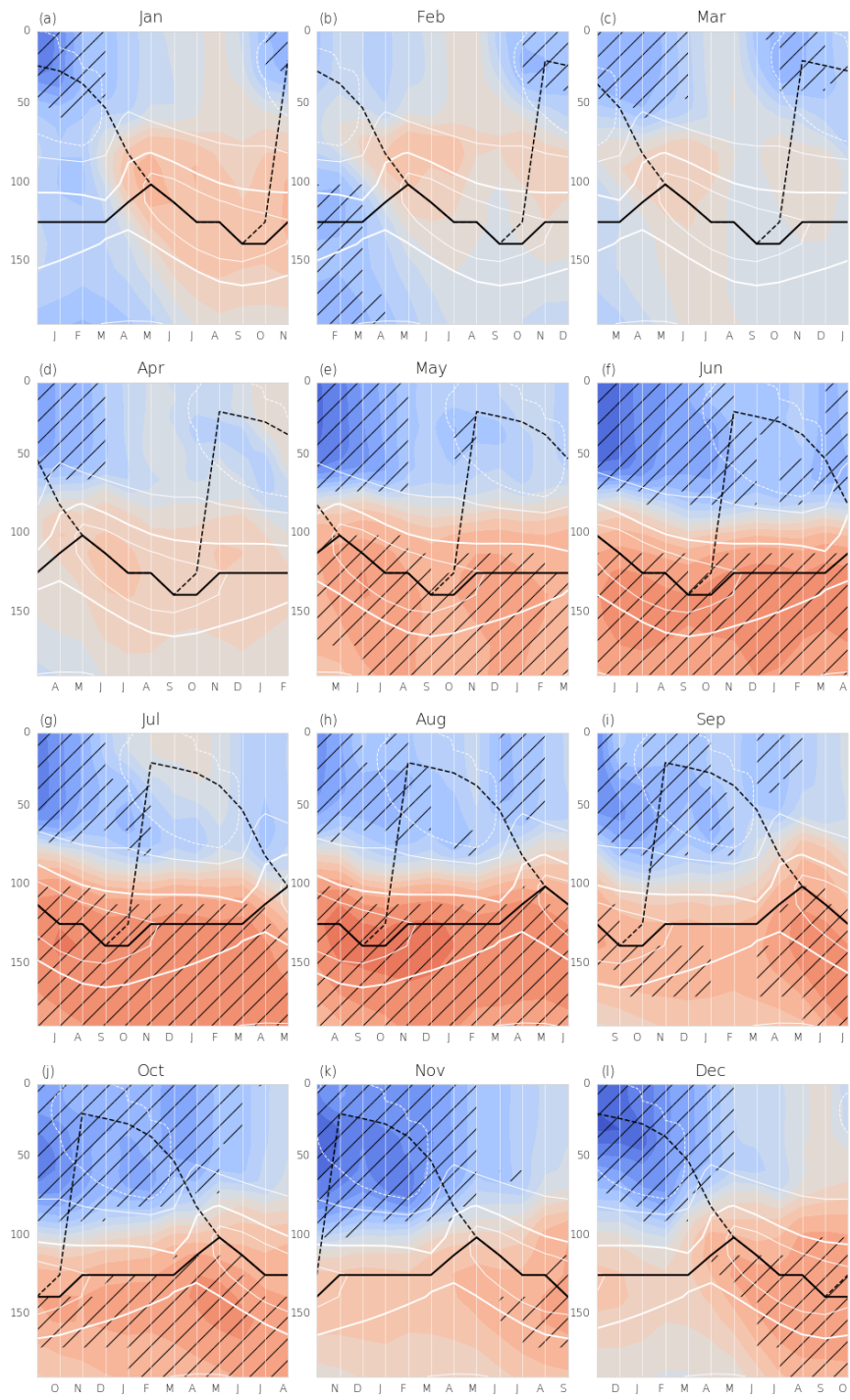


Figure Sx3: Same as Sx1, but for the Ross Sea sector.

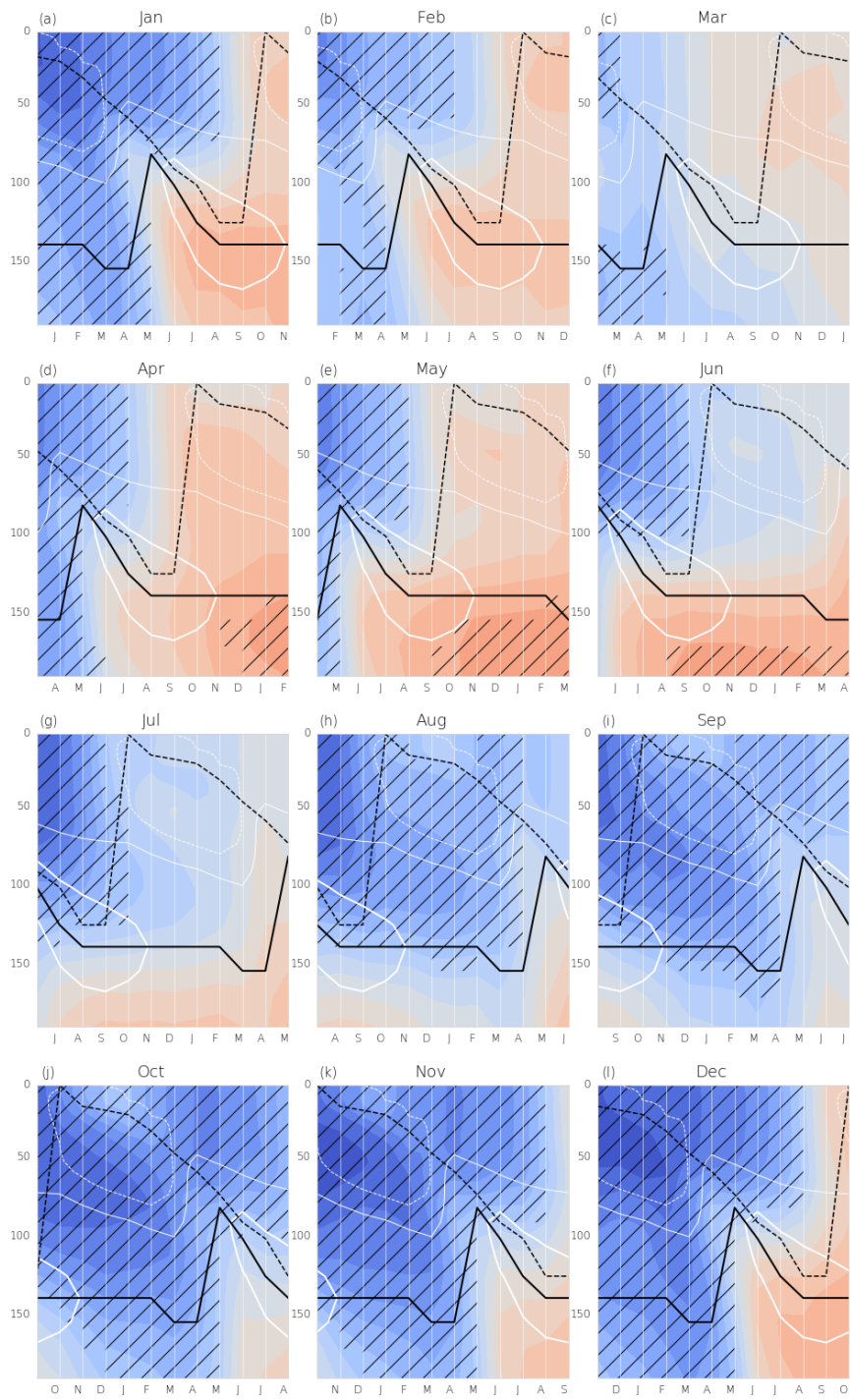


Figure Sx4: Same as Sx1, but for the Amundsen Sea sector.

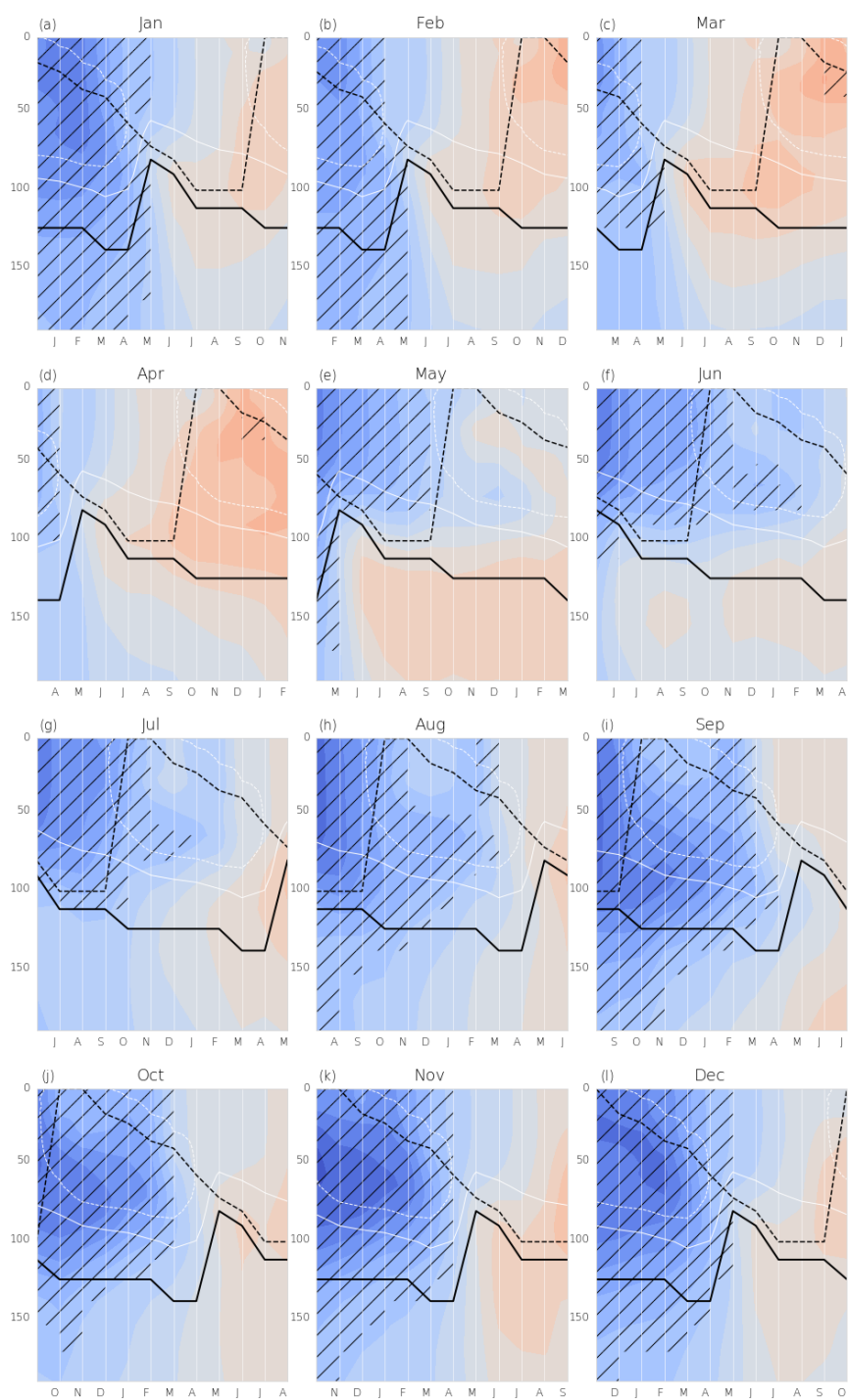


Figure Sx5: Same as Sx1, but for the Bellingshausen Sea sector.

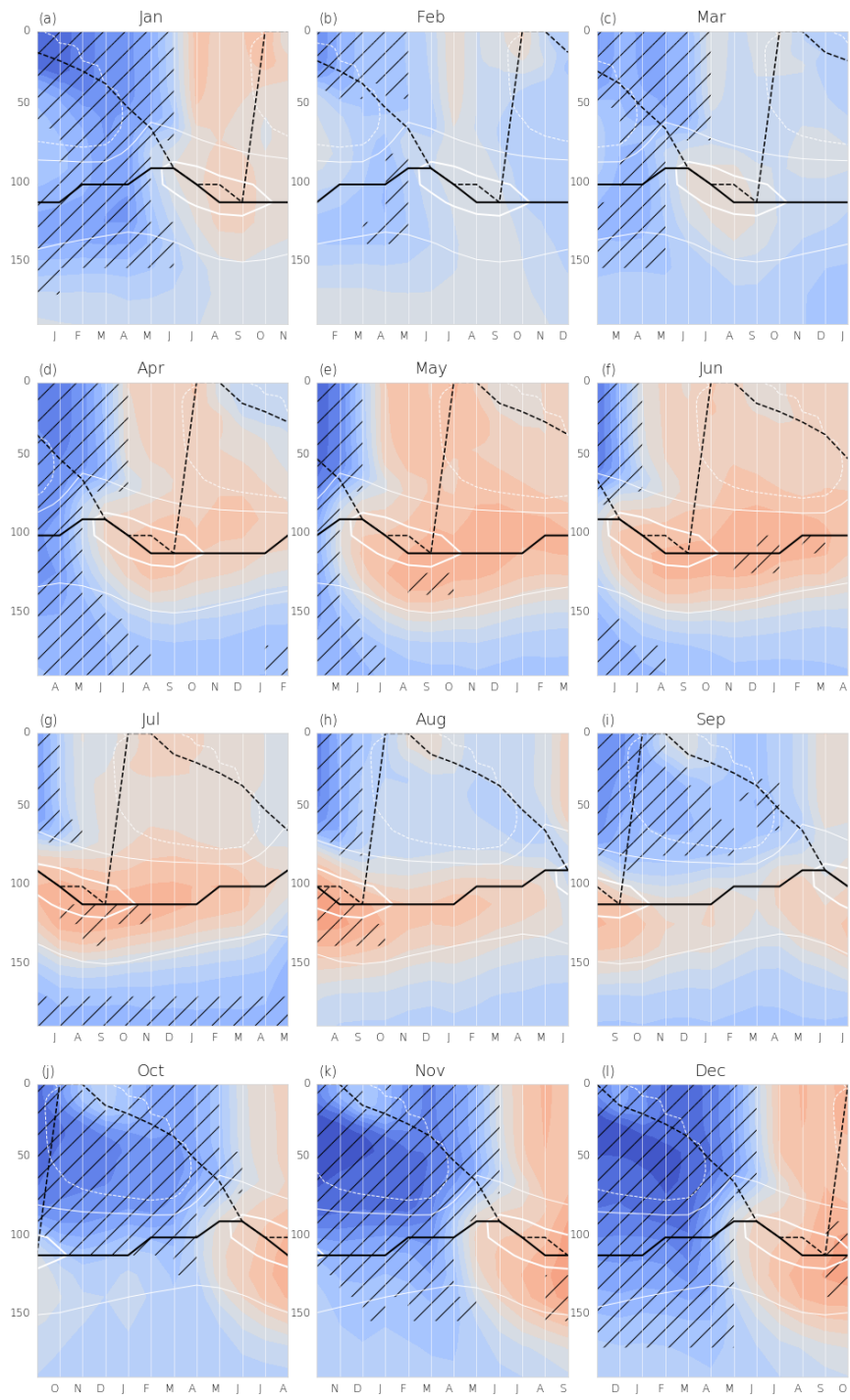


Figure Sx6: Same as Sx1, but for the Weddell Sea sector.

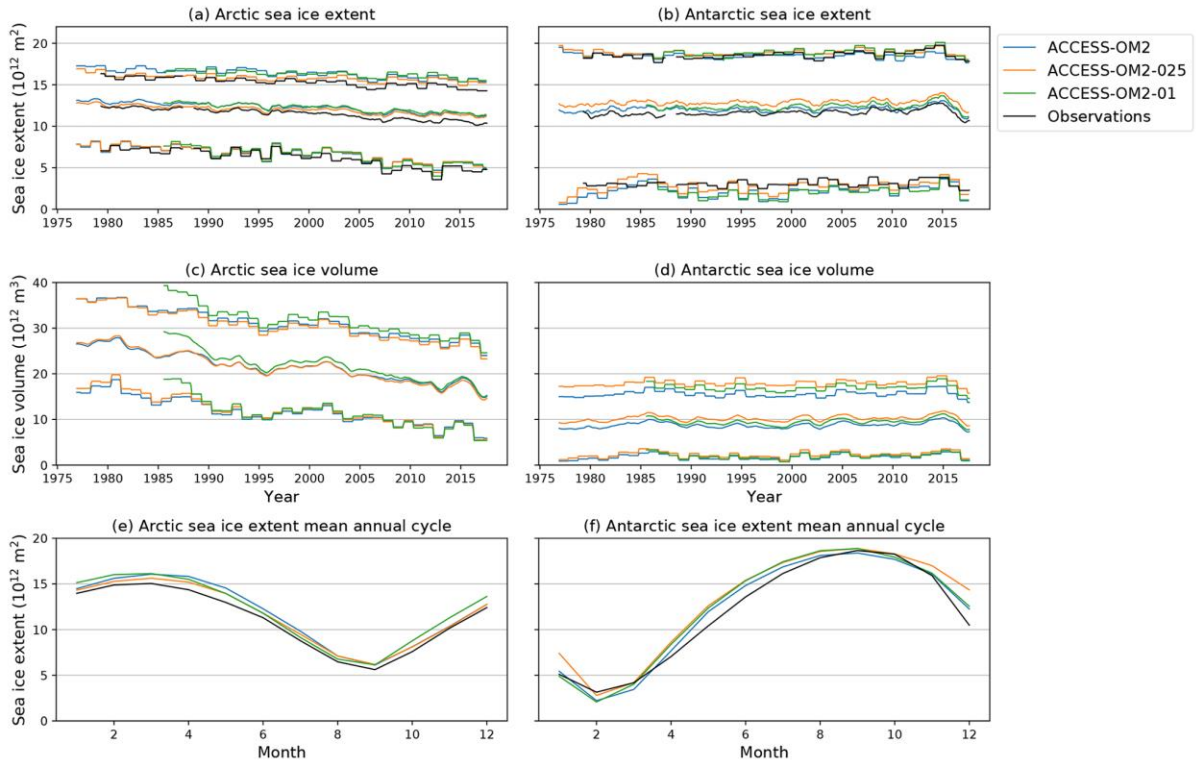


Figure Sx7: Evaluation of Arctic and Antarctic sea ice simulated in the ACCESS-OM2 model against the observed sea ice record. (a–b) Sea ice extent, (c–d) volume, and (e–f) 1993–2017 mean annual cycle of the extent in the (a, c, e) Arctic and (b, d, f) Antarctic in the three configurations. Panels (a)–(d) show running 12-month minima, means, and maxima. Observational estimates are based on passive microwave retrievals (NSIDC Sea Ice Index version 3, Fetterer et al., 2017). Note that the three models and observations use different land masks, so the time series would not be expected to agree perfectly in magnitude. Figure extracted from (Kiss et al., 2020) with permission from the author.

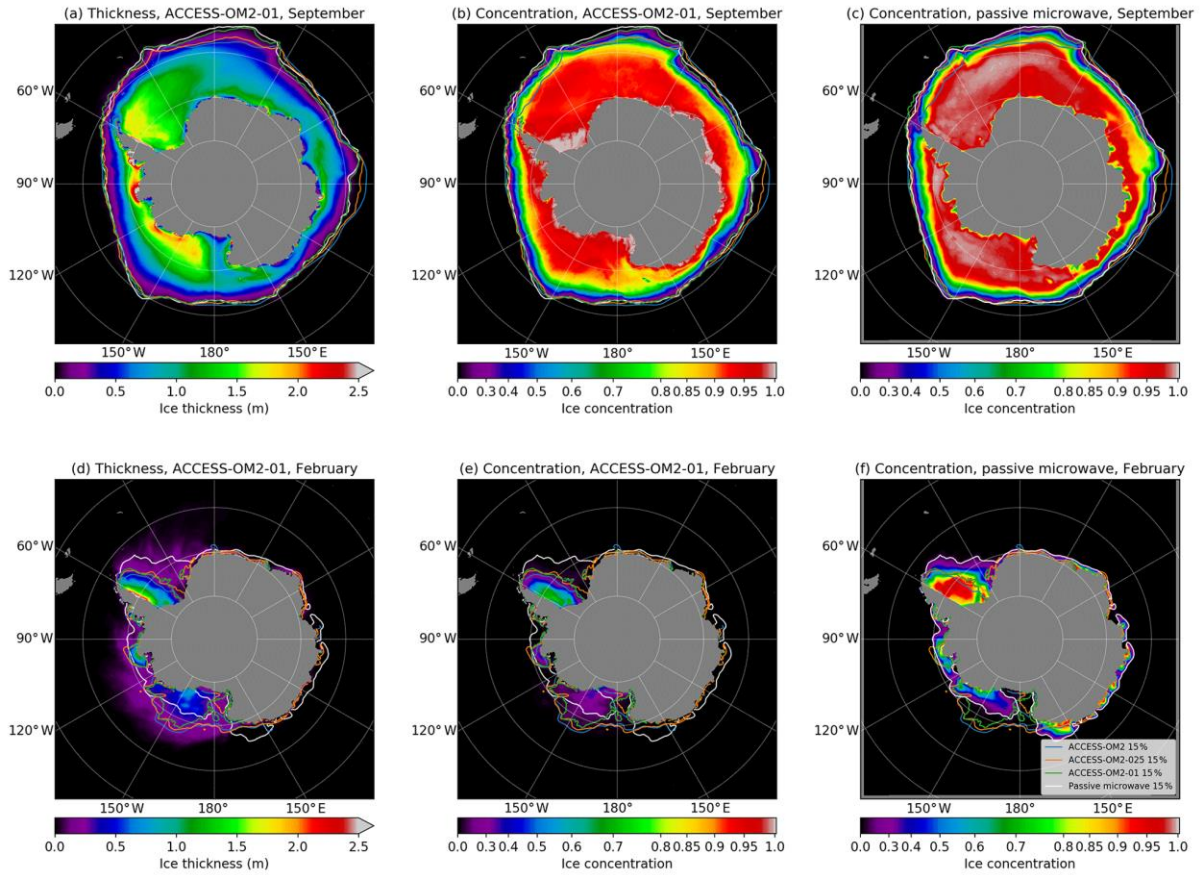


Figure Sx8: The 1993–2017 mean Antarctic sea ice thickness (ice volume per unit area; a, d) and concentration (ice area per unit area; b, e) in ACCESS-OM2-01 and concentration estimates from NOAA G02202 V3 passive microwave Goddard merged monthly data (Peng et al., 2013; Meier et al., 2017, c, f) for (a)–(c) September and (d)–(f) February. The concentration scale is nonlinear to highlight differences at high concentrations. Contour lines show the sea ice extent (defined as the 15 % concentration contour) in all three model configurations and observations. Figure extracted from (Kiss et al., 2020) with permission from the author.

Appendix C: Supplementary information for chapter 4

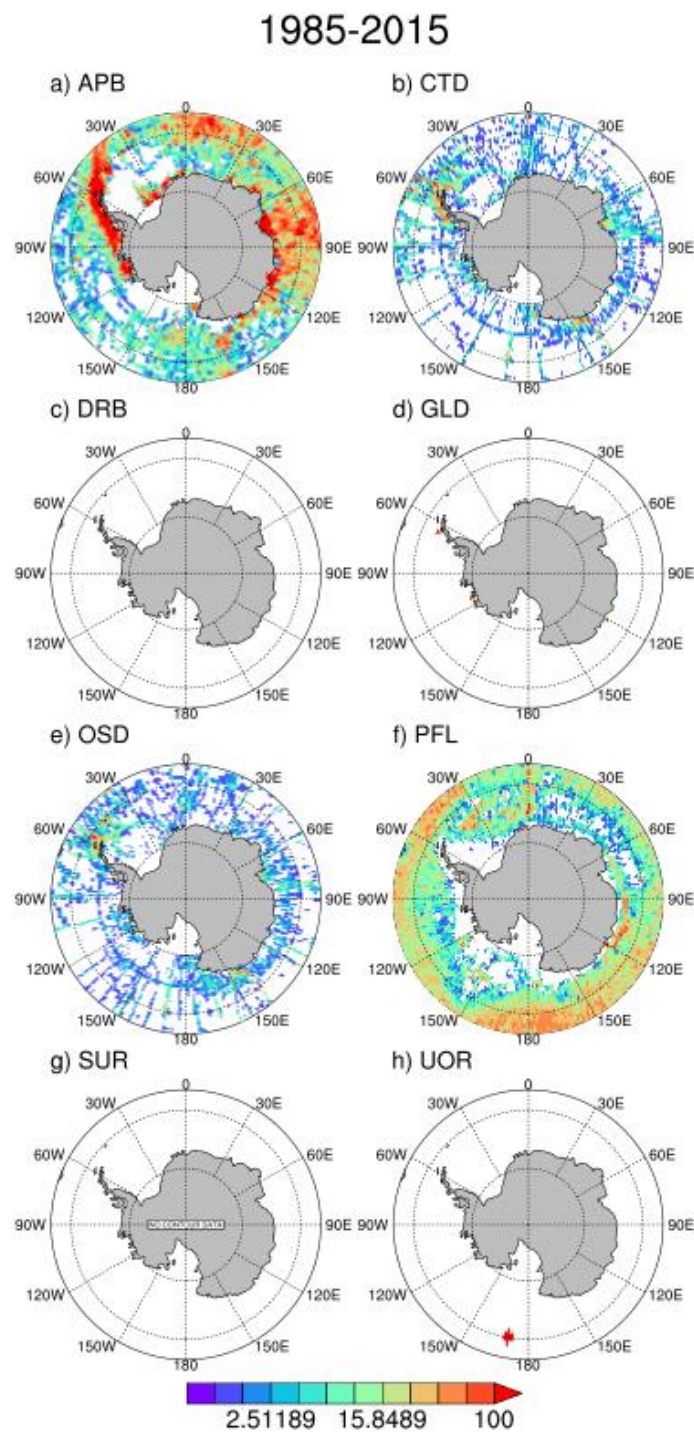


Figure Sz1: The spatial distribution of ocean profiles; (a) instrumented seal data, (b) CTD data, (c) drifting Buoy data, (d) glider data, (e) ocean station data, (f) Argo float data, (g) moored data and (h) Undulating Oceanographic Recorder (Towed CTD) data.

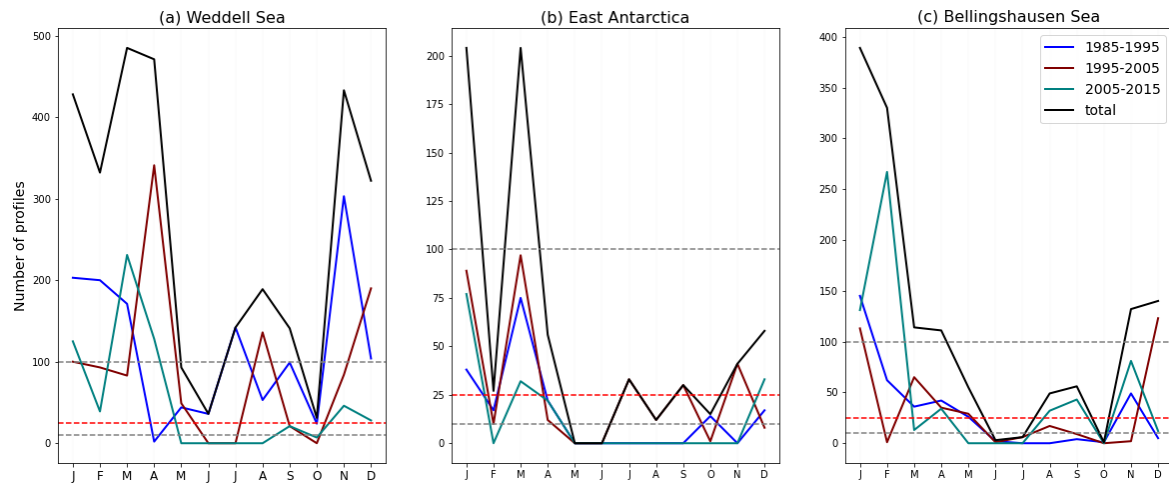


Figure Sz2: Decomposition of the number of observed ocean profiles in the (a) Weddell Sea, (b) East Antarctic and (c) Bellingshausen Sea sectors for the three decades of the study period.

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